

UNIVERSITY OF EDUCATION, WINNEBA
FACULTY OF TECHNICAL EDUCATION
DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
EDUCATION

DESIGN AND IMPLEMENTATION OF A GHANA SENIOR HIGH
SCHOOL LEARNING MANAGEMENT SYSTEM (GSLMS)

A PROJECT REPORT SUBMITTED TO THE DEPARTMENT OF INFORMATION AND
COMMUNICATION TECHNOLOGY EDUCATION, UNIVERSITY OF EDUCATION,
WINNEBA, IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF
THE DEGREE OF BACHELOR OF SCIENCE (BSc) IN ICT EDUCATION

BY
GROUP ONE (1)

2026

DECLARATION

We hereby declare that this project report titled “DESIGN AND IMPLEMENTATION OF A GHANA SENIOR HIGH SCHOOL LEARNING MANAGEMENT SYSTEM (GSLMS)” is our original work and has not been submitted, in whole or in part, to any other institution for the award of a degree or any other academic qualification.

We further declare that all sources of information used in this study have been duly acknowledged and referenced in accordance with the required academic referencing standards. We accept responsibility for the accuracy and originality of the work presented in this project report.

GROUP ONE (1) LEADER

Signature:

Date:

CERTIFICATION

I hereby certify that this project report titled “**DESIGN AND IMPLEMENTATION OF A GHANA SENIOR HIGH SCHOOL LEARNING MANAGEMENT SYSTEM (GSLMS)**” was carried out by **GROUP ONE (1)** under my supervision and has been submitted to the Department of Information and Communication Technology Education, University of Education, Winneba, in partial fulfilment of the requirements for the award of the degree of Bachelor of Science (BSc) in ICT Education.

To the best of my knowledge, this project report is the original work of the students and is considered satisfactory for submission for examination.

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DEDICATION

We dedicate this project to our beloved families for their unwavering love, encouragement, sacrifices, and support throughout our academic journey.

We also dedicate this work to our lecturers, colleagues, and friends whose guidance, motivation, and encouragement contributed to the successful completion of this project.

Finally, we dedicate this project to all educators and learners who continue to embrace technology as a means of improving teaching, learning, and academic management.

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GROUP ONE (1)

ABSTRACT

The study focused on the design and implementation of a **Ghana Senior High School Learning Management System (GSLMS)** to address challenges associated with traditional and fragmented academic management processes in Senior High Schools. The analysis of the existing system identified several challenges, including manual record keeping, fragmented academic information, difficulties in accessing learning materials, delays in assignment submission and grading, inefficient communication, challenges in monitoring student performance, and difficulties in generating academic reports. These challenges can affect the efficiency of academic administration and the timely availability of educational information to authorised users.

To address these challenges, a web-based Learning Management System was designed and implemented to provide a centralised platform for administrators, teachers, and students. The system provides functionality for user management, course management, learning material management, assignment creation and submission, assessment and grading, attendance management, academic results, communication and notifications, and report generation. The system adopts an API-first architecture consisting of a frontend developed using **HTML5, CSS3, and JavaScript**, a **PHP-based RESTful API**, and a **MySQL database**. Communication between the frontend and backend is facilitated through HTTP requests using the **Fetch API**, while data is exchanged in **JSON** format.

The developed system was evaluated through functional, access-control, API, and database testing. The test results documented in the study indicate that the tested system functions produced their expected outcomes. The implementation therefore demonstrates the feasibility of providing an integrated platform for managing major academic activities within a Senior High School environment.

The study concludes that the GSLMS provides a practical approach to reducing dependence on manual and fragmented academic processes while improving the centralisation, accessibility, and management of educational information. However, wider institutional deployment would require further testing, user training, adequate infrastructure, reliable internet connectivity, and continuous system maintenance.

Keywords: Ghana Senior High School, Learning Management System, Academic Management, RESTful API, MySQL, E-Learning, Web-Based System.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CSS	Cascading Style Sheets
CRUD	Create, Read, Update and Delete
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
GSLMS	Ghana Senior High School Learning Management System
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
ICT	Information and Communication Technology
JSON	JavaScript Object Notation
JWT	JSON Web Token
LMS	Learning Management System
MVC	Model–View–Controller
PHP	Hypertext Preprocessor
REST	Representational State Transfer
REST API	Representational State Transfer Application Programming Interface
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
XAMPP	Cross-Platform, Apache, MySQL, PHP and Perl
MySQL	My Structured Query Language
RBAC	Role-Based Access Control
CRUD	Create, Read, Update and Delete
AJAX	Asynchronous JavaScript and XML
CORS	Cross-Origin Resource Sharing
MIME	Multipurpose Internet Mail Extensions
SMTP	Simple Mail Transfer Protocol
PDF	Portable Document Format
CSV	Comma-Separated Values
GPA	Grade Point Average
SHS	Senior High School
WAEC	West African Examinations Council

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

The rapid advancement of Information and Communication Technology (ICT) has significantly transformed the educational sector worldwide. Educational institutions increasingly rely on digital technologies to enhance teaching, learning, communication, and administrative processes. One of the most important technological innovations in education is the Learning Management System (LMS), which provides a platform for delivering, managing, tracking, and evaluating educational content and activities.

Traditionally, many educational institutions have relied on face-to-face classroom instruction and manual methods for managing academic activities such as course registration, assignment submission, grading, attendance monitoring, and communication between lecturers and students. These manual processes are often time-consuming, prone to errors, difficult to monitor, and inefficient in handling large volumes of academic information. The emergence of web-based learning systems has provided educational institutions with opportunities to overcome these challenges through automation and digitalisation.

A Learning Management System is a software application that facilitates the administration, documentation, tracking, reporting, and delivery of educational courses and training programmes. LMS platforms provide a centralised environment where administrators, lecturers, and students can interact effectively. Through such systems,

lecturers can upload learning materials, create assignments, conduct assessments, and monitor student progress, while students can access course content, submit assignments, participate in discussions, and view academic results from any location with internet access (Al-Busaidi & Al-Shihi, 2010).

Several institutions currently use popular LMS platforms such as Moodle, Canvas, and Blackboard Learn. Although these systems offer numerous functionalities, many educational institutions require customised solutions tailored to their unique academic structures, administrative procedures, and user requirements. Furthermore, some existing systems may be complex for users, contain unnecessary features, or fail to address specific institutional challenges.

This project therefore proposes the design and implementation of a School Learning Management System that provides a user-friendly platform for managing academic activities. The proposed system will enable administrators to manage users, courses, departments, and academic records, while teachers can upload learning materials, create assignments, record grades, and communicate with students. Students will be able to access learning resources, submit coursework, monitor their academic performance, and interact with instructors through an integrated digital platform.

The implementation of this system is expected to improve educational service delivery, increase operational efficiency, reduce administrative workload, and enhance the overall teaching and learning experience.

1.2 Statement of the Problem

Many schools and educational institutions continue to face challenges in managing academic activities effectively. Traditional methods involving paper-based records and fragmented digital systems often result in inefficiencies, duplication of data, delayed communication, and difficulties in tracking student progress. Integrated learning management systems have been identified as effective tools for centralising academic information, supporting communication, and improving the monitoring of teaching and learning activities (Coates et al., 2005; Watson & Watson, 2007). Some of the major problems identified include:

1. Difficulty in managing large volumes of student and course information.
2. Delays in assignment submission, grading, and feedback processes.
3. Limited access to learning materials outside the classroom environment.
4. Poor communication between students, teachers, and administrators.
5. Challenges in generating academic reports and monitoring student performance.
6. Inadequate centralisation of educational resources and records.
7. Dependence on physical presence for many academic activities.

These challenges negatively affect educational administration and learning outcomes.

There is therefore a need for an integrated Learning Management System that can automate academic processes and improve educational service delivery.

1.3 Aim of the Study

The aim of this study is to design and implement a web-based School Learning Management System that facilitates the management of academic activities, improves communication among stakeholders, and enhances the teaching and learning process.

1.4 Objectives of the Study

The objectives of this study are:

1. To analyse the challenges associated with traditional academic management systems.
2. To design a user-friendly Learning Management System for educational institutions.
3. To develop a platform for managing students, teachers, courses, and academic records.
4. To provide facilities for uploading and accessing learning materials online.
5. To enable assignment creation, submission, and grading electronically.
6. To facilitate communication between students, teachers, and administrators.
7. To improve the efficiency and effectiveness of educational administration.

1.5 Research Questions

The study seeks to answer the following questions:

1. What challenges are associated with traditional academic management and learning systems in educational institutions?
2. How can a Learning Management System improve the management of academic activities, learning resources, and assignment processes?
3. How can a Learning Management System enhance communication and collaboration among administrators, teachers, students, and parents?
4. How can a Learning Management System facilitate efficient monitoring and evaluation of student academic performance?

1.6 Significance of the Study

The significance of this study lies in its potential contribution to improving educational management and learning experiences within educational institutions.

Students

The system will provide students with easy access to learning materials, assignments, grades, and academic information from any location.

Teachers

Teachers will benefit from simplified course management, assignment distribution, grading processes, and communication tools.

School Administrators

Administrators will have access to centralised academic records, reporting tools, and user management facilities, thereby improving institutional efficiency.

Educational Institutions

The implementation of the system will support digital transformation initiatives and contribute to improved educational service delivery.

Researchers

The study will serve as a reference material for future researchers interested in Learning Management Systems and educational technology.

1.7 Scope of the Study

This project focuses on the design and implementation of a web-based Learning Management System for educational institutions.

The system will provide functionalities including:

- User authentication and authorisation.
- Student management.
- Teacher management.
- Course management.
- Learning material management.
- Assignment management.
- Grade management.
- Academic reporting.

- Communication and notifications.
- Role-based access control.

The system will primarily support three categories of users:

- Administrators
- Teachers
- Students

The study will concentrate on the development and implementation of the software system and will not cover broader institutional policies or curriculum development processes.

1.8 Limitations of the Study

The study is subject to the following limitations:

1. Time constraints associated with the project development period.
2. Financial limitations affecting access to advanced development tools and infrastructure.
3. Dependence on internet connectivity for system accessibility.
4. Limited testing within selected educational environments.
5. Potential resistance to technological adoption by some users.

Despite these limitations, the system is expected to achieve its intended objectives effectively.

1.9 Organisation of the Study

This study is organised into five chapters.

Chapter One – Introduction

This chapter presents the background of the study, statement of the problem, aim and objectives, research questions, significance, scope, limitations, and organisation of the study.

Chapter Two – Literature Review

This chapter reviews relevant literature relating to Learning Management Systems, educational technologies, web-based learning environments, and existing LMS platforms.

Chapter Three – System Analysis and Feasibility Study

This chapter discusses the analysis of the existing system, requirements gathering, feasibility studies, system requirements, and system modelling diagrams.

Chapter Four – System Design and Implementation

This chapter presents the design and implementation of the proposed Learning Management System, including database design, system architecture, user interface design, and implementation details.

Chapter Five – Summary, Conclusion and Recommendations

This chapter provides a summary of the study, conclusions drawn from the findings, and recommendations for future improvements and further research.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews relevant literature related to Learning Management Systems (LMS), e-learning technologies, web-based educational systems, and digital learning environments. The review provides a theoretical and empirical foundation for the design and implementation of the proposed School Learning Management System. It examines key concepts, existing systems, technological trends, and previous studies relevant to educational technology and online learning.

2.2 Concept of Learning Management Systems

A Learning Management System (LMS) is a software application designed to facilitate the administration, documentation, delivery, tracking, reporting, and evaluation of educational courses and training programmes. LMS platforms provide a centralized environment where learners, instructors, and administrators can interact and access educational resources efficiently (Watson & Watson, 2007; Ellis, 2009). The emergence of Learning Management Systems has transformed traditional educational practices by enabling institutions to digitize learning activities and improve educational service delivery. Through an LMS, educational content can be distributed electronically, assessments can be conducted online, and communication between stakeholders can be enhanced regardless of geographical location (Ellis, 2009).

Modern LMS platforms support a variety of functionalities including:

- User management
- Course management
- Assignment management
- Online assessments
- Grade management
- Discussion forums
- Learning analytics
- Communication and notifications

The increasing demand for flexible learning opportunities has further accelerated the adoption of LMS solutions in schools, colleges, universities, and corporate training environments (Coates et al., 2005; Islam, 2013).

2.3 Electronic Learning (E-Learning)

Electronic learning, commonly referred to as e-learning, involves the use of digital technologies to facilitate learning and instructional delivery. E-learning enables students to access educational materials using computers, mobile devices, and internet-based platforms (Clark & Mayer, 2016).

According to Moore and Kearsley (2012), e-learning creates opportunities for learners to engage in educational activities independent of time and geographical constraints.

Educational institutions have increasingly adopted e-learning technologies to support distance education, blended learning, and fully online programs.

According to Sun et al. (2008) and Selim (2007), the major benefits of e-learning include:

- Increased accessibility to educational resources.
- Flexibility in learning schedules.
- Improved communication between learners and instructors.
- Reduction in administrative workload.
- Enhanced monitoring of learner progress.

Despite these advantages, e-learning systems may encounter challenges such as technological barriers, internet connectivity issues, inadequate ICT infrastructure, and resistance to technology adoption (Tarus et al., 2015; Romiszowski, 2004).

2.4 Web-Based Learning Systems

Web-based learning systems utilize internet technologies to provide educational content and facilitate interaction among learners and instructors. These systems allow users to access educational materials through web browsers without requiring specialized software installations (Al-Busaidi & Al-Shihi, 2010).

Web-based learning environments offer several advantages over traditional classroom methods. They provide centralized content management, real-time communication capabilities, and opportunities for collaborative learning. Furthermore, web-based

systems enable institutions to maintain academic records digitally and automate many administrative processes.

The growth of cloud computing and web technologies has significantly improved the capabilities of modern learning management systems. Institutions can now deploy scalable educational platforms that support thousands of concurrent users while maintaining data security and performance.

Web-based educational platforms have become increasingly important in supporting blended and distance learning models. These systems provide learners with continuous access to educational resources while enabling institutions to improve administrative efficiency and academic monitoring (Dalsgaard, 2006; Weller, 2007).

2.5 Components of a Learning Management System

A standard Learning Management System consists of several interconnected components that work together to support educational activities.

2.5.1 User Management Module

The user management module manages the registration, authentication, authorization, and management of users within the system. Different user roles such as administrators, teachers, and students are assigned specific permissions and responsibilities.

2.5.2 Course Management Module

The course management module enables administrators and teachers to create, update, organize, and manage academic courses. Course information such as descriptions, schedules, instructors, and learning materials is maintained within this module.

2.5.3 Learning Materials Module

This module allows instructors to upload and distribute educational resources including lecture notes, presentations, videos, and supplementary reading materials.

2.5.4 Assessment and Assignment Module

The assessment module supports the creation and management of assignments, quizzes, examinations, and other evaluation activities. Students can submit coursework electronically while instructors can review and grade submissions online.

2.5.5 Communication Module

Communication features facilitate interaction between users through announcements, messaging systems, notifications, discussion forums, and collaborative learning tools.

2.5.6 Reporting and Analytics Module

The reporting module provides administrators and instructors with insights into student performance, course participation, attendance, and overall academic progress.

The integration of these modules enable educational institutions to manage teaching, learning, assessment, and communication activities through a single platform. Effective integration contributes significantly to user satisfaction and system effectiveness (Naveh et al., 2010; Ozkan & Koseler, 2009).

2.6 Theoretical Framework

2.6.1 Technology Acceptance Model (TAM)

Fred Davis developed the Technology Acceptance Model (TAM) to explain how users accept and utilize technology systems. The model suggests that two primary factors influence technology adoption:

- Perceived Usefulness (PU)
- Perceived Ease of Use (PEOU)

The **Technology Acceptance Model (TAM)** remains one of the most widely used frameworks for explaining users' acceptance of information systems. Recent studies have shown that users are more likely to adopt a Learning Management System when they perceive it as useful, easy to use, and supportive of their teaching and learning activities (Almaiah, Al-Khasawneh, & Althunibat, 2020; Salloum et al., 2019). The proposed Learning Management System is therefore designed with user-friendliness, accessibility, and functionality in mind to encourage acceptance among administrators, teachers, and students.

2.6.2 Constructivist Learning Theory

Constructivist Learning Theory suggests that learners actively construct knowledge through interaction, collaboration, and engagement with learning experiences. According to Vygotsky (1978), learning is enhanced through social interaction and participation.

Learning Management Systems support constructivist principles by providing tools for collaboration, communication, resource sharing, and interactive learning activities.

2.7 Review of Existing Learning Management Systems

Previous studies indicate that the selection of an LMS depends on institutional requirements, budget constraints, scalability needs, and technical expertise available within the institution. While Moodle provides flexibility and cost-effectiveness, Canvas offers a modern user experience, and Blackboard delivers enterprise-level functionality for large educational institutions (Bradford et al., 2007; Machado & Tao, 2007).

2.7.1 Moodle

Moodle is one of the most widely used open-source Learning Management Systems worldwide. It provides extensive functionality for course management, online assessments, discussion forums, gradebooks, and collaborative learning activities.

Advantages

- Open-source and customizable.
- Large user community.
- Extensive plugin ecosystem.
- Dedicated support for online learning.

Limitations

- Complex interface for some users.
- Requires technical expertise for administration.
- Some features may be unnecessary for smaller institutions.

2.7.2 Canvas

Canvas is a cloud-based LMS that provides modern educational tools for teaching and learning. It supports mobile learning, online assessments, collaboration, and integration with third-party applications.

Advantages

- Modern user interface.
- Mobile-friendly design.
- Strong integration capabilities.
- Comprehensive assessment tools.

Limitations

- Subscription costs may be high.
- Limited customization compared to open-source systems.

2.7.3 Blackboard Learn

Blackboard Learn is a commercial LMS widely used in higher education institutions. It offers course management, assessment, collaboration, and reporting functionalities.

Advantages

- Comprehensive educational tools.

- Robust reporting capabilities.
- Enterprise-level scalability.

Limitations

- High licensing costs.
- Complex administration requirements.
- Resource-intensive deployment.

2.7.4 Comparative Analysis of Existing Learning Management Systems

Table 2.1: Comparison of Moodle, Canvas and Blackboard Learn

Feature	Moodle	Canvas	Blackboard Learn
Developer	Moodle Pty Ltd	Instructure	Anthology Inc.
System Type	Open Source	Cloud-Based Commercial	Commercial
Cost	Free (hosting costs apply)	Subscription-Based	Subscription-Based
Ease of Use	Moderate	High	Moderate
Customisation	Very High	Moderate	Limited
Mobile Support	Available	Excellent	Good
Assignment Management	Excellent	Excellent	Excellent

Assessment Tools	Extensive	Extensive	Extensive
Communication Tools	Forums, Messaging, Announcements	Messaging, Discussions, Conferences	Messaging, Discussions, Announcements
Technical Support	Community-Based	Vendor Support	Vendor Support
Scalability	High	High	Very High
Suitable for Small Schools	Highly Suitable	Suitable	Less Suitable
Suitable for Large Institutions	Suitable	Highly Suitable	Highly Suitable

Source: Adapted from Bradford et al. (2007), Machado and Tao (2007), and Dougiamas and Taylor (2003).

2.8 Empirical Review

Several studies have examined the effectiveness of Learning Management Systems in educational environments.

Al-Busaidi and Al-Shihi (2010) investigated factors influencing LMS adoption among instructors and found that perceived usefulness and ease of use significantly affected system acceptance.

Coates, James, and Baldwin (2005) examined the impact of Learning Management Systems on teaching and learning practices and concluded that LMS platforms improve communication, resource accessibility, and educational management.

Ozkan and Koseler (2009) evaluated student perceptions of e-learning systems and reported that system quality, information quality, and service quality significantly influence user satisfaction.

Watson and Watson (2007) argued that Learning Management Systems play a critical role in supporting modern educational environments by providing centralized management of learning activities and academic resources.

These studies demonstrate the importance of LMS solutions in enhancing educational processes and justify the development of a customized system tailored to institutional requirements.

Furthermore, Islam (2013) found that effective LMS implementation positively influences student engagement, learning outcomes, and academic satisfaction. Similarly, Almarashdeh (2016) reported that user satisfaction is strongly influenced by system usability, reliability, and accessibility. These findings reinforce the importance of developing a user-friendly and reliable Learning Management System.

2.9 Summary of Literature Review

The literature reviewed demonstrates that Learning Management Systems (LMSs) have become essential tools in modern education. They facilitate course delivery, academic management, communication, assessment, and student performance monitoring. Existing systems such as Moodle, Canvas, and Blackboard provide valuable functionality but may not fully address the specific operational, financial, and institutional needs of all educational institutions.

The review also highlights the relevance of the Technology Acceptance Model and Constructivist Learning Theory in explaining user acceptance and effective learning practices. Previous studies indicate that user-friendly, accessible, secure, and functional LMS platforms significantly improve educational outcomes, learner engagement, and administrative efficiency (Almaiah et al., 2020; Salloum et al., 2019).

The literature further identified several challenges affecting educational institutions, including reliance on paper-based academic processes, fragmented record management, delayed communication between administrators, teachers and students, limited access to learning materials outside the classroom, inefficient assignment submission and grading processes, and difficulties in monitoring student academic performance. These challenges often lead to administrative inefficiencies, duplication of records, delayed decision-making, and reduced quality of educational service delivery.

Based on these findings, the development of the proposed School Learning Management System is justified as a means of providing a centralised, secure, and user-friendly platform that streamlines academic administration, enhances communication, improves access to learning resources, supports online assessments and assignment management, and facilitates efficient monitoring of student performance.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the analysis of the proposed **School Learning Management System (LMS)**. System analysis is an important phase in the Software Development Life Cycle (SDLC) because it involves examining the existing educational management processes, identifying their limitations, and determining the functional and non-functional requirements of the proposed system. The analysis provides a clear understanding of the problem domain and serves as the foundation for the system design and implementation presented in the subsequent chapter.

The existing system used by many educational institutions relies heavily on manual processes and fragmented digital solutions for managing academic activities such as student registration, course management, assignment submission, grading, communication, and record keeping. These approaches often result in inefficiencies, duplication of data, delayed dissemination of information, limited accessibility to learning resources, and challenges in monitoring students' academic progress. Such limitations negatively affect the quality of teaching, learning, and administrative decision-making.

The proposed School Learning Management System seeks to address these shortcomings by providing a centralized, web-based platform that integrates the major academic and administrative functions of an educational institution. The system enables administrators to manage users, departments, courses, and academic records; teachers to upload learning

materials, create assignments and quizzes, record attendance, and assess students; and students to access course resources, submit assignments, participate in online assessments, and monitor their academic performance through a secure and user-friendly interface.

This chapter evaluates the feasibility of the proposed system from technical, operational, economic, and legal perspectives. It also identifies the functional and non-functional requirements that guide the development of the system. Furthermore, the chapter presents the system modelling techniques used to describe the behaviour and structure of the proposed LMS, including the Context Diagram, Data Flow Diagram (DFD), and Use Case Diagram. These models provide a visual representation of the interactions between users and the system, as well as the movement of data throughout the application.

The outcome of this analysis establishes the foundation for the system architecture, database design, user interface design, and implementation procedures discussed in **Chapter Four**. By thoroughly analysing the requirements of the proposed Learning Management System, the study ensures that the final solution effectively addresses the identified problems while meeting the needs of administrators, teachers, students, and other stakeholders within the educational environment.

3.2 Analysis of the Existing System

The existing academic management system used by many educational institutions is characterised by a combination of manual procedures and isolated digital applications. While some institutions utilise basic computerised systems for student registration and record keeping, many critical academic processes remain fragmented and inefficient. This

lack of integration makes it difficult to manage academic activities effectively and limits timely access to information by administrators, teachers, and students. Studies have shown that fragmented educational information systems contribute to administrative inefficiencies, poor data accessibility, reduced collaboration, and challenges in managing teaching and learning activities (Almaiah et al., 2020; Salloum et al., 2019; Turnbull et al., 2021).

The existing system was analysed based on the major functional areas of academic management, as discussed below.

3.2.1 User Management

In the existing system, user accounts for administrators, teachers, and students are often created and managed manually. User information is frequently stored across multiple files or separate systems, making it difficult to maintain consistency and accuracy. The absence of automated Role-Based Access Control (RBAC) increases the risk of unauthorized access to sensitive information and complicates user administration.

Research indicates that ineffective user management and weak access control mechanisms reduce system security, increase administrative workload, and negatively affect the adoption and usability of educational information systems (Almaiah et al., 2020; Salloum et al., 2019). Furthermore, integrated user management with clearly defined roles improves security, accountability, and operational efficiency in Learning Management Systems (Turnbull et al., 2021).

Weaknesses

- Manual creation and updating of user accounts.
- Inconsistent user information across different systems.

- Limited access control mechanisms.
- Time-consuming user administration.

3.2.2 Student Records Management

Student records such as personal information, academic history, enrolment details, attendance records, and examination results are often maintained using paper documents or spreadsheets. This approach makes retrieving, updating, and maintaining records difficult, especially as student numbers increase. Manual record management also increases the possibility of data duplication, loss, and human error. Studies have shown that reliance on manual and fragmented record-keeping systems reduces administrative efficiency, limits timely access to student information, and hinders effective academic decision-making (Bond et al., 2021; Almaiah et al., 2020). Furthermore, integrated Learning Management Systems provide centralized storage and management of student records, improving data accuracy, accessibility, and institutional efficiency (Turnbull et al., 2021).

Weaknesses

- Difficulty retrieving historical records.
- High risk of data duplication and inconsistencies.
- Increased possibility of human error.
- Limited backup and recovery mechanisms.

3.2.3 Course Management

Course creation, allocation of teachers, and student enrolment are commonly managed manually or through separate applications. Consequently, administrators spend considerable time updating course information, while students may experience delays in accessing newly assigned courses and learning resources. The use of disconnected systems for course management often results in inconsistent data, inefficient coordination, and increased administrative workload. Research has shown that integrated Learning Management Systems streamline course administration by centralizing course information, automating enrolment processes, and providing timely access to learning resources for both instructors and students (Turnbull et al., 2021; Salloum et al., 2019). Furthermore, effective course management within an LMS enhances institutional efficiency and improves the overall learning experience (Almaiah et al., 2020).

Weaknesses

- Manual course allocation.
- Delayed course updates.
- Lack of centralized course information.
- Inefficient monitoring of course enrolment.

3.2.4 Assignment Management

Assignment distribution and submission are frequently conducted using printed documents, email, or messaging applications. These methods make it difficult to monitor submission deadlines, organize student submissions, and provide timely feedback. Teachers also spend additional time sorting and recording submitted assignments manually. Such fragmented approaches reduce the efficiency of assessment management

and increase the likelihood of delayed grading, misplaced submissions, and inconsistent record keeping. Research indicates that Learning Management Systems improve assignment management by providing a centralized platform for creating, distributing, submitting, and tracking assignments, thereby enhancing both teaching efficiency and student engagement (Almaiah et al., 2020; Turnbull et al., 2021). Similarly, Salloum et al. (2019) reported that integrated assessment features contribute significantly to the effective adoption and use of e-learning systems.

Weaknesses

- Manual assignment submission.
- Difficulty tracking submission deadlines.
- Increased risk of misplaced assignments.
- Delayed grading and feedback.

3.2.5 Attendance Management

Attendance is commonly recorded manually using attendance registers or spreadsheets. This process is repetitive, susceptible to human error, and requires additional effort when preparing attendance reports. Manual attendance management also makes it difficult to identify absenteeism trends and monitor student participation over time. The reliance on paper-based or disconnected attendance systems reduces administrative efficiency and limits the timely availability of attendance information for academic decision-making. Studies have shown that integrated Learning Management Systems automate attendance recording, improve data accuracy, facilitate real-time monitoring, and simplify the generation of attendance reports (Bond et al., 2021; Turnbull et al., 2021). Additionally,

Almaiah et al. (2020) noted that the automation of routine academic processes enhances institutional efficiency and supports effective educational management.

Weaknesses

- Paper-based attendance recording.
- Time-consuming attendance compilation.
- Difficulty generating attendance reports.
- Increased likelihood of recording errors.

3.2.6 Communication Management

Communication between administrators, teachers, and students is often achieved through notice boards, printed notices, emails, or social media platforms. Since these communication channels are not integrated into a single academic system, important announcements may not reach all intended recipients promptly, resulting in missed deadlines and reduced collaboration. Research has shown that fragmented communication mechanisms hinder effective information sharing, reduce learner engagement, and create challenges in coordinating academic activities. Integrated Learning Management Systems address these issues by providing centralized communication tools, including announcements, notifications, discussion forums, and messaging services that facilitate timely and efficient interaction among stakeholders (Salloum et al., 2019; Turnbull et al., 2021). Similarly, Almaiah et al. (2020) reported that effective communication features are among the critical factors influencing the successful adoption and use of e-learning systems.

Weaknesses

- Fragmented communication channels.
- Delayed dissemination of information.
- Poor coordination among stakeholders.
- Lack of centralized notifications.

3.2.7 Report Generation

Generating academic reports such as student performance reports, attendance summaries, course statistics, and administrative reports often require manual compilation of information from multiple sources. This process is labor-intensive and increases the possibility of inaccuracies in the generated reports. Manual reporting also delays decision-making because administrators and teachers cannot readily access real-time academic information. Studies have shown that integrated Learning Management Systems improve reporting by automatically collecting, processing, and presenting academic data, thereby enhancing data accuracy, reducing administrative workload, and supporting evidence-based decision-making (Turnbull et al., 2021; Bond et al., 2021). Furthermore, Almaiah et al. (2020) noted that automated reporting and data management functionalities contribute significantly to the effectiveness and adoption of e-learning systems.

Weaknesses

- Manual report preparation.
- Slow report generation.
- High probability of calculation errors.
- Limited availability of real-time statistics.

3.2.8 Security

The existing system provides limited security mechanisms for protecting academic information. Manual records can easily be damaged, misplaced, or accessed by unauthorized individuals, while simple digital systems may lack adequate authentication, authorization, and audit mechanisms. Consequently, confidential academic information is at greater risk of unauthorized access, data breaches, and accidental loss. Research indicates that robust security features—including secure authentication, Role-Based Access Control (RBAC), audit logging, and data protection mechanisms—are essential for safeguarding educational information systems and ensuring user trust (Almaiah et al., 2020; Salloum et al., 2019). Furthermore, the adoption of comprehensive security controls within Learning Management Systems improves data confidentiality, integrity, and accountability while supporting compliance with institutional information security requirements (Turnbull et al., 2021).

Weaknesses

- Weak authentication mechanisms.
- Limited role-based access control.
- Poor data protection practices.
- Insufficient backup and recovery procedures.
- Lack of audit trails for monitoring system activities.

3.2.9 Summary of the Existing System Analysis

The analysis of the existing system reveals several shortcomings that affect the efficient management of academic activities. Manual processes, fragmented information systems, delayed communication, inefficient record management, inadequate security, and limited reporting capabilities reduce institutional productivity and negatively impact teaching and learning. These findings are consistent with previous studies, which indicate that the lack of integrated educational management systems hinders effective administration, reduces operational efficiency, and limits the accessibility of academic information (Almaiah et al., 2020; Salloum et al., 2019; Turnbull et al., 2021). Furthermore, integrated Learning Management Systems have been shown to improve collaboration, streamline academic processes, enhance data security, and support informed decision-making through centralized information management (Bond et al., 2021).

These limitations demonstrate the need for a comprehensive, integrated Learning Management System that centralizes academic operations, automates routine processes, improves communication, strengthens data security, and provides timely access to educational information for all authorized users. The proposed School Learning Management System is designed to address these challenges by providing a secure, web-based platform that enhances the efficiency and effectiveness of educational management.

3.3 Existing System Workflow

The workflow of the existing academic management system is largely manual and involves several disconnected processes performed by different stakeholders. Students submit requests, assignments, and other academic documents physically or through

separate communication platforms. Teachers process these documents manually before forwarding relevant information to departmental offices or the academic administration for further action. Records are then stored in filing cabinets, spreadsheets, or independent databases. Such fragmented workflows are common in institutions that have not fully adopted integrated Learning Management Systems, resulting in inefficiencies in information management and academic administration (Almaiah et al., 2020; Turnbull et al., 2021).

This workflow requires multiple manual interventions, resulting in delays, duplication of effort, and increased chances of errors. Information often passes through several departments before reaching its destination, making it difficult to monitor academic activities efficiently and retrieve records promptly. Studies have shown that manual and disconnected academic processes reduce institutional productivity, hinder timely decision-making, and limit access to accurate educational information, whereas integrated LMS platforms improve process automation, information sharing, and operational efficiency (Bond et al., 2021; Salloum et al., 2019).

Figure 3.1 Illustrates the workflow of the existing academic management system.

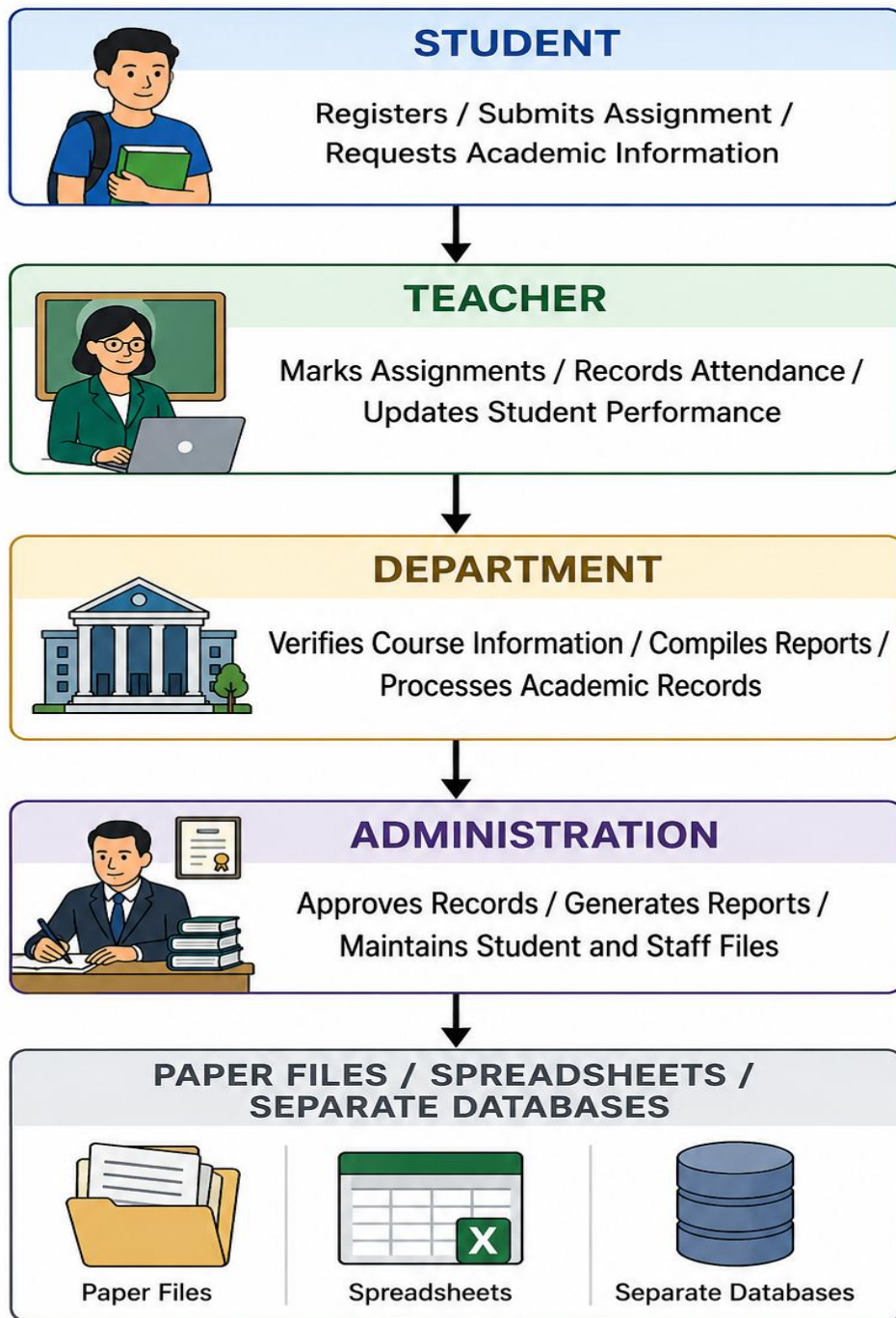


Figure 3.1: Workflow of the Existing System

Explanation

The existing workflow begins with students submitting academic requests, assignments, or registration information to their teachers. Teachers manually process these submissions by recording attendance, assessing assignments, and updating students' academic performance. The processed information is subsequently forwarded to the relevant department, where it is verified and consolidated before being sent to the administration for approval and record keeping. Such sequential and manual workflows are common in institutions that rely on traditional academic management practices, often resulting in inefficient coordination and delays in service delivery (Almaiah et al., 2020; Turnbull et al., 2021).

Since most of these activities depend on manual documentation or isolated software applications, information is repeatedly transferred between offices before it is permanently stored. This approach increases processing time, creates opportunities for data duplication and human error, and makes retrieving academic records difficult. Furthermore, communication between administrators, teachers, and students is often delayed because there is no centralized platform for sharing academic information. Studies have shown that fragmented educational systems reduce administrative efficiency, hinder timely access to information, and negatively affect collaboration among stakeholders (Salloum et al., 2019; Bond et al., 2021).

The identified weaknesses of this workflow justify the development of a fully integrated Learning Management System that automates academic processes, centralizes data management, improves communication, and provides authorised users with real-time access to educational information. Such integrated systems have been found to enhance

institutional efficiency, improve data accuracy, and support more effective teaching, learning, and academic administration (Turnbull et al., 2021; Almaiah et al., 2020).

3.4 Proposed System Workflow

The proposed School Learning Management System (LMS) introduces a centralized and automated workflow that integrates all major academic and administrative activities into a single web-based platform. Unlike the existing manual system, users interact directly with the system through a secure login interface, where access to features is determined by their assigned roles. The system processes user requests, communicates with the database through the backend API, and provides real-time responses to authorised users. The workflow is designed to reduce manual intervention, improve communication, enhance data security, and ensure that academic information is available whenever required. Administrators, teachers, and students each access customised dashboards containing only the functions relevant to their responsibilities.

Figure 3.2: Workflow of the Proposed Learning Management System

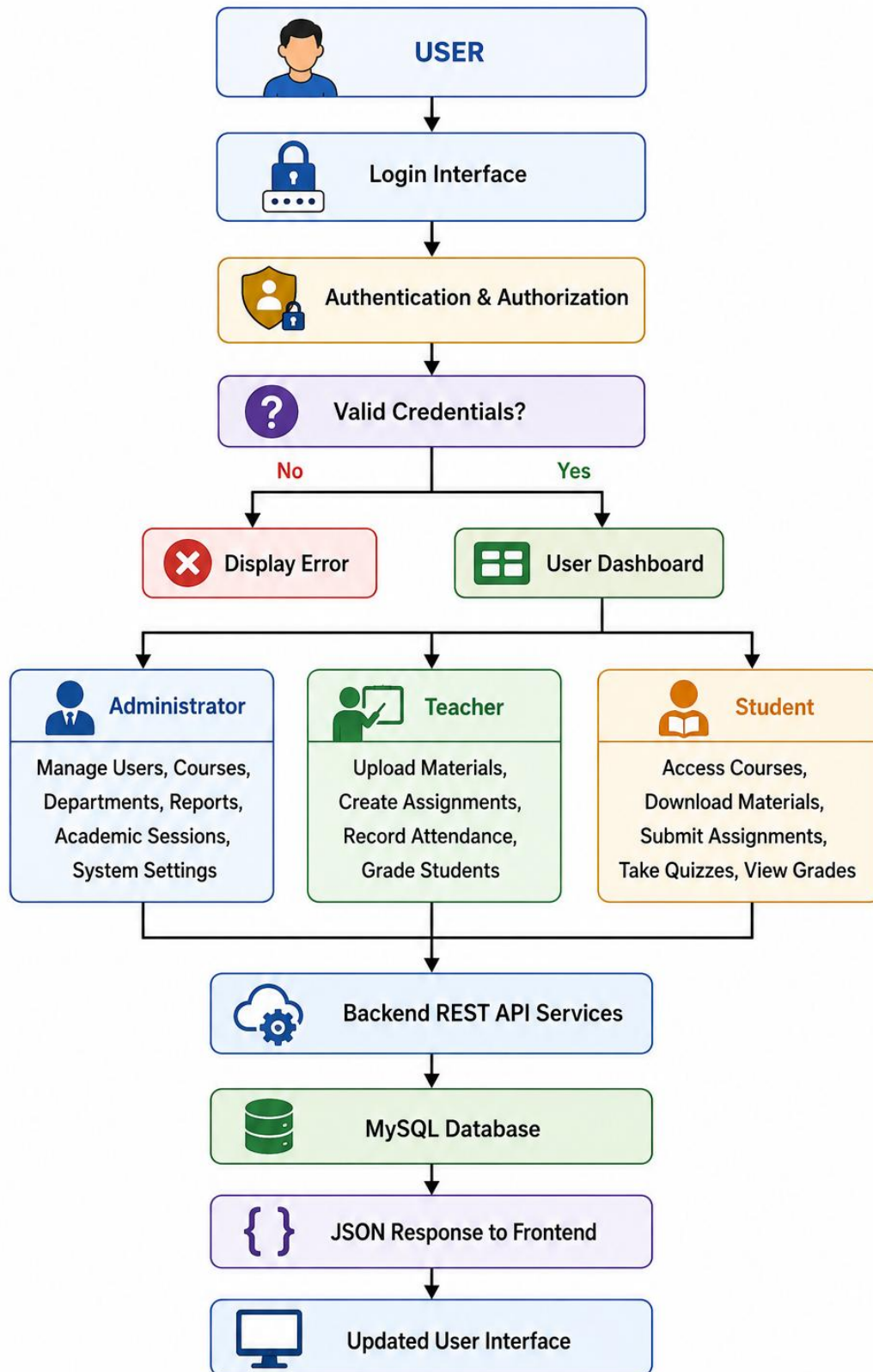


Figure 3.2: Workflow of the Proposed Learning Management System

Explanation of the Proposed Workflow

The workflow begins when a user accesses the system through the login page and enters valid authentication credentials. The backend verifies the credentials, checks the user's role, and establishes a secure session. Once authenticated, the user is redirected to a role-specific dashboard.

Administrators are provided with tools for managing users, departments, academic sessions, courses, reports, and other institutional resources. Teachers can create courses, upload learning materials, create assignments and quizzes, record attendance, assess students, and monitor academic performance. Students can access enrolled courses, download learning materials, submit assignments, participate in online assessments, view grades, and receive announcements.

Every action performed by a user is transmitted to the backend through secure RESTful API endpoints. The backend validates the request, performs the necessary business logic, and interacts with the MySQL database using prepared statements to maintain data integrity and security. The API returns responses in JSON format, which are processed by the frontend to update the user interface dynamically without exposing backend implementation details.

Compared to the existing workflow, the proposed system significantly reduces paperwork, eliminates duplication of records, improves communication among stakeholders, enhances data security through authentication and role-based access control, and enables faster retrieval of academic information. The automated workflow also supports real-time processing of academic activities, thereby improving the efficiency and effectiveness of educational administration.

3.5 Functional Requirements

Functional requirements describe the specific operations and services that the proposed School Learning Management System (LMS) must provide to satisfy the needs of its users. The system is designed using role-based access control, ensuring that each category of user has access only to the functions relevant to their responsibilities. The major functional requirements are grouped according to the three primary user roles: Administrator, Teacher, and Student.

3.5.1 Functional Requirements for the Administrator

The administrator is responsible for managing the overall operation of the Learning Management System. The system shall enable the administrator to perform the following functions:

- Securely log into and log out of the system.
- Access an administrative dashboard displaying key system statistics.
- Create, update, deactivate, and manage user accounts.
- Assign user roles and permissions.
- Manage departments and academic programmes.
- Create, update, and manage courses.
- Assign teachers to courses.
- Manage academic years, semesters, and sessions.
- View student enrolment information.
- Generate academic and administrative reports.
- View system audit logs and user activities.

- Configure system settings.
 - Perform database backup and recovery operations.
 - Manage announcements and institutional notifications.
 - Monitor overall system performance and usage.
-

3.5.2 Functional Requirements for the Teacher

Teachers are responsible for delivering course content and assessing students. The proposed system shall enable teachers to perform the following functions:

- Securely log into and log out of the system.
- Access a personalised teacher dashboard.
- View assigned courses.
- Create and update course information where permitted.
- Upload, edit, archive, and delete learning materials.
- Create assignments with submission deadlines.
- Create and manage online quizzes and examinations.
- Grade assignments and quizzes.
- Record and manage student attendance.
- View enrolled students for each course.
- Publish announcements to students.
- View student submissions.
- Provide feedback on submitted assignments.
- Monitor student academic performance.
- Generate course-related reports.

3.5.3 Functional Requirements for the Student

Students interact with the system primarily to access learning resources and participate in academic activities. The system shall enable students to perform the following functions:

- Securely log into and log out of the system.
- Access a personalised student dashboard.
- View registered or enrolled courses.
- Access and download learning materials.
- Submit assignments electronically.
- Participate in online quizzes and examinations.
- View assignment deadlines.
- Receive announcements and notifications.
- View grades and academic performance.
- Monitor attendance records.
- Update permitted profile information.
- Communicate with instructors through the system where applicable.

3.5.4 General Functional Requirements

In addition to user-specific functions, the system shall provide the following general capabilities:

- Authenticate users before granting system access.
- Enforce role-based access control to protect sensitive information.
- Validate user input before processing requests.

- Store academic information securely in a centralised database.
- Support secure file uploads and downloads.
- Record important system activities through audit logs.
- Generate reports in real time.
- Maintain data integrity through database validation and constraints.
- Support RESTful API communication between the frontend and backend.
- Provide consistent JSON responses for all API requests.
- Automatically manage user sessions and log users out after periods of inactivity.

Summary

The functional requirements define the essential services that the proposed School Learning Management System must provide to support effective academic administration and online learning. By separating system functions according to user roles, the proposed solution ensures that administrators, teachers, and students are provided with appropriate tools and permissions while maintaining the security, integrity, and efficiency of the overall system. These requirements serve as the foundation for the system design and implementation presented in the next chapter.

3.6 Non-Functional Requirements

Non-functional requirements describe the quality attributes and operational characteristics that the proposed School Learning Management System (LMS) must possess to ensure efficient, reliable, secure, and user-friendly operation. Unlike functional requirements, which define what the system should do, non-functional requirements specify how well the system should perform under different conditions.

3.6.1 Performance Requirements

The proposed LMS shall provide efficient and timely processing of user requests to ensure a satisfactory user experience.

The system shall:

- Load web pages within **three (3) seconds** under normal network conditions.
 - Process API requests within **two (2) seconds** for standard operations.
 - Support at least **500 concurrent users** without significant degradation in performance.
 - Allow users to upload learning materials of up to **100 MB** per file.
 - Process assignment submissions immediately after upload.
 - Generate reports within **five (5) seconds** for standard datasets.
 - Minimise server response time through efficient database queries and caching where applicable.
-

3.6.2 Security Requirements

Since the system stores confidential academic information, appropriate security mechanisms shall be implemented to protect user data and system resources.

The system shall:

- Encrypt user passwords using the **bcrypt hashing algorithm**.
- Implement **Role-Based Access Control (RBAC)** to restrict access based on user roles.
- Validate and sanitise all user inputs to prevent malicious attacks.
- Use **prepared SQL statements** to prevent SQL injection.

- Protect against **Cross-Site Scripting (XSS)** attacks through input validation and output encoding.
 - Incorporate Cross-Site Request Forgery (CSRF) protection for state-changing requests through the use of security tokens where applicable to the authentication and session architecture.
 - Automatically terminate inactive user sessions after a predefined timeout period.
 - Record login attempts and critical user activities in audit logs.
 - Support secure communication using **HTTPS** when deployed to a production environment.
 - Restrict file uploads to approved file types and sizes.
-

3.6.3 Reliability Requirements

The system shall operate consistently and maintain data integrity during normal operation and unexpected failures.

The system shall:

- Maintain a minimum availability of **99% uptime** during normal operation.
- Automatically back up the database at scheduled intervals.
- Maintain transaction integrity during database operations.
- Recover quickly from unexpected system failures.
- Log application errors for troubleshooting and maintenance.
- Prevent data corruption through proper database constraints and validation.
- Ensure uploaded files remain accessible unless intentionally removed by authorised users.

3.6.4 Usability Requirements

The system shall provide an intuitive and user-friendly interface that supports users with varying levels of computer literacy.

The system shall:

- Provide a responsive interface that functions correctly on desktops, tablets, and smartphones.
- Maintain consistent navigation throughout all system pages.
- Display clear labels, icons, and instructions for all major functions.
- Present informative validation and error messages.
- Minimise the number of steps required to complete common tasks.
- Support accessibility features such as readable fonts, sufficient colour contrast, and keyboard navigation where applicable.
- Provide dashboards customised to the responsibilities of administrators, teachers, and students.

3.6.5 Maintainability Requirements

The proposed LMS shall be designed to simplify future maintenance, upgrades, and feature enhancements.

The system shall:

- Follow a modular software architecture.
- Separate the frontend from the backend through RESTful APIs.
- Use reusable components to reduce code duplication.

- Maintain clear documentation for APIs, database structures, and source code.
 - Record application logs to facilitate debugging.
 - Support future integration with mobile applications and third-party educational services.
-

3.6.6 Scalability Requirements

The system shall support future institutional growth without requiring significant architectural changes.

The system shall:

- Allow additional users, departments, and courses to be added without affecting system performance.
 - Support expansion to multiple campuses or educational institutions.
 - Accommodate increasing numbers of uploaded learning materials and assignments.
 - Allow additional functional modules, such as online payments, virtual classrooms, and library management, to be integrated in future versions.
-

3.6.7 Compatibility Requirements

The LMS shall operate consistently across commonly used hardware and software platforms.

The system shall:

- Support modern web browsers including **Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.**

- Operate on Windows, Linux, macOS, and Android devices through a web browser.
 - Integrate with MySQL database servers.
 - Support RESTful API communication using JSON data exchange.
-

3.6.8 Availability Requirements

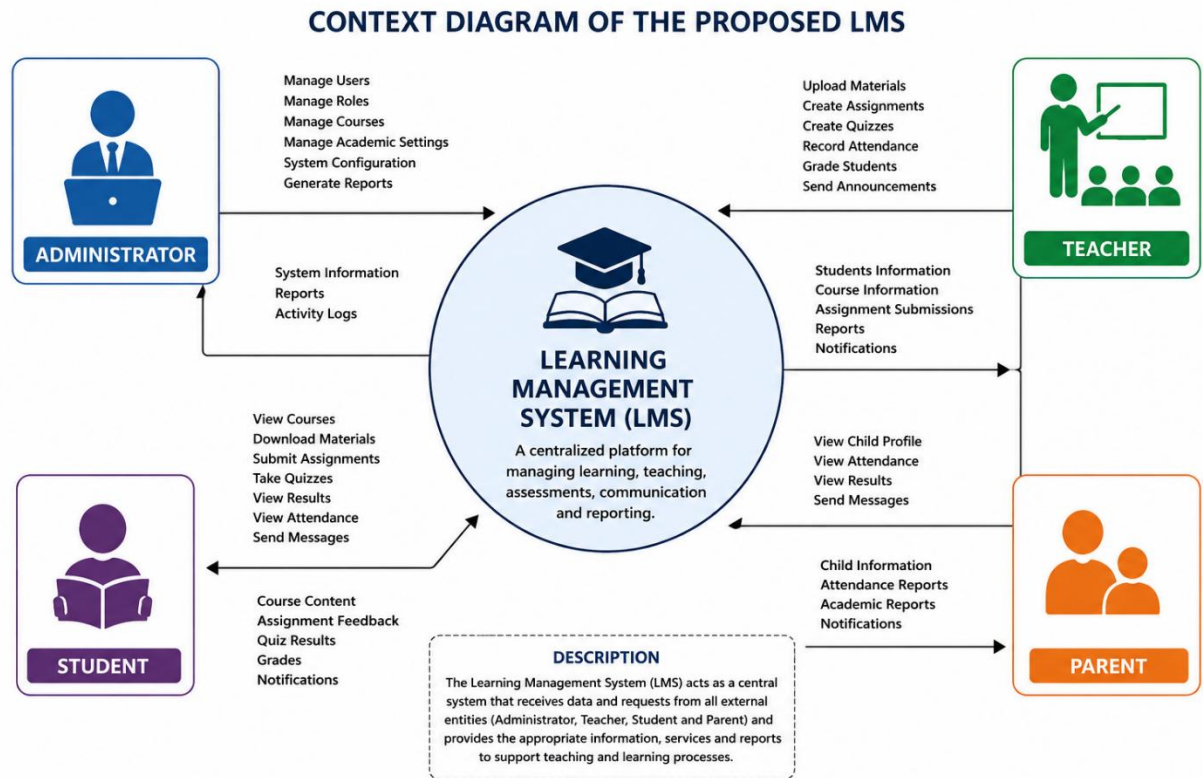
The system shall be accessible whenever authorised users require its services.

The system shall:

- Be available **24 hours a day and 7 days a week**, except during scheduled maintenance.
 - Notify users in advance of planned maintenance activities.
 - Resume normal operation promptly after maintenance or unexpected outages.
-

3.6.9 Context Diagram

Figure 3.3: Context Diagram of the Proposed LMS

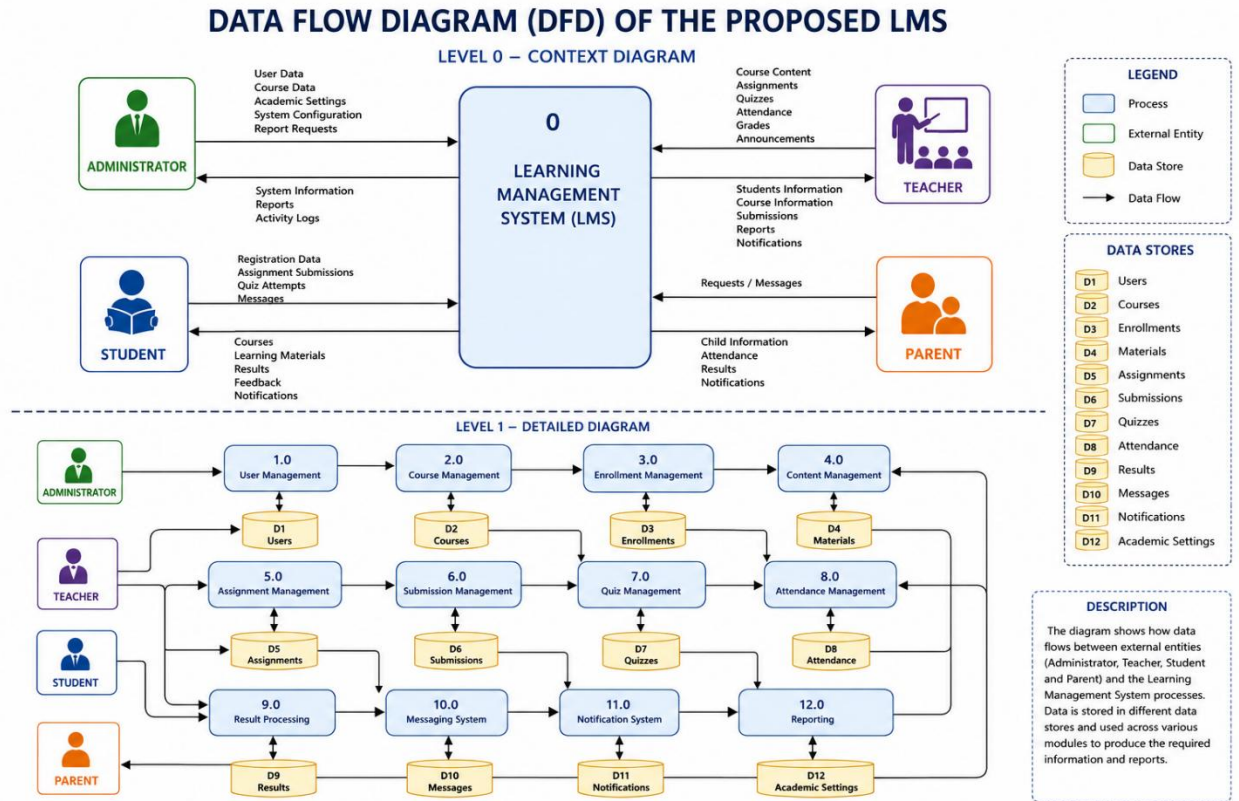


Explanation

The Context Diagram illustrates the interaction between external entities and the Learning Management System. Administrators manage users, courses, and reports. Teachers upload learning materials, create assignments, and record grades. Students access learning resources, submit assignments, and view academic results. The LMS acts as the central processing system that facilitates these interactions.

3.6.10 Data Flow Diagram (Level 0)

Figure 3.4: Data Flow Diagram (DFD Level 0)



Explanation

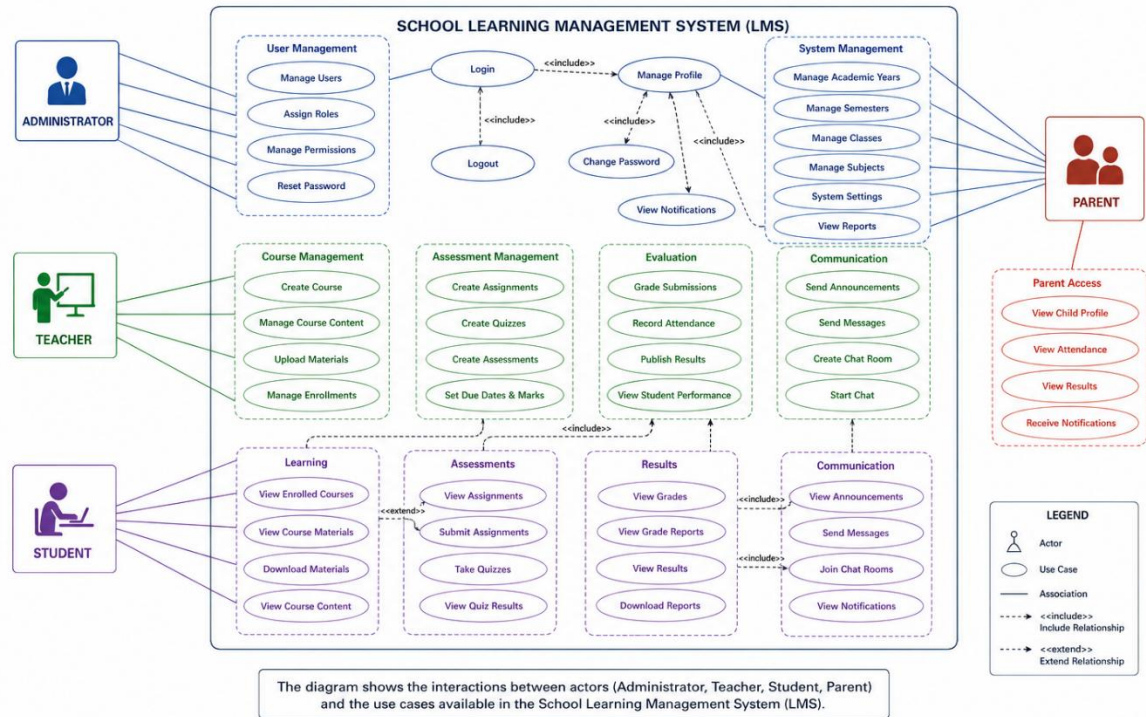
The Data Flow Diagram illustrates how information flows within the proposed system.

Administrators manage users and courses, teachers upload learning materials and assessments, and students access educational resources and submit assignments. All transactions are stored and retrieved from the LMS database.

3.6.11 Use Case Diagram

Figure 3.5: Use Case Diagram of the Proposed LMS

USE CASE DIAGRAM SCHOOL LEARNING MANAGEMENT SYSTEM (LMS)



Explanation

The Use Case Diagram identifies the major interactions between users and the system.

Administrators perform management and reporting functions, teachers manage academic activities, and students access learning resources and monitor their academic progress.

Summary

The non-functional requirements establish the quality standards that the proposed School Learning Management System must satisfy to operate effectively. These requirements address critical aspects such as performance, security, reliability, usability, maintainability, scalability, compatibility, and availability. Collectively, they ensure that the system not only performs its intended functions but also delivers a secure, efficient,

reliable, and user-friendly platform capable of supporting the academic and administrative activities of the institution while accommodating future growth and technological advancements.

3.7 System Requirements

The successful implementation of the proposed School Learning Management System (LMS) depends on the availability of appropriate hardware and software resources. These requirements ensure that the system operates efficiently, securely, and reliably while supporting multiple users simultaneously. The requirements are categorised into hardware requirements and software requirements.

3.7.1 Hardware Requirements

The hardware requirements specify the minimum and recommended computing resources needed to host and access the proposed LMS.

Server Hardware Requirements

The server hosts the backend application, RESTful APIs, database, and uploaded learning resources. To ensure efficient system performance, the following minimum hardware specifications are recommended.

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i5 (8th Generation) or AMD Ryzen 5	Intel Core i7 or AMD Ryzen 7

Memory (RAM)	8 GB	16 GB or higher
Storage	100 GB SSD	500 GB SSD
Network	Broadband Internet Connection	High-speed Fibre Internet
Backup Storage	External Hard Drive	Cloud Backup Service
Power Supply	UPS	UPS with Generator Backup

Client Hardware Requirements

Client computers are used by administrators, teachers, and students to access the LMS through a web browser.

Component	Minimum Requirement
Processor	Dual-Core Processor (2.0 GHz or higher)
Memory (RAM)	4 GB or higher
Storage	20 GB Free Disk Space
Display	1366 × 768 Resolution or Higher
Internet Connection	Stable Broadband Connection
Input Devices	Keyboard and Mouse or Touchscreen

3.7.2 Software Requirements

The software requirements describe the applications, operating systems, databases, and development tools required for developing, deploying, and operating the proposed system.

Server Software Requirements

Software	Version / Requirement
Operating System	Windows 10/11, Ubuntu Linux, Windows Server 2019/2022/2025 or another compatible Linux distribution
Web Server	Apache HTTP Server
Programming Language	PHP 8.x
Database Management System	MySQL 8.x
API Format	RESTful API using JSON
Local Development Environment	XAMPP
Database Administration	phpMyAdmin

Client Software Requirements

Users access the LMS using a standard web browser without installing additional software.

Software	Requirement
Web Browser	Google Chrome (latest version)
Web Browser	Microsoft Edge (latest version)
Web Browser	Mozilla Firefox (latest version)

Web Browser	Safari (latest version for macOS users)
Internet Connection	Stable Internet Access

Development Software Requirements

The system was developed using modern software engineering tools that support version control, API development, debugging, and collaborative development.

Software	Purpose
Visual Studio Code	Source code development
Git	Version control
GitHub	Source code repository and collaboration
Postman	API testing and documentation
XAMPP	Local web server and database environment
phpMyAdmin	Database administration
Composer	PHP dependency management
Google Chrome Developer Tools	Frontend debugging and API inspection

3.7.3 Network Requirements

The Learning Management System is designed as a web-based application; therefore, reliable network connectivity is essential for both server and client operations.

The system requires:

- A stable broadband Internet connection.
 - Secure communication using the HTTP/HTTPS protocol.
 - Local Area Network (LAN) or Wireless LAN (WLAN) support within the institution.
 - Continuous network availability to ensure uninterrupted access to learning resources and academic services.
-

3.7.4 Browser Compatibility

To ensure accessibility across different computing platforms, the proposed LMS is compatible with modern web browsers, including:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari
- Opera

The user interface is responsive and adapts automatically to desktops, laptops, tablets, and smartphones, providing a consistent user experience across different screen sizes.

3.7.5 Summary

The proposed School Learning Management System requires modest hardware resources and widely available software technologies, making it suitable for deployment in educational institutions of varying sizes. The use of PHP, MySQL, Apache, RESTful APIs, and modern web browsers ensures compatibility, portability, and ease of

maintenance. Furthermore, the adoption of development tools such as Visual Studio Code, Git, GitHub, and Postman supports efficient software development, testing, and future system enhancement. These requirements provide the technical foundation necessary for the successful implementation and long-term operation of the proposed Learning Management System.

3.8 Database Requirements

The database is one of the most important components of the proposed School Learning Management System (LMS). It serves as the central repository for storing, organising, and managing all academic and administrative information generated by the system. The database maintains records relating to users, courses, departments, enrolments, learning materials, assignments, grades, attendance, announcements, and other institutional data. A well-designed database ensures data consistency, efficient retrieval of information, and secure management of academic records.

3.8.1 Database Management System Selection

The proposed Learning Management System uses **MySQL** as its Database Management System (DBMS). MySQL was selected because it is an open-source, reliable, and high-performance relational database management system that integrates effectively with PHP and supports web-based application development.

The selection of MySQL was based on the following advantages:

- It is open-source and freely available, making it cost-effective for educational institutions.

- It provides high-speed data processing and efficient query execution.
 - It integrates seamlessly with PHP and Apache through the XAMPP development environment.
 - It supports relational database structures suitable for complex academic information.
 - It offers strong security features such as user authentication and access privileges.
 - It supports backup, recovery, and transaction management.
 - It is widely adopted in educational and enterprise web applications, ensuring long-term community support and documentation.
-

3.8.2 Database Normalisation

Database normalisation is the process of organising data into related tables to minimise redundancy and improve data integrity. The database for the proposed LMS was designed following the principles of relational database normalisation.

The database satisfies the following normal forms:

First Normal Form (1NF)

The database ensures that:

- Each table contains atomic (indivisible) values.
- Every record is uniquely identified by a primary key.
- Repeating groups of data have been eliminated.

Second Normal Form (2NF)

The database satisfies Second Normal Form by ensuring that:

- All non-key attributes depend entirely on the primary key.

- Partial dependencies have been removed.
- Related information is stored in separate tables.

Third Normal Form (3NF)

The database satisfies Third Normal Form by ensuring that:

- There are no transitive dependencies.
- Every non-key attribute depends only on the primary key.
- Data duplication is minimised.
- Data consistency is maintained throughout the database.

The application of these normalisation principles improves storage efficiency, simplifies maintenance, and reduces the possibility of update anomalies.

3.8.3 Database Relationships

The proposed database follows a relational model in which related tables are connected using **primary keys** and **foreign keys**. These relationships maintain referential integrity and ensure consistency among the various academic records.

Some of the major relationships in the database include:

Parent Table	Child Table	Relationship
Roles	Users	One-to-Many
Departments	Courses	One-to-Many
Users (Teacher)	Courses	One-to-Many
Courses	Materials	One-to-Many
Courses	Assignments	One-to-Many
Courses	Quizzes	One-to-Many

Courses	Attendance	One-to-Many
Students	Enrolments	One-to-Many
Courses	Enrolments	One-to-Many
Assignments	Submissions	One-to-Many
Students	Submissions	One-to-Many
Students	Grades	One-to-Many
Courses	Grades	One-to-Many
Users	Audit Logs	One-to-Many

These relationships enable the system to retrieve related academic information efficiently while maintaining data accuracy and consistency.

3.8.4 Database Security

Since the database stores confidential institutional information, several security mechanisms have been incorporated to protect data from unauthorised access and malicious activities.

The proposed system implements the following database security measures:

- User authentication before accessing protected resources.
- Role-Based Access Control (RBAC) to restrict database operations according to user responsibilities.
- Password encryption using the **bcrypt** hashing algorithm.
- Prepared SQL statements to prevent SQL injection attacks.
- Input validation and sanitisation before database operations.

- Foreign key constraints to preserve referential integrity.
- Database transaction management to maintain consistency during updates.
- Automatic recording of important activities in audit logs.
- Scheduled database backups to minimise data loss.
- Recovery procedures for restoring data in the event of system failure.
- Restricted administrative privileges to authorised personnel only.

These measures help ensure the confidentiality, integrity, and availability of academic information stored within the database.

3.8.5 Database Entities

The proposed Learning Management System database consists of several interconnected entities that support the various academic and administrative functions of the system. The major entities include:

- Users
- Roles
- Departments
- Courses
- Student Enrolments
- Learning Materials
- Assignments
- Assignment Submissions
- Quizzes
- Quiz Attempts

- Grades
- Attendance Records
- Announcements
- Academic Sessions
- Audit Logs

Each entity represents a specific aspect of the educational process and is linked to other entities through well-defined relationships.

3.8.6 Database Design Considerations

The database was designed with the following objectives:

- Eliminate unnecessary data redundancy.
 - Ensure high data integrity through relational constraints.
 - Support efficient searching, sorting, and reporting.
 - Provide scalability for future institutional growth.
 - Facilitate secure storage of academic records.
 - Support RESTful API operations through efficient query execution.
 - Enable easy maintenance and future database expansion.
-

Summary

The proposed School Learning Management System employs **MySQL**, a robust relational database management system, to provide efficient and secure storage of academic and administrative information. The database has been normalised up to the **Third Normal Form (3NF)** to minimise redundancy and improve consistency.

Relationships among entities are maintained using primary and foreign keys, while comprehensive security mechanisms—including authentication, role-based access control, prepared statements, password hashing, and scheduled backups—protect sensitive institutional data. Consequently, the database provides a reliable and scalable foundation for supporting the functional requirements of the proposed Learning Management System.

3.9 API Architecture

The proposed School Learning Management System (LMS) adopts an **API-first architecture**, in which the frontend and backend are developed as separate components that communicate through **Representational State Transfer (REST)** APIs. This architectural approach promotes modularity, scalability, maintainability, and reusability by ensuring that the user interface is independent of the application logic and database operations.

In the proposed architecture, users interact with the system through a web-based frontend developed using HTML5, CSS3, and JavaScript. The frontend sends requests to the backend through RESTful API endpoints using standard HTTP methods. The backend, developed with PHP, processes these requests, performs authentication and business logic, communicates with the MySQL database, and returns responses in JavaScript Object Notation (JSON) format.

This separation of concerns enables the backend APIs to serve multiple client applications, including web, mobile, and desktop applications, without modifying the underlying business logic.

aFigure 3.6: API Architecture of the Proposed Learning Management System

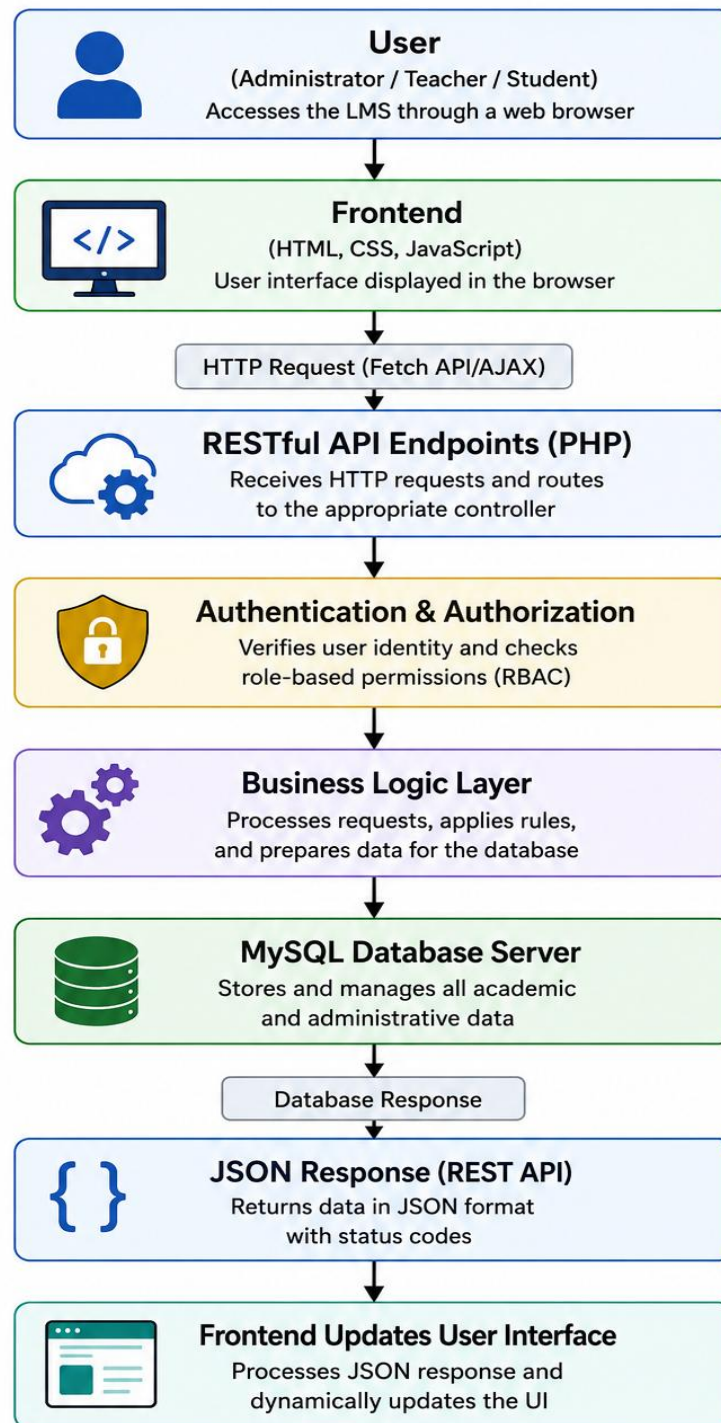


Figure 3.3: API Architecture of the Proposed Learning Management System

3.9.1 RESTful API Design

The proposed LMS follows the principles of **Representational State Transfer (REST)**.

REST is an architectural style for designing web services that enables communication between distributed software components using standard HTTP protocols.

Each resource within the system, such as users, courses, assignments, learning materials, attendance records, and grades, is represented by a unique API endpoint. These endpoints perform specific operations without exposing the internal implementation of the application.

Examples of API resources include:

- /api/auth/login
- /api/users
- /api/courses
- /api/materials
- /api/assignments
- /api/quizzes
- /api/grades
- /api/attendance

This resource-oriented approach simplifies integration, maintenance, and future system expansion.

3.9.2 HTTP Methods

The RESTful API uses standard HTTP methods to perform different operations on system resources.

HTTP Method	Purpose	Example
GET	Retrieve information	Retrieve course materials
POST	Create new resources	Create a course or upload material
PUT	Update existing resources	Update course details or grades
DELETE	Remove resources	Delete a course or learning material

Using standard HTTP methods improves consistency and makes the API easier to understand and integrate with different client applications.

3.9.3 JSON Data Exchange

The proposed system uses **JavaScript Object Notation (JSON)** as the standard data exchange format between the frontend and backend.

When a user performs an operation, the frontend sends data to the backend as a JSON request. After processing the request, the backend returns a JSON response containing the requested information or an appropriate error message.

A typical API response includes:

- Operation status
- Response message

- Requested data
- HTTP status code
- Timestamp

JSON was selected because it is lightweight, easy to read, language-independent, and supported by modern web browsers and programming languages.

3.9.4 Stateless Communication

The RESTful APIs are designed using the principle of **stateless communication**. Under this approach, every client request contains all the information required for the server to process that request. The server does not rely on previous requests to determine the current request's context.

The benefits of stateless communication include:

- Improved scalability.
- Reduced server memory usage.
- Easier load balancing.
- Better fault tolerance.
- Simplified API maintenance.

Although user authentication is managed through secure sessions, each API request is independently validated before processing to ensure that only authorised users can access protected resources.

3.9.5 Authentication and Authorization

Before granting access to protected resources, every API request passes through an authentication and authorisation layer.

The authentication component verifies the user's identity using secure login credentials and establishes a valid user session. Once authenticated, the authorisation component determines whether the user has sufficient privileges to perform the requested operation based on **Role-Based Access Control (RBAC)**.

The system defines three primary user roles:

- Administrator
- Teacher
- Student

Each role is granted access only to the resources and operations relevant to its responsibilities, thereby enhancing system security and protecting confidential academic information.

3.9.6 Advantages of the API-First Architecture

The API-first approach provides several advantages over traditional monolithic web applications in which the frontend and backend are tightly coupled.

The proposed architecture offers the following benefits:

- Clear separation between the frontend and backend.
- Improved maintainability through modular development.
- Easier testing of backend services using tools such as Postman.
- Reusability of APIs across web, mobile, and desktop applications.
- Simplified integration with third-party educational systems.

- Improved scalability through independent deployment of frontend and backend components.
 - Faster development by allowing frontend and backend teams to work independently.
 - Consistent communication through standard JSON responses.
-

Summary

The proposed School Learning Management System adopts an **API-first architecture** that separates the presentation layer from the business logic and database layers.

Communication between the frontend and backend is achieved through **RESTful APIs**, **HTTP methods**, and **JSON** data exchange, while **stateless request processing** improves scalability and maintainability. Authentication and role-based authorisation ensure that only authorised users can access protected resources. This architecture provides a secure, flexible, and scalable foundation capable of supporting future enhancements, including mobile applications and integration with external educational platforms.

3.10 Module Description

The proposed School Learning Management System (LMS) is organised into several functional modules, each responsible for performing a specific set of academic or administrative tasks. This modular design improves maintainability, scalability, and system organisation by separating related functionalities into independent components. Each module communicates with the backend through RESTful APIs and interacts with the MySQL database to perform authorised operations.

3.10.1 Authentication Module

The Authentication Module is responsible for verifying user identities before granting access to the system. It ensures that only authorised users can access protected resources while maintaining secure user sessions.

Functions

- User login
- User logout
- Session management
- Password verification
- Role-Based Access Control (RBAC)
- User authentication
- Session timeout management
- Password encryption using bcrypt

3.10.2 User Management Module

This module enables administrators to manage users within the system. It maintains records for administrators, teachers, and students while controlling user roles and permissions.

Functions

- Create user accounts
- Update user information
- Delete or deactivate users

- Assign user roles
 - Manage departments
 - Reset passwords
 - Search and filter users
 - View user activity
-

3.10.3 Course Management Module

The Course Management Module manages all academic courses offered by the institution. It allows administrators and authorised teachers to create and organise courses while assigning instructors and enrolling students.

Functions

- Create courses
 - Edit course information
 - Delete or archive courses
 - Assign teachers to courses
 - Manage course enrolments
 - View course details
 - Search available courses
 - Organise courses by department and academic session
-

3.10.4 Learning Materials Module

This module manages digital learning resources uploaded by teachers. Students can access and download materials for their enrolled courses.

Functions

- Upload learning materials
 - Download learning materials
 - Archive or restore materials
 - Delete learning materials
 - Organise materials by course
 - Search learning resources
 - Preview supported files
 - Manage uploaded documents
-

3.10.5 Assignment Management Module

The Assignment Management Module enables teachers to create coursework while allowing students to submit assignments electronically.

Functions

- Create assignments
- Edit assignment details
- Set submission deadlines
- Upload assignment instructions
- Submit assignments
- View student submissions
- Grade assignments
- Provide feedback
- Monitor submission status

3.10.6 Quiz and Assessment Module

This module supports online quizzes and assessments that evaluate students' understanding of course content.

Functions

- Create quizzes
- Add objective and subjective questions
- Configure quiz duration
- Schedule quizzes
- Submit quiz attempts
- Automatically score objective questions
- Record quiz results
- Generate assessment reports

3.10.7 Attendance Management Module

The Attendance Module enables teachers to record and monitor student attendance for each course.

Functions

- Record attendance
- Update attendance records
- View attendance history
- Generate attendance reports
- Monitor absenteeism

- Export attendance records
-

3.10.8 Grade Management Module

The Grade Management Module manages student assessment results and overall academic performance.

Functions

- Record grades
 - Update grades
 - Calculate final scores
 - Generate grade reports
 - View student performance
 - Export academic results
-

3.10.9 Announcement and Notification Module

This module facilitates communication between administrators, teachers, and students by distributing academic announcements and notifications.

Functions

- Publish announcements
- Send notifications
- Display important notices
- Notify assignment deadlines
- Notify quiz schedules
- Notify grade publication

- Broadcast institutional messages
-

3.10.10 Reporting Module

The Reporting Module provides administrators and teachers with analytical reports that support academic planning and decision-making.

Functions

- Generate student reports
 - Generate course reports
 - Generate attendance reports
 - Generate grade reports
 - Produce enrolment statistics
 - Export reports in printable formats
 - Monitor overall academic performance
-

3.10.11 Audit and Activity Logging Module

The Audit Module records significant user activities performed within the system. These records improve accountability, troubleshooting, and system security.

Functions

- Record login activities
- Record user operations
- Monitor system access
- Record administrative actions
- Track failed login attempts

- Generate audit reports
-

3.10.12 API Integration Module

The API Integration Module serves as the communication layer between the frontend application and the backend services. It processes requests, validates data, and returns JSON responses.

Functions

- Process RESTful API requests
 - Validate incoming requests
 - Return JSON responses
 - Handle HTTP methods (GET, POST, PUT, DELETE)
 - Communicate with the database
 - Handle API errors
 - Manage data exchange between frontend and backend
-

3.10.13 Database Management Module

This module manages all database operations performed by the system and ensures that academic information is stored securely and consistently.

Functions

- Store system data
- Retrieve records
- Update information
- Delete records

- Manage database relationships
 - Maintain referential integrity
 - Perform database backup and recovery
 - Optimise database performance
-

Summary

The proposed School Learning Management System is organised into specialised modules that collectively support academic administration, teaching, and learning. The Authentication, User Management, Course Management, Learning Materials, Assignment, Quiz, Attendance, Grade, Announcement, Reporting, Audit, API Integration, and Database Management modules work together to provide a secure, scalable, and efficient platform. This modular architecture simplifies system maintenance, enhances scalability, promotes code reusability, and allows future expansion without significantly affecting existing functionality.

3.11 Data Dictionary

The data dictionary provides a description of the major data elements used in the proposed School Learning Management System (LMS). It defines the attributes, data types, and purposes of important fields within the database. The data dictionary serves as a reference for developers, database administrators, and system users by promoting consistency in database design and implementation.

3.11.1 Users Table

Field	Data Type	Description
user_id	INT	Unique identifier for each user
role_id	INT	References the user's role
department_id	INT	References the department
username	VARCHAR(50)	Username for login
email	VARCHAR(100)	User email address
password	VARCHAR(255)	Encrypted password
first_name	VARCHAR(100)	User's first name
last_name	VARCHAR(100)	User's last name
phone	VARCHAR(20)	Contact number
status	ENUM	Account status (Active/Inactive)
created_at	TIMESTAMP	Date account was created

3.11.2 Roles Table

Field	Data Type	Description
role_id	INT	Unique role identifier
role_name	VARCHAR(50)	Name of the role
description	TEXT	Role description

3.11.3 Departments Table

Field	Data Type	Description
department_id	INT	Unique department identifier
department_name	VARCHAR(100)	Department name
department_code	VARCHAR(20)	Department code
description	TEXT	Department description

3.11.4 Courses Table

Field	Data Type	Description
course_id	INT	Unique course identifier
department_id	INT	Department offering the course
teacher_id	INT	Assigned teacher
course_code	VARCHAR(20)	Course code
course_name	VARCHAR(150)	Course title
description	TEXT	Course description
credit_hours	INT	Credit hours
semester	VARCHAR(20)	Academic semester
academic_year	VARCHAR(20)	Academic year
status	ENUM	Course status

3.11.5 Student Enrolments Table

Field	Data Type	Description
enrolment_id	INT	Enrolment identifier
student_id	INT	Registered student
course_id	INT	Enrolled course
enrolment_date	DATE	Date of enrolment
status	ENUM	Enrolment status

3.11.6 Learning Materials Table

Field	Data Type	Description
material_id	INT	Material identifier
course_id	INT	Associated course
teacher_id	INT	Uploading teacher
title	VARCHAR(200)	Material title
material_type	VARCHAR(50)	Material category
file_path	VARCHAR(255)	Storage location
file_size	BIGINT	File size
upload_date	DATETIME	Upload date
is_archived	BOOLEAN	Archive status

3.11.7 Assignments Table

Field	Data Type	Description
assignment_id	INT	Assignment identifier
course_id	INT	Related course
teacher_id	INT	Assignment creator
title	VARCHAR(200)	Assignment title
description	TEXT	Assignment details
due_date	DATETIME	Submission deadline
total_marks	DECIMAL(5,2)	Maximum marks
created_at	TIMESTAMP	Date created

3.11.8 Assignment Submissions Table

Field	Data Type	Description
submission_id	INT	Submission identifier
assignment_id	INT	Related assignment
student_id	INT	Student submitting
file_path	VARCHAR(255)	Submitted file
submission_date	DATETIME	Submission date
marks	DECIMAL(5,2)	Marks awarded
feedback	TEXT	Teacher comments
status	ENUM	Submission status

3.11.9 Quizzes Table

Field	Data Type	Description
quiz_id	INT	Quiz identifier
course_id	INT	Associated course
teacher_id	INT	Quiz creator
title	VARCHAR(200)	Quiz title
duration	INT	Time limit (minutes)
total_marks	DECIMAL(5,2)	Total marks
availability_date	DATETIME	Quiz opening date

3.11.10 Grades Table

Field	Data Type	Description
grade_id	INT	Grade identifier
student_id	INT	Student receiving grade
course_id	INT	Course
assessment_type	VARCHAR(50)	Assignment, Quiz, Exam
score	DECIMAL(5,2)	Marks obtained
grade	VARCHAR(5)	Letter grade
remarks	VARCHAR(100)	Performance remarks

3.11.11 Attendance Table

Field	Data Type	Description
attendance_id	INT	Attendance identifier
student_id	INT	Student
course_id	INT	Course
attendance_date	DATE	Attendance date
status	ENUM	Present, Absent, Late
remarks	VARCHAR(100)	Attendance remarks

3.11.12 Announcements Table

Field	Data Type	Description
announcement_id	INT	Announcement identifier
user_id	INT	Publisher
title	VARCHAR(200)	Announcement title
content	TEXT	Announcement details
publish_date	DATETIME	Publication date
expiry_date	DATETIME	Expiry date

3.11.13 Audit Logs Table

Field	Data Type	Description
log_id	INT	Log identifier

user_id	INT	User performing activity
action	VARCHAR(150)	Performed action
module	VARCHAR(100)	System module
ip_address	VARCHAR(50)	Client IP address
created_at	DATETIME	Activity timestamp

Summary

The data dictionary defines the major entities and attributes used in the proposed School Learning Management System database. It establishes a common reference for developers and database administrators, ensuring consistency in naming conventions, data types, and relationships. The documented fields support secure storage, efficient retrieval, and accurate management of academic and administrative information while providing a foundation for system implementation and future maintenance.

3.12 Decision Table

The proposed School Learning Management System implements Role-Based Access Control (RBAC) to ensure that users can only perform functions authorised for their respective roles. The decision table below summarises the permissions assigned to Administrators, Teachers, and Students.

Table 3.14: Decision Table for User Roles and Permissions

System Function	Administrator	Teacher	Student
Login	✓	✓	✓

Logout	✓	✓	✓
View Dashboard	✓	✓	✓
Manage User Accounts	✓	✗	✗
Manage Roles & Permissions	✓	✗	✗
Manage Departments	✓	✗	✗
Create Courses	✓	✓*	✗
Edit Courses	✓	✓*	✗
Delete Courses	✓	✗	✗
View Courses	✓	✓	✓
Enrol Students	✓	✗	✓**
Upload Learning Materials	✗	✓	✗
Download Learning Materials	✓	✓	✓
Archive Learning Materials	✗	✓	✗
Create Assignments	✗	✓	✗

Submit Assignments	X	X	✓
Grade Assignments	X	✓	X
Create Quizzes	X	✓	X
Attempt Quizzes	X	X	✓
Record Attendance	X	✓	X
View Attendance	✓	✓	✓
Publish Announcements	✓	✓	X
View Announcements	✓	✓	✓
View Grades	✓	✓	✓***
Generate Reports	✓	✓*****	X
View Audit Logs	✓	X	X
Backup Database	✓	X	X
Manage System Settings	✓	X	X

Key

- ✓ = Permission Granted
- X = Permission Denied

- $\checkmark^*$ = Teachers can create and manage only the courses assigned to them.
- $\checkmark^{**}$ = Teachers can generate reports only for their own courses.
- $\checkmark$ = Students can enrol only in courses made available to them (subject to institutional policy).
- $\checkmark^*$ = Students can view only their own grades and academic records.

Explanation

The decision table illustrates how the proposed Learning Management System enforces access control through Role-Based Access Control (RBAC). Administrators possess full privileges to manage users, departments, courses, reports, audit logs, and system configuration. Teachers are responsible for instructional activities such as managing assigned courses, uploading learning materials, creating assignments and quizzes, recording attendance, grading student work, and publishing announcements. Students have limited access, allowing them to view course information, access learning materials, submit assignments, participate in quizzes, monitor attendance, and view their individual academic results.

By restricting users to only the functions necessary for their responsibilities, the system enhances security, protects sensitive academic information, and maintains the integrity of institutional data.

3.13 Use Case Description

The Use Case Description explains the major interactions between the users of the proposed School Learning Management System (LMS) and the system itself. Each use

case represents a specific function performed by an actor to achieve a particular objective. The system has three primary actors: **Administrator**, **Teacher**, and **Student**, each interacting with the system according to their assigned roles and privileges.

Table 3.15: Use Case Description

Use Case	Primary Actor	Description
Login	Administrator, Teacher, Student	Authenticates users before granting access to the system.
Logout	Administrator, Teacher, Student	Ends the current user session securely.
View Dashboard	Administrator, Teacher, Student	Displays role-specific information and system activities.
Manage Users	Administrator	Creates, updates, deactivates, and manages user accounts and roles.
Manage Departments	Administrator	Creates, updates, and manages academic departments.
Create Course	Administrator, Teacher	Creates a new course and assigns course information.
Update Course	Administrator, Teacher	Modifies course details and settings.

View Courses	Administrator, Teacher, Student	Displays available courses according to user privileges.
Enrol Student	Administrator	Registers students for selected courses.
Upload Learning Material	Teacher	Uploads lecture notes, presentations, videos, and other learning resources.
Download Learning Material	Teacher, Student	Downloads learning materials from enrolled courses.
Archive Learning Material	Teacher	Archives outdated or inactive learning materials.
Create Assignment	Teacher	Creates assignments for students enrolled in a course.
Submit Assignment	Student	Uploads completed assignments before the submission deadline.
Grade Assignment	Teacher	Evaluates submitted assignments and records marks.

Create Quiz	Teacher	Creates online quizzes for assessment.
Attempt Quiz	Student	Completes quizzes within the allocated time.
Record Attendance	Teacher	Records students' attendance during lectures.
View Attendance	Teacher, Student	Displays attendance records for courses.
Record Grades	Teacher	Records examination and assessment scores.
View Grades	Student, Teacher	Displays student academic performance.
Publish Announcement	Administrator, Teacher	Posts announcements and important academic information.
View Announcements	Administrator, Teacher, Student	Displays institutional and course announcements.
Generate Reports	Administrator, Teacher	Produces reports on academic activities, attendance, grades, and users.

View Audit Logs	Administrator	Reviews user activities and system events for security and monitoring.
Backup Database	Administrator	Performs database backup to prevent data loss.

Description of Major Use Cases

Login

Actor: Administrator, Teacher, Student

Purpose:

To authenticate users and grant secure access to the Learning Management System.

Precondition:

- User account exists.
- Valid username and password are available.

Main Flow:

1. User enters login credentials.
2. System validates the credentials.
3. User is authenticated.
4. Appropriate dashboard is displayed.

Postcondition:

- User session is created.
- Access permissions are assigned according to the user's role.

Upload Learning Material

Actor: Teacher

Purpose:

To provide students with lecture notes, presentations, videos, and other educational resources.

Precondition:

- Teacher is logged into the system.
- Teacher is assigned to the selected course.

Main Flow:

1. Teacher selects a course.
2. Teacher enters material details.
3. Teacher uploads the file.
4. System validates the file.
5. Material is stored in the database and file storage.
6. Students can access the uploaded material.

Postcondition:

- Learning material becomes available to authorised students.
-

Submit Assignment

Actor: Student

Purpose:

To submit completed coursework electronically.

Precondition:

- Student is enrolled in the course.
- Assignment is available and submission deadline has not expired.

Main Flow:

1. Student selects an assignment.
2. Student uploads the completed work.
3. System validates the uploaded file.
4. Submission is stored successfully.
5. Confirmation message is displayed.

Postcondition:

- Assignment is recorded and becomes available for grading.
-

Grade Assignment

Actor: Teacher

Purpose:

To evaluate submitted assignments and provide marks and feedback.

Precondition:

- Student submissions are available.
- Teacher has permission to access the course.

Main Flow:

1. Teacher views submitted assignments.
2. Teacher reviews the submission.
3. Teacher enters marks and feedback.

4. System saves the assessment.

Postcondition:

- Student grades are updated and become available for viewing.
-

Generate Reports

Actor: Administrator, Teacher

Purpose:

To produce academic and administrative reports for decision-making.

Precondition:

- User is authenticated.
- User has permission to generate reports.

Main Flow:

1. User selects report type.
2. User specifies report parameters.
3. System retrieves relevant data.
4. Report is generated.
5. User views or downloads the report.

Postcondition:

- Report is available for printing or export.
-

Summary

The use case descriptions complement the Use Case Diagram by providing detailed explanations of how users interact with the proposed School Learning Management

System. They define the actors involved, the objectives of each interaction, the conditions required before execution, the sequence of activities performed, and the expected outcomes. Together, the Use Case Diagram and Use Case Descriptions provide a comprehensive specification of the system's functional behaviour and support the implementation of a secure, role-based, and user-centred Learning Management System.

3.14 Data Flow Diagram (DFD) Explanation

The Data Flow Diagram (DFD) illustrates how data moves through the proposed School Learning Management System (LMS). It identifies the major processes, data stores, external entities, and the flow of information among them. The DFD demonstrates how users interact with the system and how information is processed, stored, and retrieved. The major processes of the proposed system are explained below.

3.14.1 Process 1: User Authentication

The User Authentication process verifies the identity of users before granting access to the Learning Management System. All users, including administrators, teachers, and students, must successfully authenticate before accessing protected resources.

Input

- Username
- Password

Processing

- Validate user credentials.
- Compare encrypted password with the stored password.

- Verify account status.
- Determine user role.
- Create user session.
- Record login activity.

Output

- Successful login confirmation.
- User dashboard.
- Error message for invalid credentials.
- User session information.

Data Store

- Users Table
 - Roles Table
 - Audit Logs Table
-

3.14.2 Process 2: Course Management

The Course Management process allows authorised users to create, update, manage, and retrieve course information.

Input

- Course code
- Course title
- Course description
- Teacher assignment
- Department information

- Academic year
- Semester

Processing

- Validate course information.
- Create new course.
- Update existing course.
- Assign teachers.
- Save course information.

Output

- Course successfully created.
- Updated course details.
- List of available courses.

Data Store

- Courses Table
- Departments Table
- Users Table

3.14.3 Process 3: Learning Material Management

This process enables teachers to upload, organise, archive, and manage digital learning resources.

Input

- Course selection
- Material title

- Material type
- Uploaded file

Processing

- Validate uploaded file.
- Store file on the server.
- Save metadata.
- Associate material with the selected course.
- Archive materials when necessary.

Output

- Upload confirmation.
- Updated material list.
- Downloadable learning resources.

Data Store

- Learning Materials Table
 - Courses Table
-

3.14.4 Process 4: Assignment Management

The Assignment Management process supports assignment creation, submission, grading, and feedback.

Input

Teacher

- Assignment details
- Due date

- Instructions

Student

- Assignment file
- Submission comments

Processing

- Create assignments.
- Validate submissions.
- Store uploaded files.
- Grade assignments.
- Record feedback.

Output

- Assignment created.
- Submission confirmation.
- Student grades.
- Teacher feedback.

Data Store

- Assignments Table
- Assignment Submissions Table
- Grades Table

3.14.5 Process 5: Quiz Management

This process manages online quizzes used for continuous assessment.

Input

- Quiz information
- Questions
- Student answers

Processing

- Create quizzes.
- Display questions.
- Record answers.
- Automatically calculate objective scores.
- Store quiz results.

Output

- Quiz results.
- Student scores.
- Performance reports.

Data Store

- Quizzes Table
- Quiz Attempts Table
- Grades Table

3.14.6 Process 6: Attendance Management

The Attendance Management process records and maintains student attendance for each course.

Input

- Student list

- Attendance status
- Attendance date

Processing

- Validate attendance records.
- Save attendance information.
- Update attendance history.

Output

- Attendance confirmation.
- Attendance reports.
- Attendance statistics.

Data Store

- Attendance Table
- Courses Table
- Users Table

3.14.7 Process 7: Grade Management

The Grade Management process records and manages students' academic performance.

Input

- Student marks
- Assessment type
- Course information

Processing

- Validate grades.

- Calculate totals.
- Compute final grades.
- Store academic results.

Output

- Student grades.
- Academic transcript.
- Grade reports.

Data Store

- Grades Table
 - Students Table
 - Courses Table
-

3.14.8 Process 8: Announcement Management

This process enables administrators and teachers to communicate important academic information to users.

Input

- Announcement title
- Announcement content
- Publication date
- Expiry date

Processing

- Validate announcement.
- Store announcement.

- Publish to relevant users.

Output

- Published announcement.
- User notifications.

Data Store

- Announcements Table
-

3.14.9 Process 9: Report Generation

The Report Generation process provides academic and administrative reports that support institutional decision-making.

Input

- Report type
- Search criteria
- Date range

Processing

- Retrieve relevant data.
- Analyse records.
- Generate reports.
- Format report output.

Output

- Student reports.
- Attendance reports.
- Grade reports.

- Course reports.
- Statistical summaries.

Data Store

- Users Table
 - Courses Table
 - Attendance Table
 - Grades Table
 - Assignments Table
 - Learning Materials Table
-

3.14.10 Process 10: Audit Logging

The Audit Logging process records significant activities performed within the system to enhance accountability and security.

Input

- User activity
- Login information
- System events

Processing

- Record user action.
- Store timestamp.
- Save IP address.
- Save activity description.

Output

- Audit trail.
- Security reports.
- User activity history.

Data Store

- Audit Logs Table
-

Summary

The Data Flow Diagram illustrates the movement of information throughout the proposed School Learning Management System. Each process transforms user inputs into meaningful outputs while interacting with the appropriate database tables. Authentication controls access to the system, course and learning material management support teaching and learning activities, assignment, quiz, attendance, and grade management facilitate academic evaluation, while announcements, reporting, and audit logging improve communication, administration, and accountability. Together, these processes ensure efficient, secure, and reliable management of academic information within the Learning Management System.

3.15 Technology Selection

The successful implementation of the proposed School Learning Management System (LMS) depends on selecting appropriate technologies that provide reliability, scalability, maintainability, and cost-effectiveness. The technologies used were selected based on their compatibility, performance, ease of development, community support, and suitability for web-based educational applications.

3.15.1 HTML5

HyperText Markup Language version 5 (HTML5) was selected as the primary markup language for developing the frontend interface of the proposed Learning Management System.

HTML5 was chosen because it:

- Provides a standard structure for web applications.
- Supports semantic elements that improve accessibility.
- Is compatible with all modern web browsers.
- Supports multimedia content without requiring additional plugins.
- Integrates easily with CSS3 and JavaScript.

HTML5 enables the development of responsive and well-structured web pages that are accessible across different devices.

3.15.2 CSS3

Cascading Style Sheets version 3 (CSS3) was selected to design and style the graphical user interface of the system.

CSS3 provides several advantages, including:

- Responsive web design.
- Flexible layouts using Flexbox and CSS Grid.
- Attractive user interfaces.
- Animation and transition effects.
- Improved user experience across desktop and mobile devices.

CSS3 separates presentation from application logic, making the system easier to maintain and update.

3.15.3 JavaScript

JavaScript was selected to provide interactivity and dynamic functionality within the frontend of the application.

JavaScript enables the system to:

- Validate user input before submission.
- Communicate with backend APIs asynchronously.
- Update page content without full-page reloads.
- Display notifications and alerts.
- Improve responsiveness and user experience.

Its widespread browser support and integration with modern web technologies make it suitable for developing interactive educational applications.

3.15.4 PHP 8

PHP version 8 was selected as the backend programming language responsible for implementing the business logic and RESTful APIs of the Learning Management System.

PHP 8 was chosen because it:

- Integrates efficiently with MySQL databases.
- Supports object-oriented programming.
- Provides high execution performance.
- Includes improved security features.

- Supports modern programming practices.
- Has extensive documentation and community support.

PHP processes client requests, manages authentication, performs database operations, and returns JSON responses to the frontend.

3.15.5 MySQL

MySQL was selected as the relational database management system for storing and managing academic and administrative data.

The reasons for selecting MySQL include:

- High performance.
- Reliability.
- Open-source licensing.
- Strong security features.
- Support for relational database design.
- Compatibility with PHP.
- Efficient data indexing and query optimisation.

MySQL provides secure storage for users, courses, assignments, learning materials, grades, attendance records, and other institutional data.

3.15.6 REST API

The proposed Learning Management System adopts a **Representational State Transfer (REST)** architecture for communication between the frontend and backend.

REST APIs were selected because they:

- Separate the frontend from the backend.
- Improve scalability.
- Support multiple client applications.
- Simplify integration with mobile and desktop applications.
- Use standard HTTP protocols.
- Facilitate modular software development.

RESTful architecture allows the frontend to communicate with backend services through well-defined API endpoints.

3.15.7 JSON

JavaScript Object Notation (JSON) was selected as the standard format for exchanging information between the frontend and backend.

JSON was chosen because it:

- Is lightweight.
- Is easy to read and write.
- Is language independent.
- Reduces network bandwidth usage.
- Is supported by almost every programming language.

All API responses in the proposed system are transmitted in JSON format, enabling efficient communication between application components.

3.15.8 Fetch API

The **Fetch API** was selected as the mechanism for sending asynchronous HTTP requests from the frontend to the backend REST APIs.

Its advantages include:

- Simpler syntax than traditional AJAX.
- Native browser support.
- Promise-based asynchronous programming.
- Improved readability.
- Efficient handling of JSON data.
- Better error handling.

The Fetch API enables users to perform system operations without reloading web pages, resulting in a smoother user experience.

3.15.9 XAMPP

XAMPP was selected as the local development environment for implementing and testing the proposed Learning Management System.

XAMPP provides:

- Apache Web Server.
- PHP Runtime Environment.
- MySQL Database Server.
- phpMyAdmin Database Administration Tool.

The development environment simplifies installation, configuration, and testing by integrating all required software into a single package.

3.15.10 Visual Studio Code (VS Code)

Visual Studio Code was selected as the primary Integrated Development Environment (IDE) for developing the proposed system.

The reasons for selecting VS Code include:

- Lightweight and high-performance editor.
- Excellent support for PHP, HTML, CSS, and JavaScript.
- Integrated Git and GitHub support.
- Intelligent code completion and debugging.
- Extensive extension marketplace.
- Cross-platform compatibility.

VS Code improved developer productivity through features such as syntax highlighting, code navigation, integrated terminal, and version control integration.

3.15.11 Git and GitHub

Git was used as the version control system, while **GitHub** served as the remote repository for source code management and collaboration.

The combination of Git and GitHub provides:

- Version tracking.
- Source code backup.
- Collaboration among developers.
- Branch management.
- Change history.
- Easy deployment and recovery.

These tools helped ensure effective project management throughout the software development process.

Summary

The technologies selected for the proposed School Learning Management System were chosen based on their reliability, compatibility, scalability, security, and ease of development. HTML5, CSS3, and JavaScript provide a responsive and interactive user interface, while PHP 8 and MySQL deliver a robust backend capable of securely managing academic data. RESTful APIs, JSON, and the Fetch API enable efficient communication between the frontend and backend, supporting the project's API-first architecture. XAMPP provides a convenient local development environment, whereas Visual Studio Code, Git, and GitHub facilitate efficient software development and version control. Collectively, these technologies provide a modern, secure, and scalable platform capable of meeting the functional and non-functional requirements of the proposed Learning Management System.

3.16 Security Considerations

Security is a critical aspect of the proposed School Learning Management System (LMS) because the system stores sensitive academic and personal information, including user credentials, grades, attendance records, assignments, and learning materials. The system incorporates multiple security mechanisms to protect data confidentiality, integrity, and availability while preventing unauthorized access and common web-based attacks. The major security measures implemented in the proposed system are described below.

3.16.1 Password Hashing

Passwords are not stored in plain text within the database. Instead, the system uses the **bcrypt** hashing algorithm provided by PHP's `password_hash()` function to securely encrypt user passwords before storage.

During login, the entered password is verified using the `password_verify()` function rather than comparing plain-text passwords.

The use of bcrypt provides the following benefits:

- Prevents exposure of user passwords if the database is compromised.
- Uses one-way encryption that cannot easily be reversed.
- Includes automatic salting to strengthen password security.
- Protects against brute-force and rainbow table attacks.

3.16.2 User Authentication

Authentication ensures that only legitimate users gain access to the system.

The authentication process involves:

1. User enters a username or email address and password.
2. The system validates the submitted credentials.
3. The password hash is verified.
4. User identity is confirmed.
5. User role is retrieved.
6. Access is granted based on assigned permissions.

If authentication fails, an appropriate error message is returned without revealing sensitive system information.

Only authenticated users are permitted to access protected API endpoints.

3.16.3 Session Management

The proposed LMS uses secure PHP sessions to maintain authenticated user sessions after successful login.

The session management mechanism includes:

- Automatic session creation after login.
- Session regeneration after successful authentication to prevent session fixation attacks.
- Automatic session timeout after a period of inactivity.
- Session destruction during logout.
- Validation of active sessions before processing protected requests.
- Storage of user identity and role within the session.

Proper session management prevents unauthorized users from accessing protected resources after a session has expired.

3.16.4 Input Validation and Sanitisation

All user input is validated before being processed by the application.

Validation includes:

- Required field verification.
- Data type validation.

- Email format validation.
- File type validation.
- File size validation.
- Length restrictions.
- Numeric range validation.

Input sanitisation removes malicious characters before storing or displaying data.

These validation mechanisms help prevent:

- Invalid data entry.
- Cross-Site Scripting (XSS).
- Malicious file uploads.
- Unexpected application errors.

3.16.5 Prepared Statements

The proposed system uses **prepared SQL statements** with parameter binding for all database operations.

Prepared statements separate SQL commands from user-supplied data, preventing malicious SQL code from being executed.

Example database operations include:

- User authentication.
- Course creation.
- Assignment submission.
- Grade recording.
- Attendance management.

Prepared statements significantly reduce the risk of **SQL Injection attacks**, which are among the most common web application vulnerabilities.

3.16.6 Role-Based Access Control (RBAC)

The proposed Learning Management System implements **Role-Based Access Control (RBAC)** to regulate user permissions.

Three primary user roles are defined:

- Administrator
- Teacher
- Student

Each role is granted only the permissions required to perform its assigned responsibilities.

Examples include:

Administrator

- Manage users
- Manage departments
- Manage courses
- View reports
- Manage system settings
- Backup database
- View audit logs

Teacher

- Upload learning materials

- Create assignments
- Create quizzes
- Record attendance
- Grade students
- Publish announcements

Student

- View enrolled courses
- Download learning materials
- Submit assignments
- Attempt quizzes
- View grades
- View attendance

RBAC enforces the principle of **least privilege**, ensuring that users cannot access functions beyond their authorised responsibilities.

3.16.7 File Upload Security

The Learning Management System allows teachers and students to upload learning materials and assignment submissions. To prevent malicious uploads, the following controls are implemented:

- Validation of permitted file types (e.g., PDF, DOCX, PPTX).
- File size restrictions.
- Renaming uploaded files with unique identifiers.
- Secure storage outside publicly writable directories where applicable.

- Scanning and validation before database registration.

These measures reduce the risk of malicious file execution and unauthorized file access.

3.16.8 API Security

Since the proposed system adopts an API-first architecture, backend services are protected through several security mechanisms.

These include:

- Authentication before accessing protected endpoints.
- Session verification for each API request.
- Role-based authorization checks.
- Validation of request parameters.
- Standard HTTP status codes for error handling.
- JSON-only responses to ensure consistent communication.

These controls ensure that API resources are accessed only by authorised users.

3.16.9 Error Logging and Audit Trails

The system maintains comprehensive logs of significant user and system activities.

Examples of logged events include:

- User login and logout.
- Failed authentication attempts.
- Course creation and updates.
- Assignment submissions.
- Grade modifications.

- Administrative actions.
- System errors.

Audit logs support troubleshooting, monitoring, and accountability while providing evidence of user activities.

3.16.10 Database Backup and Recovery

To minimise the risk of data loss, the proposed LMS incorporates database backup and recovery mechanisms.

These include:

- Scheduled database backups.
- Secure backup storage.
- Recovery procedures for restoring lost or corrupted data.
- Protection of backup files through restricted access.

Regular backups ensure business continuity in the event of hardware failure, software malfunction, or accidental data deletion.

Summary

Security is a fundamental consideration in the design and implementation of the proposed School Learning Management System. The system employs multiple layers of protection, including **bcrypt password hashing, secure authentication, session management, input validation and sanitisation, prepared SQL statements, and Role-Based Access Control (RBAC)**. Additional measures such as secure file upload handling, API protection, audit logging, and regular database backups further strengthen the system

against common security threats. Collectively, these mechanisms ensure the confidentiality, integrity, and availability of academic information while providing a secure and reliable learning environment for administrators, teachers, and students.

3.17 Constraints and Assumptions

The successful implementation and operation of the proposed School Learning Management System (LMS) depend on certain assumptions regarding the operating environment and user behaviour. Additionally, the system is subject to technical and operational constraints that may affect its performance and deployment. Understanding these assumptions and constraints provides a realistic context for evaluating the capabilities and limitations of the proposed solution.

3.17.1 Assumptions

The development and deployment of the proposed Learning Management System are based on the following assumptions:

- Users have access to a computer, laptop, tablet, or smartphone capable of accessing the system.
- Users have a stable internet connection when accessing the web application remotely.
- Each user possesses valid login credentials issued by the system administrator.
- Administrators are responsible for creating and maintaining user accounts and assigning appropriate roles.

- Teachers will upload learning materials, assignments, quizzes, and attendance records regularly.
 - Students will use the system responsibly to access learning materials, submit assignments, and participate in assessments.
 - The institution possesses the necessary server infrastructure to host the application and database.
 - The MySQL database server will remain operational and accessible whenever the application is in use.
 - Users possess basic computer literacy required to navigate the Learning Management System.
 - Appropriate security policies and backup procedures are implemented by the institution.
-

3.17.2 Constraints

Despite the advantages of the proposed system, several limitations may affect its implementation and operation.

Internet Connectivity

Since the Learning Management System is web-based, users require reliable internet connectivity to access system resources. Poor or unstable internet connections may delay login, file uploads, assignment submissions, and access to learning materials.

Browser Compatibility

The system is designed to function on modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari. Older or unsupported browsers may not correctly display some interface components or support all system features.

Server Availability

The availability of the system depends on the operational status of the web server and database server. Hardware failures, maintenance activities, or server downtime may temporarily prevent users from accessing the system.

Hardware Limitations

System performance may be affected by the processing capability, memory capacity, and storage resources of both the server and client devices. Institutions with limited hardware resources may experience reduced performance under heavy workloads.

Scalability

Although the system is designed to accommodate institutional growth, extremely large numbers of simultaneous users may require additional server resources or infrastructure upgrades to maintain acceptable performance.

Power Supply

Reliable electrical power is essential for both the server and client devices. Power outages may interrupt access to the system and affect ongoing academic activities, particularly in environments without backup power solutions.

Data Security Risks

Although comprehensive security mechanisms have been implemented, no web application can eliminate cybersecurity risks. Threats such as phishing, malware, weak

user passwords, and social engineering attacks may still compromise user accounts if appropriate security practices are not followed.

File Storage Capacity

The storage of learning materials, assignment submissions, and multimedia resources requires sufficient disk space. As the volume of uploaded content increases over time, additional storage capacity may become necessary.

Dependence on Third-Party Software

The operation of the proposed system depends on external software components such as Apache, PHP, MySQL, XAMPP, and modern web browsers. Changes, updates, or compatibility issues in these technologies may require corresponding updates to the application.

3.17.3 Mitigation Measures

To minimise the impact of the identified constraints, the following measures are recommended:

- Deploy the system on reliable server infrastructure with adequate processing power and storage.
- Perform regular database backups and system maintenance.
- Encourage users to access the system using supported web browsers.
- Implement strong password policies and user security awareness programmes.
- Monitor server performance and upgrade infrastructure as user numbers increase.
- Provide uninterrupted power supply (UPS) and backup power solutions for the server environment.

- Ensure reliable internet connectivity for both administrators and end users.
 - Regularly update software components to address security vulnerabilities and improve compatibility.
-

Summary

The proposed School Learning Management System is developed based on several operational assumptions regarding user access, institutional infrastructure, and system administration. At the same time, its implementation may be influenced by constraints such as internet connectivity, browser compatibility, server availability, hardware resources, storage capacity, and cybersecurity risks. By recognising these assumptions and constraints and implementing appropriate mitigation measures, the institution can improve the reliability, security, and long-term sustainability of the proposed Learning Management System.

3.18 Chapter Summary

This chapter presented the methodology and system design adopted for the development of the proposed School Learning Management System (LMS). It began by analysing the existing manual and partially digitised academic management processes, identifying their limitations in areas such as user management, course administration, assignment handling, communication, reporting, and security. Based on these findings, the proposed system workflow was introduced to demonstrate how the new LMS addresses the identified challenges through a web-based, API-first architecture.

The chapter further described the functional and non-functional requirements of the system, together with the hardware, software, and database requirements necessary for successful implementation. The overall system architecture, including the RESTful API architecture, was presented to illustrate the interaction between the frontend, backend, authentication layer, and MySQL database. In addition, the major system modules were described, highlighting their respective responsibilities in supporting academic and administrative activities.

The chapter also documented the database design through a data dictionary, explained the role-based decision table, and provided detailed descriptions of the system use cases and Data Flow Diagram (DFD) processes. Furthermore, the technologies selected for system development, including HTML5, CSS3, JavaScript, PHP 8, MySQL, REST APIs, JSON, the Fetch API, and XAMPP, were justified based on their suitability, compatibility, and performance.

Finally, the chapter discussed the security mechanisms incorporated into the system, including password hashing, authentication, session management, input validation, prepared SQL statements, and Role-Based Access Control (RBAC). The assumptions and constraints associated with the proposed system were also presented to provide a realistic understanding of the operational environment and potential limitations.

Overall, this chapter established the technical foundation for implementing the proposed School Learning Management System. The design decisions and methodologies discussed provide a secure, scalable, maintainable, and user-centred solution capable of improving academic administration, enhancing teaching and learning activities, and

supporting efficient educational service delivery. The next chapter presents the implementation, testing, and evaluation of the developed system.

CHAPTER FOUR

SYSTEM DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter presents the design and implementation of the proposed School Learning Management System (LMS). It translates the requirements and system analysis presented in Chapter Three into a functional web-based system. The chapter describes the system architecture, use case design, database design, data dictionary, input and output designs, user interface design, system modules, implementation technologies, and testing procedures.

The proposed Learning Management System adopts an API-first architecture in which the frontend communicates with backend services through RESTful API endpoints. The system uses HTML5, CSS3, and JavaScript for the presentation layer, PHP for the backend application and API services, and MySQL for data storage and management. The architecture enables authorised users to interact with academic information and services through a centralised platform.

The chapter also presents the design of the database and the relationships among the major entities that support user management, courses, enrolment, learning materials, assignments, assessments, attendance, results, communication, and other academic activities. The input and output designs describe how information is entered into and retrieved from the system, while the user interface design illustrates the interaction between administrators, teachers, and students and the system.

Finally, the chapter discusses the major modules implemented in the system, the technologies and development tools used, and the testing procedures applied to verify that the system functions according to the specified requirements. The testing process covers individual components, interactions between system components, complete system functionality, and user acceptance. The chapter therefore provides a detailed account of how the proposed LMS was designed, developed, and validated.

4.2 Design of the Proposed System

The proposed School Learning Management System (LMS) was designed to address the limitations identified in the existing academic management system. The design focuses on centralising academic activities, automating routine processes, improving communication among stakeholders, strengthening data security, and providing authorised users with timely access to academic information.

The system adopts a modular and API-first design approach. The frontend provides the user interface through which administrators, teachers, and students interact with the system. Requests from the frontend are sent to the backend through RESTful API endpoints, where authentication, authorisation, validation, and business logic are processed. The backend communicates with the MySQL database to store and retrieve system information, and the results are returned to the frontend in JSON format.

The proposed system provides role-specific functionality. Administrators are responsible for managing users, departments, courses, academic sessions, reports, system settings, and other administrative activities. Teachers can manage their courses, learning materials, assignments, assessments, attendance, grades, and announcements. Students can access

enrolled courses, learning materials, assignments, quizzes, attendance information, grades, and notifications.

The design also incorporates security mechanisms to protect academic information and restrict access to authorised users. Authentication and role-based authorisation are applied to ensure that users can access only the functions and information appropriate to their assigned roles. Input validation, secure password storage, prepared database statements, session management, and other security controls are incorporated into the system design.

The proposed system uses a centralised MySQL database to maintain academic and administrative information. The database is structured around related entities that support user management, courses, enrolment, learning materials, assignments, submissions, assessments, attendance, grades, announcements, notifications, and audit activities. This centralised approach reduces data duplication and facilitates efficient retrieval and management of academic information.

Overall, the design of the proposed LMS provides an integrated platform through which academic processes can be managed more efficiently. The architecture, database, modules, interfaces, and security mechanisms described in this chapter form the foundation for the implementation and testing of the proposed system.

4.3 System Architecture

The proposed School Learning Management System adopts a **three-layer, API-first architecture** consisting of the presentation layer, application/API layer, and data layer. The architecture separates the user interface from the application logic and database, making the system easier to maintain, secure, and extend. Communication between the

frontend and backend is handled through RESTful API services using HTTP requests and JSON responses.

4.3.1 Presentation Layer

The presentation layer provides the user interface through which administrators, teachers, and students interact with the system. It is developed using **HTML5, CSS3, and JavaScript**. The interface provides role-specific dashboards and forms for performing academic and administrative activities.

JavaScript and the Fetch API are used to send requests to the backend REST API without requiring the entire webpage to be reloaded. The presentation layer is responsible for displaying information received from the API and providing appropriate forms and interfaces for user interaction.

4.3.2 Application/API Layer

The application/API layer forms the central processing component of the system. It is implemented using **PHP** and provides RESTful API endpoints through which the frontend communicates with the backend.

This layer is responsible for:

- User authentication and authorisation.
- Role-based access control.
- Request validation.
- Business logic processing.

- User management.
- Course management.
- Assignment management.
- Learning material management.
- Attendance management.
- Assessment and grading.
- Notifications and communication.
- Report processing.
- Audit and activity logging.

The API receives HTTP requests from the frontend, validates and processes the requests, communicates with the database when necessary, and returns the appropriate response in **JSON format**. The API architecture separates the frontend from the backend and allows system components to communicate through well-defined endpoints.

4.3.3 Data Layer

The data layer consists of the **MySQL database**, which provides persistent storage for the information generated and managed by the LMS. The database stores information relating to users, roles, departments, courses, enrolments, learning materials, assignments, submissions, assessments, attendance, grades, announcements, notifications, and system activities.

The data layer is accessed through the backend API rather than directly by the frontend.

This separation helps protect the database from direct client-side access and allows database operations to be controlled through the application's authentication, authorisation, validation, and business logic mechanisms.

4.3.4 Communication Flow

Communication between the layers follows a request-and-response model. When a user performs an operation, the frontend sends an HTTP request through the Fetch API to the appropriate RESTful API endpoint. The backend authenticates and authorises the request, validates the submitted data, processes the required business logic, and communicates with the MySQL database where necessary. The database returns the requested information or confirms the requested operation, after which the backend sends a JSON response to the frontend. The frontend then updates the user interface based on the response.

The overall communication flow can be represented as follows:

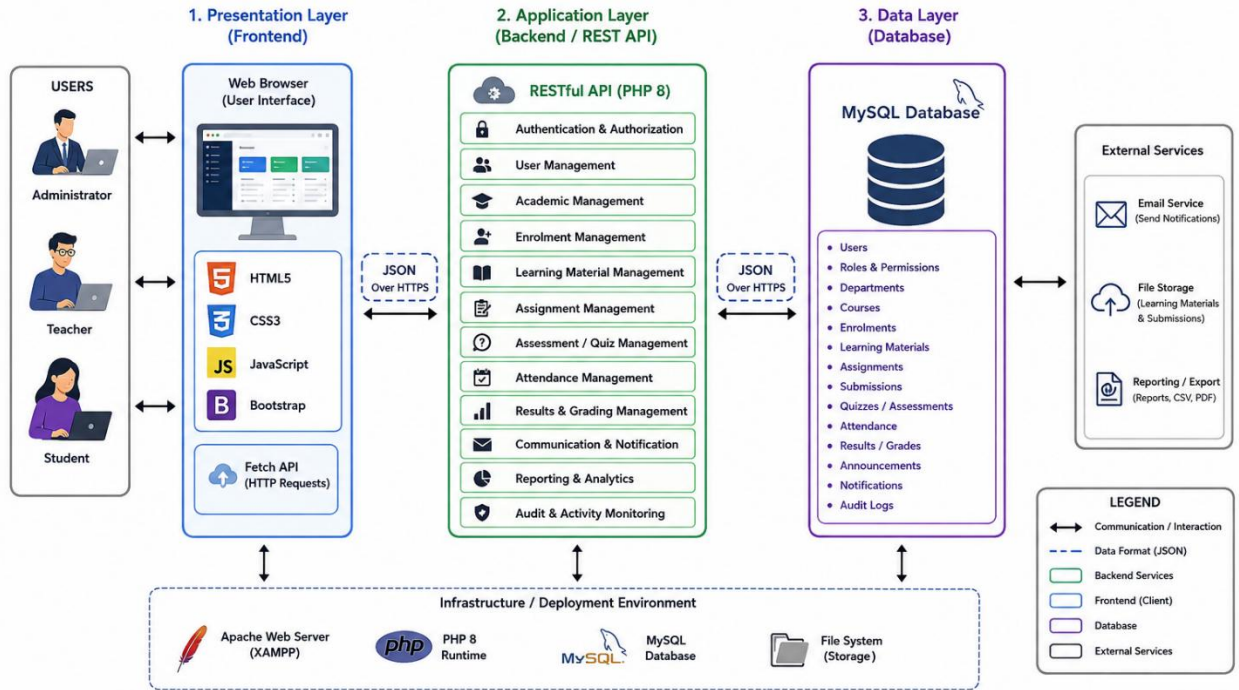
**User → Frontend → Fetch API → PHP REST API → Authentication/Authorisation
→ Business Logic → MySQL Database → JSON Response → Frontend → Updated
Interface**

This architecture provides a clear separation of responsibilities between the user interface, application services, and data storage. It also supports maintainability, scalability, security, and the future extension of the Learning Management System.

Figure 4.1 illustrates the system architecture of the proposed Learning Management System.

Proposed Learning Management System Architecture

A 3-Tier Web Application Architecture



4.4 Use Case Design

The use case design describes the interactions between the users of the proposed School Learning Management System and the functions provided by the system. It identifies the major actors, their responsibilities, and the activities they can perform within the system. The design is based on the functional requirements presented in Chapter Three and focuses on the major roles supported by the proposed LMS.

The principal actors identified for the system are **Administrator, Teacher, and Student**. Each actor is assigned specific permissions according to their responsibilities within the academic environment. Role-based access ensures that users can access only the functions relevant to their roles.

4.4.1 Administrator

The Administrator is responsible for the overall management and administration of the Learning Management System. The administrator can log into the system, manage user accounts, manage departments and courses, configure academic years or sessions, monitor system activities, generate reports, and manage system settings.

The major administrator uses cases include:

- Login and logout.
- Manage user accounts.
- Manage user roles and permissions.
- Manage departments.
- Manage courses.
- Manage academic years and sessions.
- View and generate academic reports.
- Monitor system activities and audit logs.
- Manage system settings.
- Manage system backup and related administrative functions.

4.4.2 Teacher

The Teacher interacts with the system primarily to manage teaching, learning, assessment, and student academic activities. Teachers can access assigned courses, upload learning materials, create assignments and quizzes, record attendance, assess students, enter grades, and communicate academic information.

The major teacher use cases include:

- Login and logout.
- Access assigned courses.
- Upload and manage learning materials.
- Create and manage assignments.
- Create and manage quizzes.
- Record student attendance.
- Grade assignments and assessments.
- View student performance.
- Publish announcements.
- Communicate with students.
- View relevant academic reports.

4.4.3 Student

The student uses the system to access learning resources and participate in academic activities. Students can access courses in which they are enrolled, view or download learning materials, submit assignments, take quizzes, monitor attendance, view grades, and receive academic notifications.

The major student use cases include:

- Login and logout.

- Access enrolled courses.
- View and download learning materials.
- Submit assignments.
- Take quizzes and assessments.
- View assignment results and grades.
- View attendance information.
- Receive announcements and notifications.
- Access relevant academic information.

4.4.4 Use Case Diagram

The use case design presented in Chapter Three provided the functional basis for implementing the proposed School Learning Management System. In Chapter Four, the emphasis is placed on the implementation outcomes and the impact of the developed system rather than reproducing the use case diagram.

The implemented system provides separate functionality for administrators, teachers, and students according to their assigned roles. Authentication and authorisation mechanisms ensure that users are directed to the appropriate system functions after successful login.

Administrator functionality: The administrator can manage users, courses, departments, academic sessions, system settings, reports, and other administrative activities. This centralises administrative operations and reduces the need to maintain separate records for different academic activities.

Teacher functionality: Teachers can access assigned courses, upload learning materials, create assignments and quizzes, record attendance, assess student work, and manage academic performance information. These functions reduce manual processing and improve the organisation of teaching and assessment activities.

Student functionality: Students can access their courses and learning materials, submit assignments, participate in quizzes, view grades, monitor attendance information, and receive academic notifications. This provides students with more direct and timely access to their academic information.

The implementation of these use cases demonstrates the practical impact of the proposed system. Academic activities that previously depended on paper records, spreadsheets, separate communication channels, and repeated manual processing can be performed through a centralised web-based platform. This improves accessibility, reduces duplication of effort, supports more efficient information management, and provides authorised users with timely access to academic information.

The implemented use cases therefore demonstrate that the proposed Learning Management System provides an integrated platform for supporting administrative, teaching, assessment, attendance, and student-learning activities.

4.4.5 Use Case Description

Use Case	Actor	Description
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Login	Administrator, Teacher, Student	Authenticates a user and provides access according to the assigned role.
Manage Users	Administrator	Creates, updates, views, and manages user accounts.
Manage Departments	Administrator	Creates and manages academic departments.
Manage Courses	Administrator, Teacher	Creates, updates, and manages course information according to assigned permissions.
Manage Materials	Teacher	Uploads, updates, and manages learning materials.
Create Assignment	Teacher	Creates assignments and specifies relevant academic requirements and deadlines.
Submit Assignment	Student	Submits completed academic work through the system.
Grade Assignment	Teacher	Reviews student submissions and records grades and feedback.
Manage Quizzes	Teacher	Creates and manages online quizzes and assessments.
Take Quiz	Student	Attempts available quizzes and submits answers for assessment.

Record Attendance	Teacher	Records and manages student attendance for academic activities.
View Grades	Student	Views grades and academic performance information.
Generate Reports	Administrator	Generates academic and administrative reports from stored system data.
Manage Announcements	Teacher, Administrator	Publishes academic or administrative announcements to relevant users.
View Notifications	Teacher, Student	Views system notifications and relevant academic updates.
View Audit Logs	Administrator	Reviews recorded system activities for monitoring and accountability.

The use case design provides a functional representation of how each category of user interacts with the proposed LMS. It also establishes the basis for the subsequent module implementation and testing presented later in this chapter.

4.5 Database Design

The proposed School Learning Management System uses **MySQL** as its relational database management system. The database was designed to provide a centralised and structured repository for storing, managing, and retrieving academic and administrative

information. It supports the major functions of the system, including user management, course management, enrolment, learning materials, assignments, assessments, attendance, results, communication, notifications, and system activity monitoring.

The database design follows a relational approach in which information is organised into related tables. Each table represents a specific entity or functional area of the system, while relationships between tables are established using primary keys and foreign keys. This approach reduces unnecessary duplication of data and supports consistent storage and retrieval of information.

4.5.1 Database Structure

The database is organised into several functional groups that correspond to the major modules of the LMS. These include:

- **User and Security Management:** stores user accounts, roles, permissions, authentication-related information, and system activity records.
- **Academic Management:** stores departments, academic years, sessions, courses, and related academic information.
- **Enrolment Management:** maintains information about students and their enrolment in courses.
- **Learning Material Management:** stores information about uploaded course materials and their association with courses.
- **Assignment Management:** manages assignments, submission records, deadlines, grades, and feedback.

- **Assessment and Quiz Management:** stores quizzes, questions, student attempts, and assessment results.
- **Attendance Management:** maintains student attendance records for relevant courses and academic activities.
- **Results Management:** stores student grades, performance records, and related academic results.
- **Communication Management:** supports announcements, notifications, messages, and other communication activities.
- **Audit and Activity Management:** records relevant user and system activities for monitoring and accountability.

The relationships among these entities allow information to be shared across different modules without requiring the same information to be repeatedly stored.

4.5.2 Primary and Foreign Keys

Primary keys are used to uniquely identify records within individual tables. For example, a user record is identified by a unique user identifier, while courses, assignments, materials, and submissions have their own unique identifiers.

Foreign keys establish relationships between related tables. For example, a course record can be associated with a teacher or department, while an assignment can be associated with a particular course. Similarly, assignment submissions can be associated with both the relevant assignment and the student who submitted the work.

The use of primary and foreign keys helps maintain **referential integrity** and ensures that relationships between academic records remain consistent.

4.5.3 Database Normalisation

The database is designed using relational database principles and follows normalisation practices to reduce data redundancy and improve data integrity. The design aims to organise information so that each table represents a logical entity and attributes are stored in appropriate tables.

The database design considers **Third Normal Form (3NF)**, where appropriate, to minimise unnecessary duplication and ensure that non-key attributes depend on the appropriate primary key. This improves maintainability and reduces inconsistencies when records are inserted, updated, or deleted.

4.5.4 Database Relationships

The major relationships within the proposed LMS include:

- A **user** can have an assigned role that determines the functions available to the user.
- A **department** can be associated with multiple courses.
- A **teacher** can manage or be assigned to multiple courses.
- A **student** can enrol in multiple courses.
- A **course** can have multiple enrolled students.
- A **course** can contain multiple learning materials.

- A **course** can contain multiple assignments and assessments.
- An **assignment** can have multiple student submissions.
- A **student** can submit assignments associated with their enrolled courses.
- A **course** can have attendance records for participating students.
- A **student** can have multiple academic results and performance records.
- Users can receive announcements and notifications based on their roles and activities.

These relationships provide the foundation for integrating the different academic functions of the LMS.

4.5.5 Database Security

Database security is an important consideration in the design because the system stores sensitive academic and user information. Direct access to the database is restricted to the backend application, while frontend requests are processed through the RESTful API.

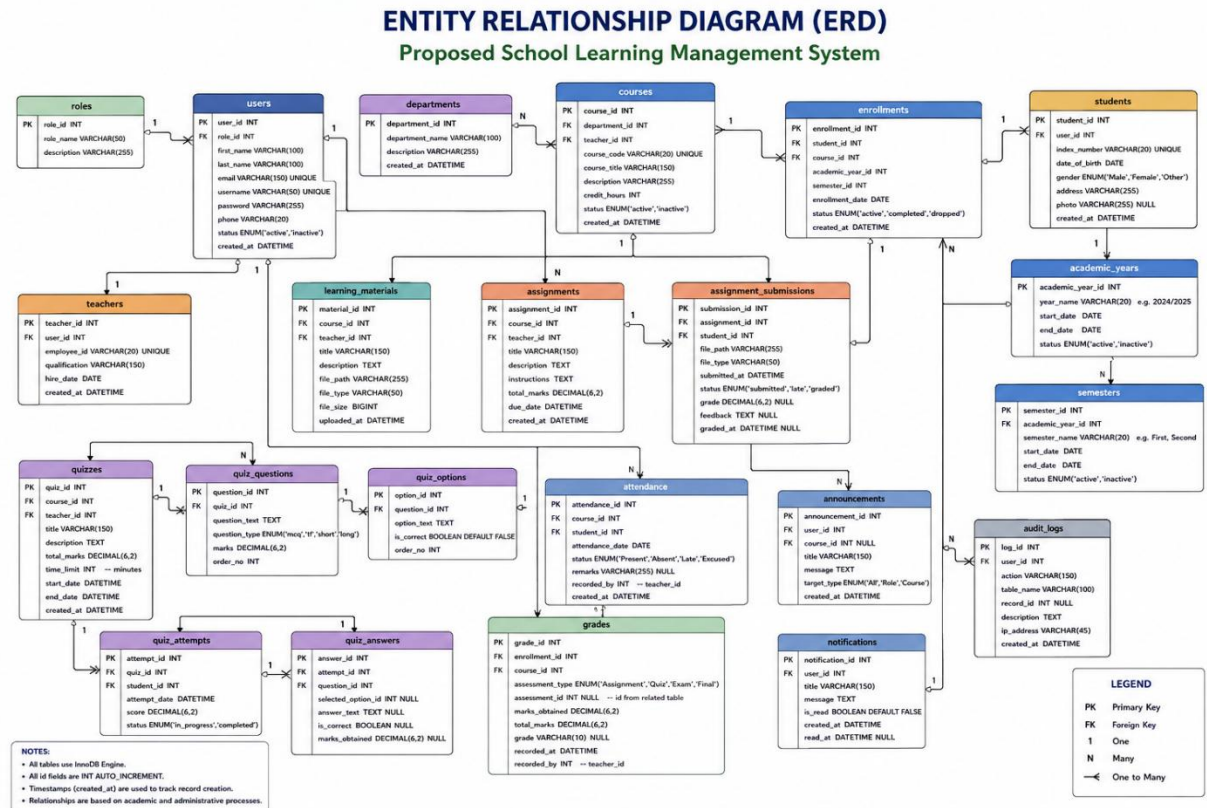
Security measures include authentication and authorisation, role-based access control, input validation, prepared SQL statements, secure password storage, and controlled database access. These measures help protect academic records from unauthorised access, manipulation, and accidental loss.

Regular database backup is also considered to support recovery in the event of hardware failure, software errors, or data loss.

4.5.6 Entity Relationship Diagram

The relationships among the major entities of the proposed Learning Management System are represented using an **Entity Relationship Diagram (ERD)**. The ERD provides a graphical representation of the database entities, their attributes, and the relationships between them.

Figure 4.2 presents the Entity Relationship Diagram of the proposed School Learning Management System. The diagram provides the basis for understanding how the major database entities interact and how academic information flows between the different modules of the system.



4.6 Data Dictionary

The data dictionary provides a detailed description of the major data elements used in the database of the proposed School Learning Management System. It defines the names,

data types, and purposes of important fields used to store and manage information within the system. The data dictionary serves as a reference for understanding the structure of the database and the relationships among its major entities.

The dictionary is organised according to the major functional areas of the LMS, including user management, academic management, learning resources, assignments, assessments, attendance, results, communication, and system monitoring.

4.6.1 User and Security Tables

Table	Field	Data Type	Description
Users	user_id	INT	Unique identifier for a system user.
Users	username	VARCHAR	Unique username used to identify the user.
Users	password	VARCHAR	Securely stored user password hash.
Users	email	VARCHAR	User's email address.
Users	role_id	INT	Identifies the role assigned to the user.
Users	status	VARCHAR	Indicates whether the user account is active or inactive.
Roles	role_id	INT	Unique identifier for a user role.

Roles	role_name	VARCHAR	Name of the assigned system role.
Permissions	permission_id	INT	Unique identifier for a system permission.
Permissions	permission_name	VARCHAR	Defines an action that a user may perform.

4.6.2 Academic Management Tables

Table	Field	Data Type	Description
Departments	department_id	INT	Unique identifier for an academic department.
Departments	department_name	VARCHAR	Name of the academic department.
Academic_Years	academic_year_id	INT	Unique identifier for an academic year.
Academic_Years	year_name	VARCHAR	Name or description of the academic year.
Courses	course_id	INT	Unique identifier for a course.
Courses	course_code	VARCHAR	Unique code assigned to the course.
Courses	course_name	VARCHAR	Name of the course.

Courses	department_id	INT	Identifies the department responsible for the course.
Courses	teacher_id	INT	Identifies the teacher assigned to the course.

4.6.3 Student Enrolment Tables

Table	Field	Data Type	Description
Students	student_id	INT	Unique identifier for a student.
Students	user_id	INT	Links the student record to a user account.
Students	student_number	VARCHAR	Unique identification number assigned to the student.
Enrolments	enrolment_id	INT	Unique identifier for an enrolment record.

Enrolments	student_id	INT	Identifies the student being enrolled.
Enrolments	course_id	INT	Identifies the course in which the student is enrolled.
Enrolments	academic_year_id	INT	Identifies the relevant academic year.

4.6.4 Learning Material Tables

Table	Field	Data Type	Description
Materials	material_id	INT	Unique identifier for a learning material.
Materials	course_id	INT	Identifies the course associated with the material.
Materials	title	VARCHAR	Title of the learning material.
Materials	file_path	VARCHAR	Location of the uploaded material.

Materials	uploaded_by	INT	Identifies the user who uploaded the material.
Materials	created_at	DATETIME	Records when the material was uploaded.

4.6.5 Assignment Tables

Table	Field	Data Type	Description
Assignments	assignment_id	INT	Unique identifier for an assignment.
Assignments	course_id	INT	Identifies the course associated with the assignment.
Assignments	title	VARCHAR	Title of the assignment.
Assignments	description	TEXT	Instructions or description of the assignment.

Assignments	due_date	DATETIME	Specifies the assignment submission deadline.
Assignment_Submissions	submission_id	INT	Unique identifier for a submission.
Assignment_Submissions	assignment_id	INT	Identifies the assignment being submitted.
Assignment_Submissions	student_id	INT	Identifies the student submitting the assignment.
Assignment_Submissions	file_path	VARCHAR	Location of the submitted file.
Assignment_Submissions	grade	DECIMAL	Grade awarded for the submission.
Assignment_Submissions	feedback	TEXT	Feedback provided by the teacher.

4.6.6 Assessment and Quiz Tables

Table	Field	Data Type	Description
Quizzes	quiz_id	INT	Unique identifier for a quiz.
Quizzes	course_id	INT	Identifies the course associated with the quiz.
Quizzes	title	VARCHAR	Title of the quiz.
Quizzes	duration	INT	Duration allocated for completing the quiz.
Quiz_Questions	question_id	INT	Unique identifier for a quiz question.
Quiz_Questions	quiz_id	INT	Identifies the quiz containing the question.
Quiz_Questions	question_text	TEXT	Text of the quiz question.
Quiz_Submissions	submission_id	INT	Unique identifier for a quiz submission.
Quiz_Submissions	quiz_id	INT	Identifies the quiz attempted.

Quiz_Submissions	student_id	INT	Identifies the student attempting the quiz.
Quiz_Submissions	score	DECIMAL	Score obtained by the student.

4.6.7 Attendance Tables

Table	Field	Data Type	Description
Attendance	attendance_id	INT	Unique identifier for an attendance record.
Attendance	student_id	INT	Identifies the student whose attendance is recorded.
Attendance	course_id	INT	Identifies the course associated with the attendance record.
Attendance	attendance_date	DATE	Date on which attendance was recorded.

Attendance	status	VARCHAR	Indicates whether the student was present, absent, or otherwise recorded.
------------	--------	---------	---

4.6.8 Results and Performance Tables

Table	Field	Data Type	Description
Results	result_id	INT	Unique identifier for an academic result.
Results	student_id	INT	Identifies the student associated with the result.
Results	course_id	INT	Identifies the course associated with the result.
Results	score	DECIMAL	Numerical score obtained by the student.
Results	grade	VARCHAR	Grade assigned based on the

			student's performance.
Grade_Categories	category_id	INT	Unique identifier for a grade category.
Grade_Categories	category_name	VARCHAR	Name of the grading category.

4.6.9 Communication and Notification Tables

Table	Field	Data Type	Description
Announcements	announcement_id	INT	Unique identifier for an announcement.
Announcements	title	VARCHAR	Title of the announcement.
Announcements	message	TEXT	Content of the announcement.
Announcements	created_by	INT	Identifies the user who created the announcement.
Notifications	notification_id	INT	Unique identifier for a notification.

Notifications	user_id	INT	Identifies the recipient of the notification.
Notifications	message	TEXT	Content of the notification.
Notifications	is_read	BOOLEAN	Indicates whether the notification has been read.

4.6.10 Audit and Activity Tables

Table	Field	Data Type	Description
Audit_Logs	audit_id	INT	Unique identifier for an audit record.
Audit_Logs	user_id	INT	Identifies the user who performed the activity.
Audit_Logs	action	VARCHAR	Describes the action performed by the user.
Audit_Logs	activity_time	DATETIME	Records the date and time of the activity.

Audit_Logs	ip_address	VARCHAR	Records the IP address associated with the activity where applicable.
------------	------------	---------	---

The data dictionary provides a structured reference for the principal data elements used by the proposed Learning Management System. Primary keys uniquely identify records, while foreign keys establish relationships between related entities. The defined data elements support the implementation of the system's major modules and provide a consistent basis for storing, retrieving, and processing academic and administrative information.

4.7 Input Design

Input design describes how data is entered into the proposed School Learning Management System and how the system validates and processes the information before storing it in the database. The design aims to ensure that data entered by administrators, teachers, and students is accurate, complete, consistent, and appropriate for the intended operation.

The system provides role-specific input forms based on the responsibilities and permissions of each user. The major input areas include user management, academic management, learning materials, assignments, assessments, attendance, results, and communication.

4.7.1 Administrator Input

The Administrator provides information required for managing the overall operation of the LMS. Major administrator input forms include:

- **User Registration Form:** captures username, name, email address, password, role, and account status.
- **Department Form:** captures department name and other relevant departmental information.
- **Course Form:** captures course code, course name, department, teacher assignment, and related course information.
- **Academic Session Form:** captures academic year, semester, and session information.
- **System Settings Form:** captures configuration information required for system operation.
- **Report Parameters** allows administrators to specify criteria such as academic year, course, department, student, or date range when generating reports.

4.7.2 Teacher Input

Teachers provide information required for teaching, assessment, attendance, and student performance management. Major teacher input forms include:

- **Course Management Form:** captures and updates relevant course information.
- **Learning Material Form:** allows teachers to provide material titles, descriptions, associated courses, and upload files.

- **Assignment Form:** captures assignment title, description, course, instructions, maximum marks, and submission deadline.
- **Quiz and Assessment Form:** captures assessment details, questions, options, correct answers, marks, and other relevant settings.
- **Attendance Form:** records the attendance status of students for a selected course and date.
- **Grading Form:** allows teachers to enter marks, grades, and feedback for student submissions.
- **Announcement Form:** captures announcement titles, messages, target users, and related academic information.

4.7.3 Student Input

Students provide information primarily when participating in learning and assessment activities. Major student input forms include:

- **Profile Form** allows students to provide or update permitted personal information.
- **Assignment Submission Form:** allows students to upload completed assignments and provide required submission information.
- **Quiz Input** allows students to select or enter answers when completing online quizzes.
- **Communication Input:** allows students to send messages or other permitted academic communication.

- **Feedback Input:** allows students to provide feedback where the relevant system functionality is available.

4.7.4 Input Validation

Input validation is incorporated into the system to reduce incorrect, incomplete, or potentially harmful data. Validation is performed before information is processed or stored in the database.

The major validation mechanisms include:

- Required fields are checked to ensure that essential information is provided.
- Email addresses are validated using appropriate formats.
- Usernames and other unique identifiers are checked for duplication.
- Password requirements are enforced during account creation or password changes.
- Numerical fields are validated to ensure that appropriate numeric values are entered.
- Dates are validated to ensure that they follow the required format and logical constraints.
- Assignment deadlines are checked against relevant dates.
- Uploaded files are validated according to permitted file types and size restrictions.
- User permissions are checked before restricted operations are processed.
- Database queries use prepared statements to reduce the risk of SQL injection.

4.7.5 Input Processing

After successful validation, the submitted information is sent from the frontend to the appropriate RESTful API endpoint. The backend authenticates and authorises the request before processing the associated business logic. Where database interaction is required, the API performs the appropriate database operation and returns a JSON response to the frontend.

The general input processing sequence is:

User Input → Frontend Form → Validation → HTTP Request → REST API → Authentication/Authorisation → Business Logic → MySQL Database → JSON Response → Updated Interface

This approach ensures that data is processed through the appropriate application layer rather than allowing the frontend to communicate directly with the database.

4.7.6 Input Design Principles

The input interfaces are designed according to the following principles:

- **Simplicity:** Forms contain only information relevant to the operation being performed.
- **Consistency:** Similar forms use consistent labels, controls, buttons, and validation messages.
- **Accuracy:** Validation mechanisms reduce incorrect or invalid data entry.
- **Security:** Sensitive operations require appropriate authentication and authorisation.

- **User feedback:** The system displays success, warning, and error messages after user actions.
- **Role-based access:** Users are presented with input functions appropriate to their assigned roles.
- **Usability:** Forms are designed to support clear navigation and efficient data entry.

The input design therefore provides a controlled mechanism for capturing information from administrators, teachers, and students while ensuring that submitted data is validated, securely processed, and stored appropriately within the LMS database.

4.8 Output Design

Output design describes how information processed by the proposed School Learning Management System is presented to administrators, teachers, and students. The system converts information stored in the database into meaningful outputs such as dashboards, academic reports, grades, attendance records, notifications, and other system responses. The design aims to ensure that outputs are accurate, relevant, clearly presented, and accessible only to authorised users.

The major outputs of the proposed LMS are described below.

4.8.1 Administrator Outputs

The Administrator receives outputs that support academic and system management.

These include:

- **Administrator Dashboard:** displays summaries of users, courses, departments, academic activities, and other relevant system statistics.

- **User Reports:** provides information about registered administrators, teachers, and students.
- **Course Reports:** presents course information, teacher assignments, and enrolment statistics.
- **Academic Reports:** provides student performance and other academic information based on selected criteria.
- **Attendance Reports:** summarises student attendance records for selected courses, classes, or periods.
- **System Activity Reports:** provides information about recorded user and system activities.
- **Audit Logs** displays relevant activities performed by users for monitoring and accountability.
- **Notifications:** informs administrators about relevant system and academic events.

4.8.2 Teacher Outputs

Teachers receive outputs that support teaching, assessment, attendance, and student performance management. These include:

- **Teacher Dashboard:** provides an overview of assigned courses, students, assignments, assessments, and other teaching activities.
- **Course Information:** displays course details and enrolled students.

- **Learning Material Status:** displays uploaded materials and their associated courses.
- **Assignment Submissions:** provides a list of student submissions and their submission status.
- **Grading Results:** displays grades and feedback recorded for student submissions.
- **Attendance Reports:** presents attendance records for students enrolled in a teacher's courses.
- **Student Performance Reports:** provides information about student academic performance.
- **Quiz and Assessment Results:** displays student scores and assessment outcomes.
- **Announcements:** displays published academic announcements and their recipients.

4.8.3 Student Outputs

Students receive outputs that provide access to learning resources and academic information. These include:

- **Student Dashboard:** displays enrolled courses, pending assignments, upcoming assessments, notifications, and other relevant academic information.

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

School Calendar

Logout

My Dashboard

69

Kwamew Osei

Learner

89

Hello, Kwamew Osei!

You have 0 pending assignment(s) and 2 class(es) today.

8

Enrolled Courses

0

Pending Assignments

36%

Attendance Rate

0

Upcoming Assessments

Next 7 days

2

Today's Classes

Scheduled today

Today's Schedule

Recent Grades

View All

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

School Calendar

Course Materials

Logout

My Dashboard

69

Kwamew Osei

Learner

Grade Trends by Subject

100%

90%

80%

70%

60%

50%

40%

30%

20%

10%

0%

Integrated Science

 Dashboard

-  My Classes
-  My Subjects
-  My Courses
-  Assignments
-  Quizzes
-  Grades
-  Announcements
-  Attendance
-  Timetable
-  School Calendar
-  Course Materials

 Logout

 My Courses



Biology
No teacher assigned

0% Complete

[Continue Learning](#)



Chemistry
No teacher assigned

0% Complete

[Continue Learning](#)



Core Mathematics3
No teacher assigned

0% Complete



Core Mathematics7
No teacher assigned

0% Complete

 Dashboard

-  My Classes
-  My Subjects
-  My Courses
-  Assignments
-  Quizzes
-  Grades
-  Announcements
-  Attendance
-  Timetable
-  School Calendar
-  Course Materials

 Logout

 Pending Assignments [View All](#)


All caught up!

 Announcements

- TASD**
__ANN_META__({"priority":"urgent","class_audience":""} TASD
84d ago
- ghhhh hy hvivyiyilkiu**
__ANN_META__({"priority":"normal"} ghhh hy hvivyiyilkiu
93d ago
- Not enough arguments. 878**
__ANN_META__({"priority":"normal","class_audience":["SHS 1 Business A","SHS 1 General Arts A","SHS 1 General Arts...
98d ago

 My Courses



GHANA EDUCATION SERVICE



GHANA EDUCATION SERVICE

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

School Calendar

Course Materials

Logout

My Dashboard

69

Kwame Osei
Learner

Today's Schedule

8:00 AM
1h 30m

Biology
SHS 1 General Science A • Ama Asante

Lab B201

10:00 AM
1h 30m

Chemistry
SHS 1 General Science A • Yaw Frimpong

Room A201

Pending Assignments

View All

✓

All caught up!

Recent Grades

View All

A1

sdshdsdshjds

Integrated Science

78%

78.00/100.00

C5

hikkp gghh

Integrated Science

56%

56.00/100.00

B2

SWELEW

Integrated Science

71%

50.00/70.00

Announcements

TASD

__ANN_META__({"priority":"urgent","class_audience":""}) TASD

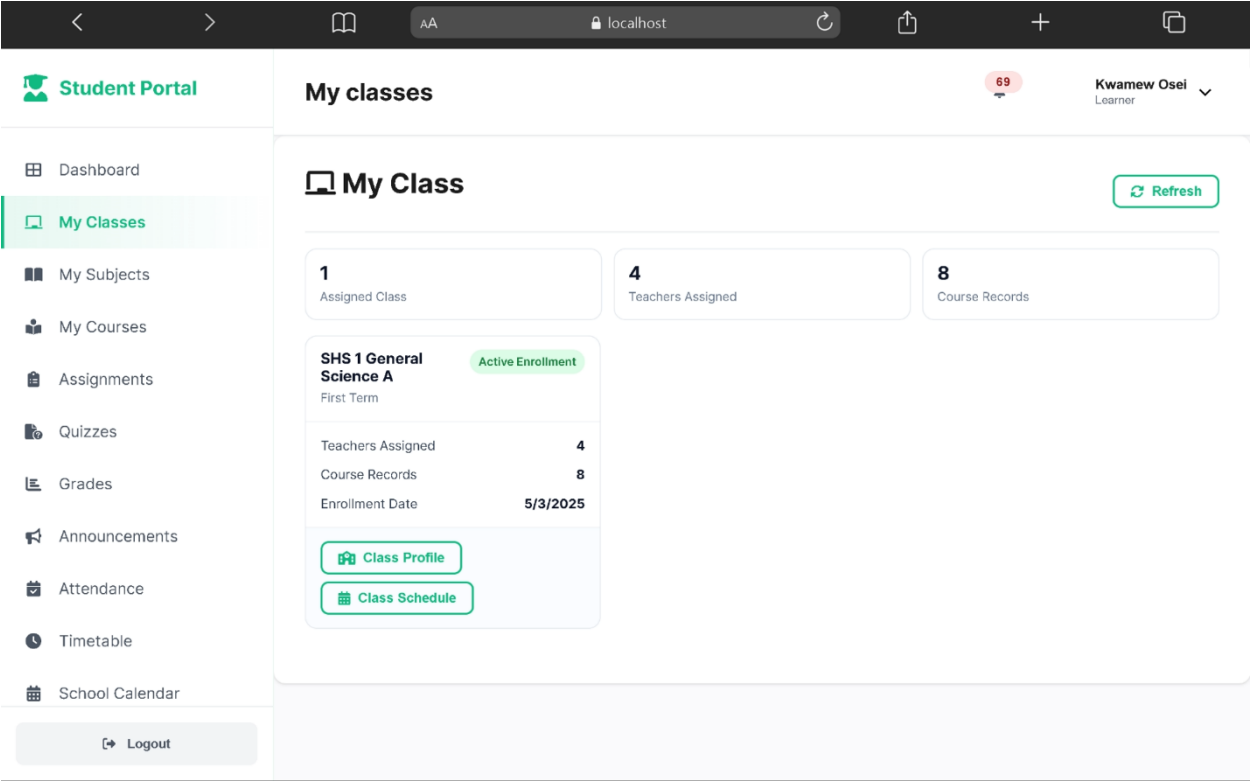
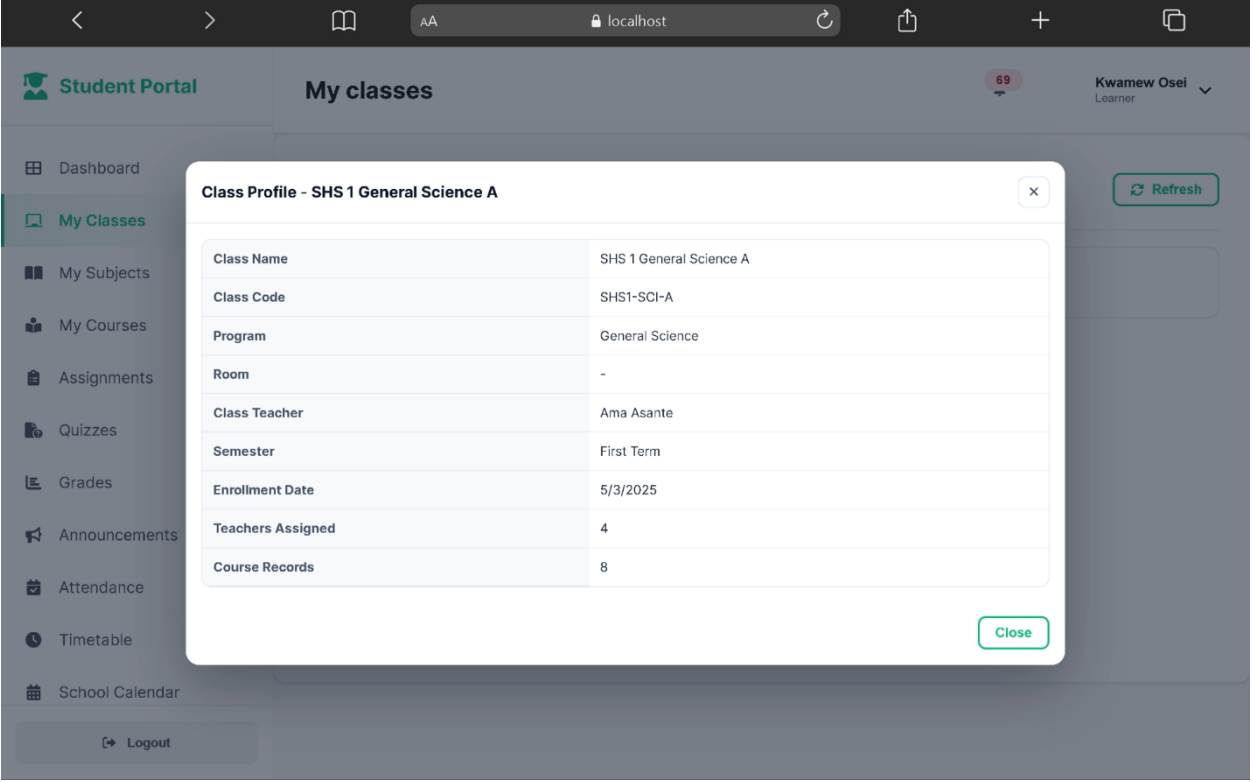
84d ago

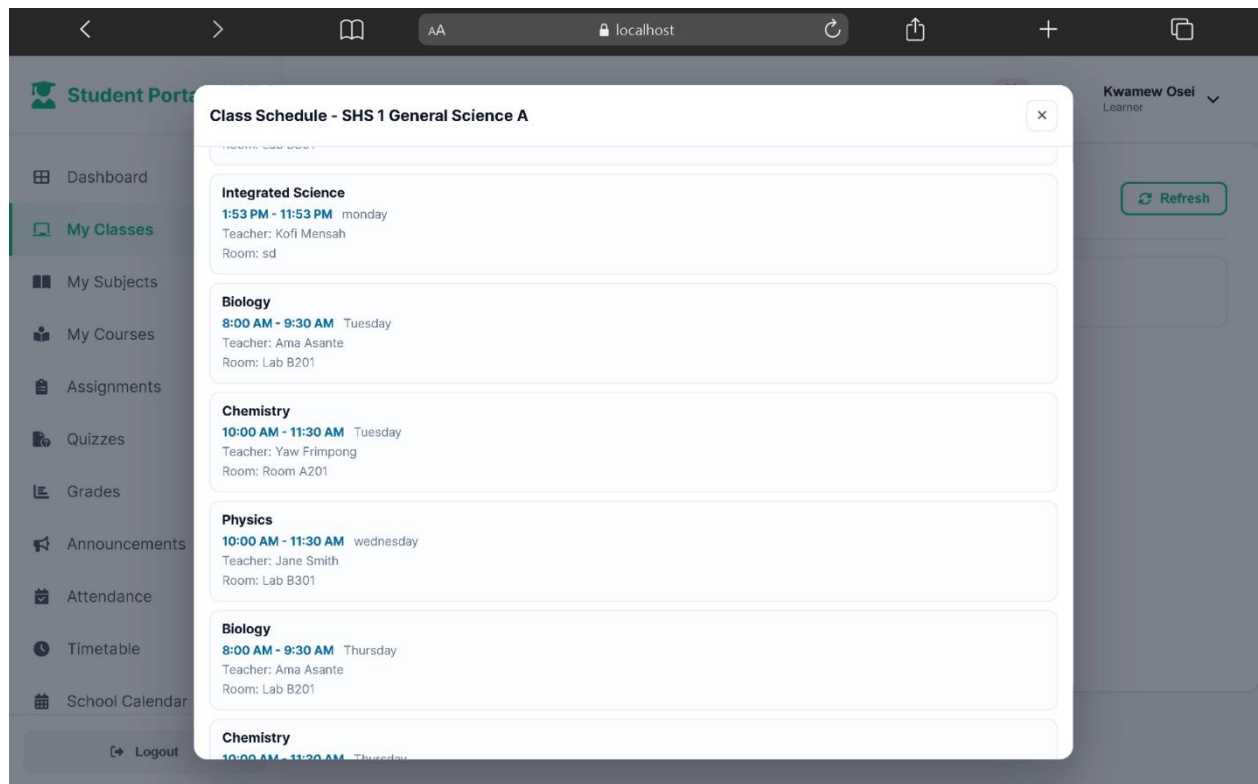
ghhhh hy hvivyiyiikiu

__ANN_META__({"priority":"normal"}) ghhhh hy hvivyiyiikiu

93d ago

- **Course Information:** displays courses in which the student is enrolled.





- **Learning Materials** provides access to available course materials.

- **Assignment Information:** displays assignment descriptions, deadlines, submission status, grades, and feedback.

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

School Calendar

Logout

Assignment Details

Close

TITLE

Assignment 2: Problem Set - INT

SUBJECT

Integrated Science

DUE DATE

8/12/2026, 8:21:00 AM

SUBMISSION TYPE

File Upload

DESCRIPTION

Solve problems from textbook section 2.3-2.5

INSTRUCTION FILE

Download Instructions

SUBMITTED FILE

View Submitted File

SUBMIT ASSIGNMENT

Choose File

No file chosen
Accepted by backend upload handler. Max size is 30MB.

Kwame Osei

Learner

Refresh

All subjects

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

School Calendar

Logout

Assignment Details

Close

TITLE

SWELEW

SUBJECT

Integrated Science

DUE DATE

3/29/2026, 11:08:00 AM

SUBMISSION TYPE

Text and/or File

DESCRIPTION

ASCASCA

INSTRUCTION FILE

Download Instructions

GRADE & FEEDBACK

Score: 50.00 / 70.00

Feedback:

tftyu

SUBMITTED FILE

Kwame Osei

Learner

Refresh

All subjects

graded

Integrated Science

5/13/2026, 11:45:00 PM

Text and/or File

On time

56.00/100.00

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

School Calendar

Logout

TITLE

Assignment 1: Chapter Overview - MATH

SUBJECT

Core Mathematics7

DUE DATE

8/30/2026, 8:19:00 AM

SUBMISSION TYPE

Text and/or File

DESCRIPTION

Complete readings from Chapter 1-2 and submit your summary.

INSTRUCTION FILE

Download Instructions

SUBMIT ASSIGNMENT

Write your response or notes here...

Choose File

No file chosen

Accepted by backend upload handler. Max size is 30MB.

Cancel

Submit Assignment

Kwamew Osei

Learner

Refresh

All subjects

1: Chapter MATH

pending

ings from Chapter

your summary.

Core Mathematics7

8/30/2026, 8:19:00 AM

Text and/or File

On time

Not Graded

Submit

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

School Calendar

Logout

Assignment Details

Close

TITLE

Assignment 1: Chapter Overview - MATH

SUBJECT

Core Mathematics7

DUE DATE

8/30/2026, 8:19:00 AM

SUBMISSION TYPE

Text and/or File

DESCRIPTION

Complete readings from Chapter 1-2 and submit your summary.

INSTRUCTION FILE

Download Instructions

SUBMIT ASSIGNMENT

Write your response or notes here...

Choose File

No file chosen

Accepted by backend upload handler. Max size is 30MB.

Kwamew Osei

Learner

Refresh

All subjects

1: Chapter MATH

pending

ings from Chapter

your summary.

Core Mathematics7

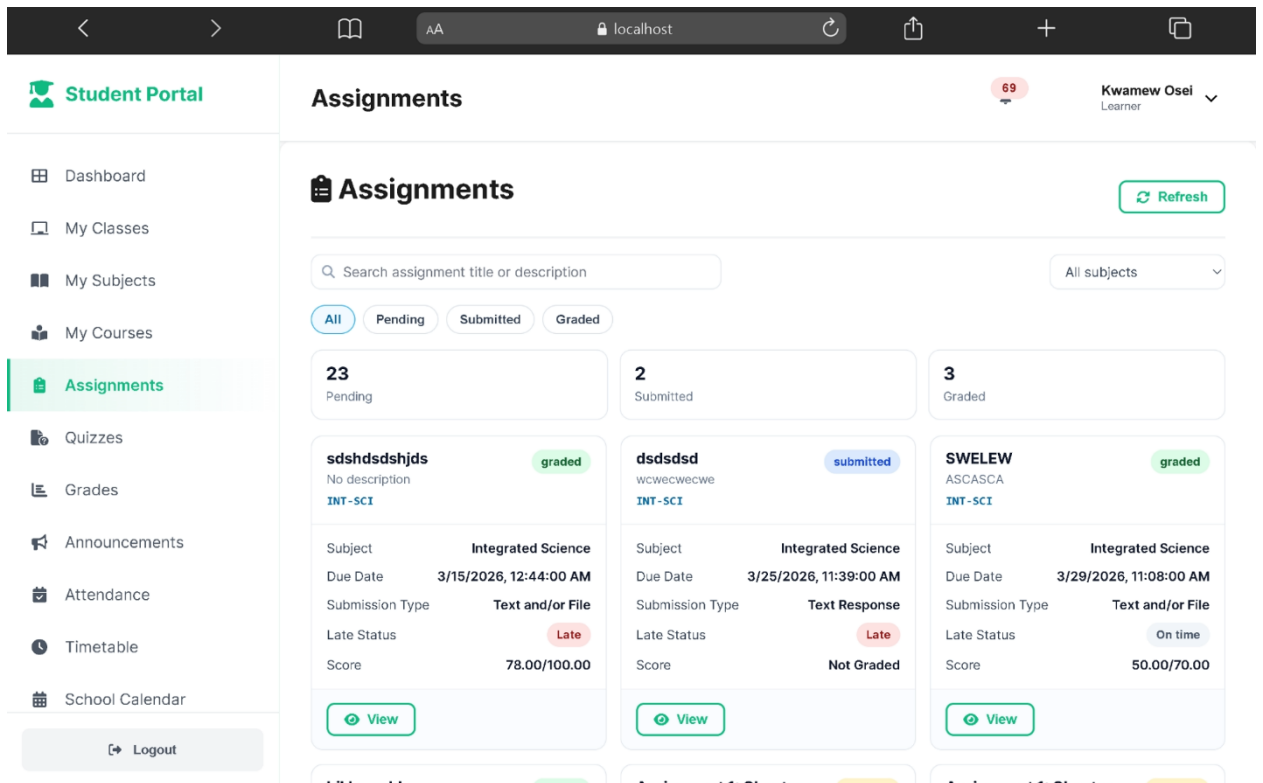
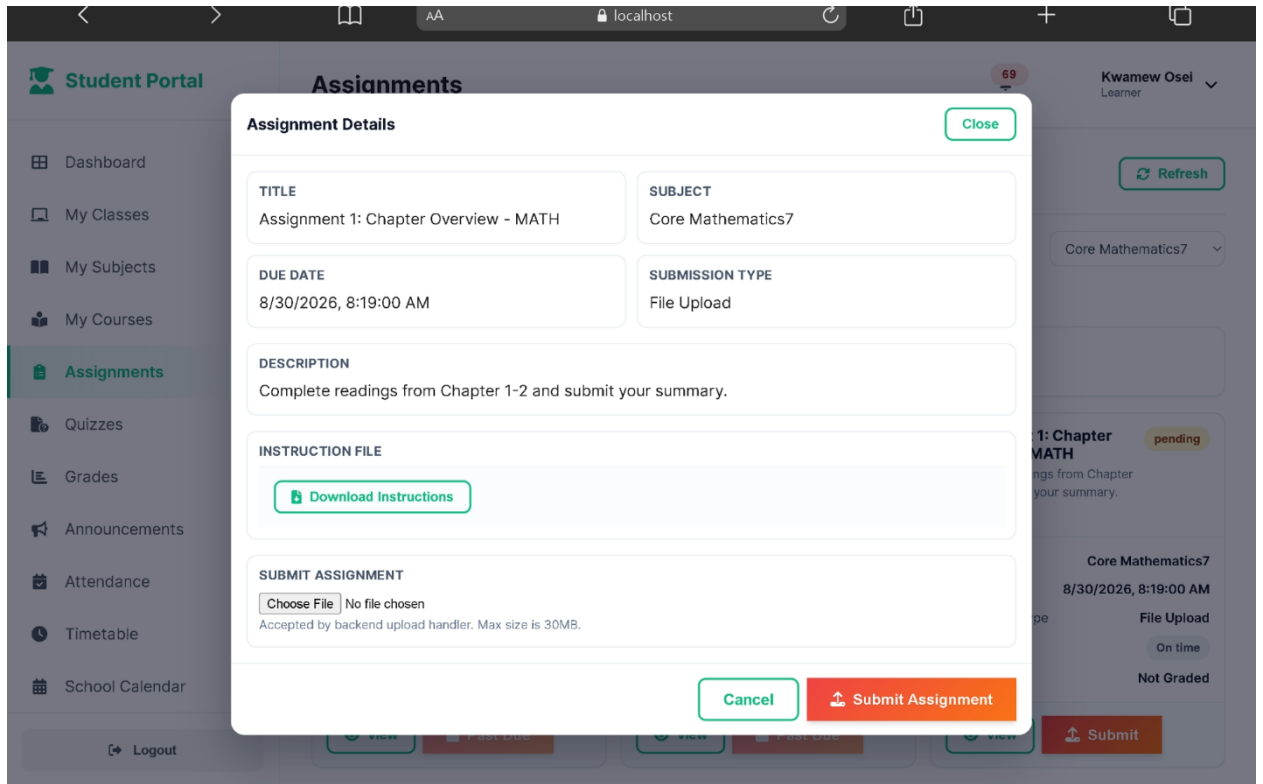
8/30/2026, 8:19:00 AM

Text and/or File

On time

Not Graded

Submit



The screenshot displays a web browser window with a 'Student Portal' interface. The browser's address bar shows 'localhost'. The portal has a dark sidebar on the left with a menu including 'Dashboard', 'My Classes', 'My Subjects', 'My Courses', 'Assignments' (highlighted), 'Quizzes', 'Grades', 'Announcements', 'Attendance', 'Timetable', and 'School Calendar'. At the bottom of the sidebar is a 'Logout' button. The main content area is a light gray box with a white border, containing the following sections:

- TITLE:** Assignment 2: Problem Set - INT
- SUBJECT:** Integrated Science
- DUE DATE:** 8/12/2026, 8:21:00 AM
- SUBMISSION TYPE:** File Upload
- DESCRIPTION:** Solve problems from textbook section 2.3-2.5
- INSTRUCTION FILE:** A button labeled 'Download Instructions' with a document icon.
- SUBMITTED FILE:** A button labeled 'View Submitted File' with a document icon.
- SUBMIT ASSIGNMENT:** A 'Choose File' button, the text 'No file chosen', and a note: 'Accepted by backend upload handler. Max size is 30MB.'

At the bottom right of the form are two buttons: 'Cancel' and 'Already Submitted' (which is disabled and has a lock icon). On the right side of the portal, the user's name 'Kwame Osei' and role 'Learner' are shown, along with a 'Refresh' button and a dropdown menu currently set to 'All subjects'.

- **Quiz Results** displays scores and relevant assessment outcomes after completion.

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

School Calendar

Logout

Quizzes

89

Kwamew Osei

Learner

Search quizzes by title, subject, class

All subjects

All status

Refresh

All

Available

Locked

Completed

Available Quizzes

0

Completed Quizzes

4

Average Score

62.3%

Best Score

153.8%

Mid sem

Integrated Science

Completed

Questions

2

Total Points

3

Duration

1 min

Attempts

9/10

Best Score

100.0%

Available From

-

Available To

-

Details

Start

bgnnbre

Integrated Science

Completed

Questions

1

Total Points

1

Duration

30 min

Attempts

2/6

Best Score

0.0%

Available From

-

Available To

-

Details

Start

ghjghghg

Integrated Science

Completed

Questions

2

Total Points

2

Duration

30 min

Attempts

1/1

Best Score

100.0%

Available From

5/7/2026, 12:00:00 AM

Available To

5/7/2026, 11:59:00 PM

Details

Max Attempts Reached

cqqewffewff

Take Quiz: Mid sem

00:37

1

2

Q1 of 2: What is 5 + 3?

Type: multiple choice • Points: 2

☐ A. 27

☒ B. 8

☐ C. 12

☐ D. 37

< Previous

Next >

Save Draft

Submit Quiz

Take Quiz: Mid sem

00:51

- 1
- 2

Q1 of 2: What is 5 + 3?

Type: multiple choice • Points: 2

☐ A. 27

☐ B. 8

☐ C. 12

☐ D. 37

< Previous

Next >

Save Draft

Submit Quiz

Result: Mid sem

Score

2 / 3

Percentage

66.7%

Attempt

9

Review Answers

Q1. What is 5 + 3?

Your Answer: B

Correct Answer: B

Status: Correct

Explanation: -

Q2. What is the function of Red?

Your Answer: -

Correct Answer: A

Status: Incorrect

Explanation: WHich is Bad

Close

Student Portal

Dashboard
My Classes
My Subjects
My Courses
Assignments
Quizzes
Grades
Announcements
Attendance
Timetable
Logout

Mid sem

Quiz Information

Subject	Integrated Science
Class	SHS 1 General Science A
Status	Completed
Description	Mid sem
Duration	1 min
Total Points	3
Questions	2
Attempts Allowed	10
Available From	-
Available To	-

Attempt Summary

Total Attempts	9
Submitted Attempts	9
Attempts Left	1
Best Score	100.0%
Average Score	59.3%

Attempt History

ATTEMPT	STATUS	SCORE	PERCENT	SUBMITTED	ACTION
9	submitted	2/3	66.7%	5/14/2026, 10:25:48 AM	View Result
8	submitted	0/2	0.0%	5/14/2026, 10:24:04 AM	View Result
7	submitted	1/3	33.3%	5/14/2026, 10:03:49 AM	View Result
6	submitted	2/2	100.0%	5/14/2026, 9:43:07 AM	View Result

69

Kwame Osei

Learner

All status Refresh

Best Score 153.8%

Completed

2

2

30 min

1/1

100.0%

5/7/2026, 12:00:00 AM

5/7/2026, 11:59:00 PM

Max Attempts Reached

Student Portal

Dashboard
My Classes
My Subjects
My Courses
Assignments
Quizzes
Grades
Announcements
Attendance
Timetable
Logout

Quizzes

Take Quiz: Mid sem

00:05

1 2

Q2 of 2: What is the function of Red?

Type: multiple choice • Points: 1

Question image

☐ A. Danger
☐ B. White
☐ C. Gold
☐ D. Damp

< Previous

Next >

Save Draft

Submit Quiz

70

Kwame Osei

Learner

Max Attempts Reached

- Attendance Information:** allows students to view their recorded attendance where permitted.

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

Logout

Attendance

30

Kwamew Osei

Learner

Monthly Attendance Trend

Date	Attendance
Feb 26	0%
Mar 26	30%
May 26	67%

Monthly Summary

Feb 2026

Attendance 0%

Present 0

Absent 0

Late 1

Mar 2026

Attendance 33%

Present 2

Absent 2

Late 1

May 2026

Attendance 67%

Present 2

Absent 1

Late 0

Attendance History

mm/dd/yyyy

mm/dd/yyyy

All Statuses

Filter

Export PDF

Date	Subject	Status	Remarks
------	---------	--------	---------

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

Logout

Attendance

30

Kwamew Osei

Learner

My Attendance

Refresh

All Subjects

All Time

36%

Attendance Rate

4

Present Days

3

Absent Days

2

Late / Tardy

2

Excused

Attendance Calendar

August 2026

Su	Mo	Tu	We	Th	Fr	Sa
		4	5	6	7	8
2	3	11	12	13	14	15
9	10		16	17	18	19
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Attendance by Subject

Subject	Total	Present	Absent	Rate
Core Mathematics3	1	1	0	100%
Core Mathematics7	6	1	1	17%
Integrated Science	4	2	2	50%

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

Logout

Attendance

30

Kwamew Osei
Learner

All Statuses

Filter

Export PDF

Date	Subject	Status	Remarks
12 May 2026	Integrated Science	absent	—
07 May 2026	Integrated Science	present	—
07 May 2026	Core Mathematics3	present	—
25 Mar 2026	Integrated Science	absent	—
16 Mar 2026	Core Mathematics7	present	—
15 Mar 2026	Core Mathematics7	absent	—
14 Mar 2026	Core Mathematics7	late	—
13 Mar 2026	Core Mathematics7	excused	—
01 Mar 2026	Integrated Science	present	—
14 Feb 2026	Core Mathematics7	excused	—
09 Feb 2026	Core Mathematics7	late	—

- **Academic Results:** displays grades and other available performance information.

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

Logout

Grades

70

Kwame Osei

Learner

12 Subject Grades

Current term subject breakdown with school grade mapping.

1 subject

SUBJECT	TOTAL SCORE	PERCENTAGE	GRADE	REMARK
Integrated Science	690.00	100%	A1	Excellent

Grade Scale

School boundary reference used for grade conversion.

80 - 100

Excellent

A1

70 - 79.99

Very Good

B2

65 - 69.99

Good

B3

60 - 64.99

Credit

C4

55 - 59.99

Credit

C5

50 - 54.99

Credit

C6

Student Portal

Dashboard

My Classes

My Subjects

My Courses

Assignments

Quizzes

Grades

Announcements

Attendance

Timetable

Logout

Grades

70

Kwame Osei

Learner

My Grades

2025-2027 1 • First Term

View subject results, semester summaries, grade trends, class ranking, and downloadable report cards for the active term or any past academic year.

Academic Year

Semester

2025-2027 1

First Term

Refresh

Download Report Card

Print

Grade data loaded successfully.

SUBJECT AVERAGE

690%

Based on current term result

OVERALL GRADE

A1

Grade point 1.00

CLASS RANK

#1

Class ranking is enabled

SUBJECTS GRADED

1

Subjects in the current report

ATTENDANCE

54.55%

Attendance computed from report data

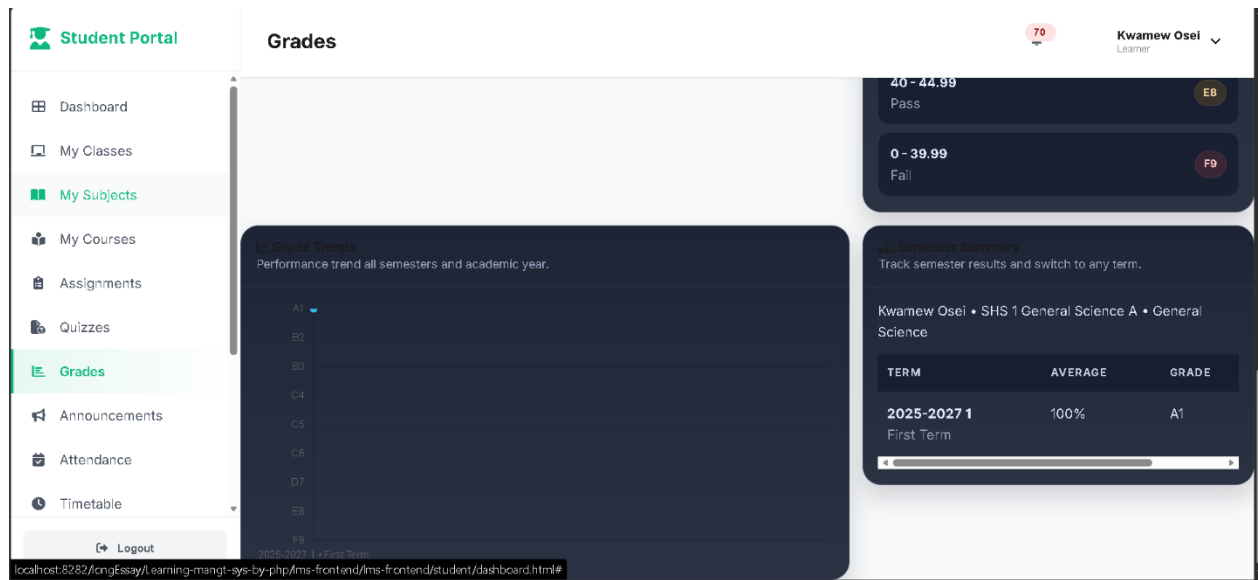
12 Subject Grades

Current term subject breakdown with school grade mapping.

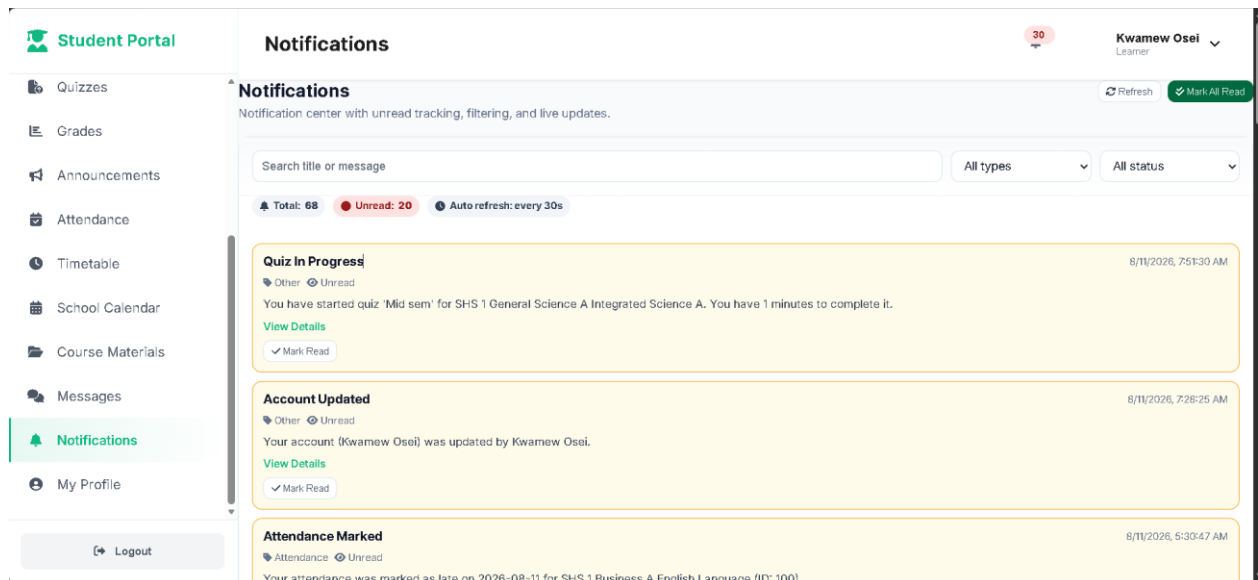
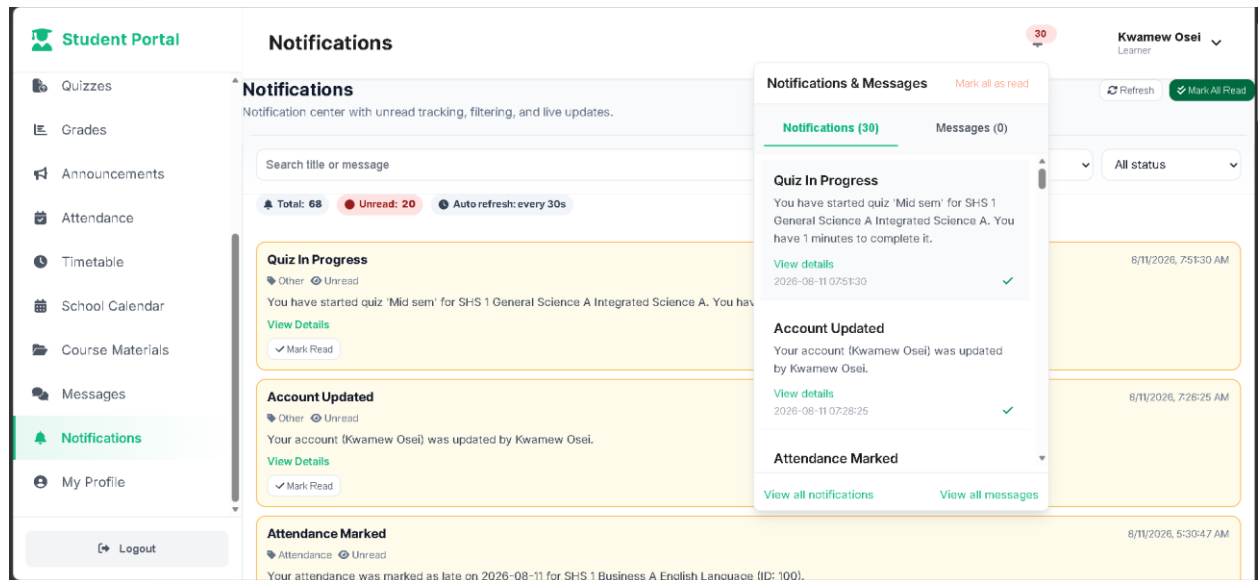
1 subject

Grade Scale

School boundary reference used for grade conversion.



- **Notifications:** displays announcements, academic reminders, and other relevant system notifications.

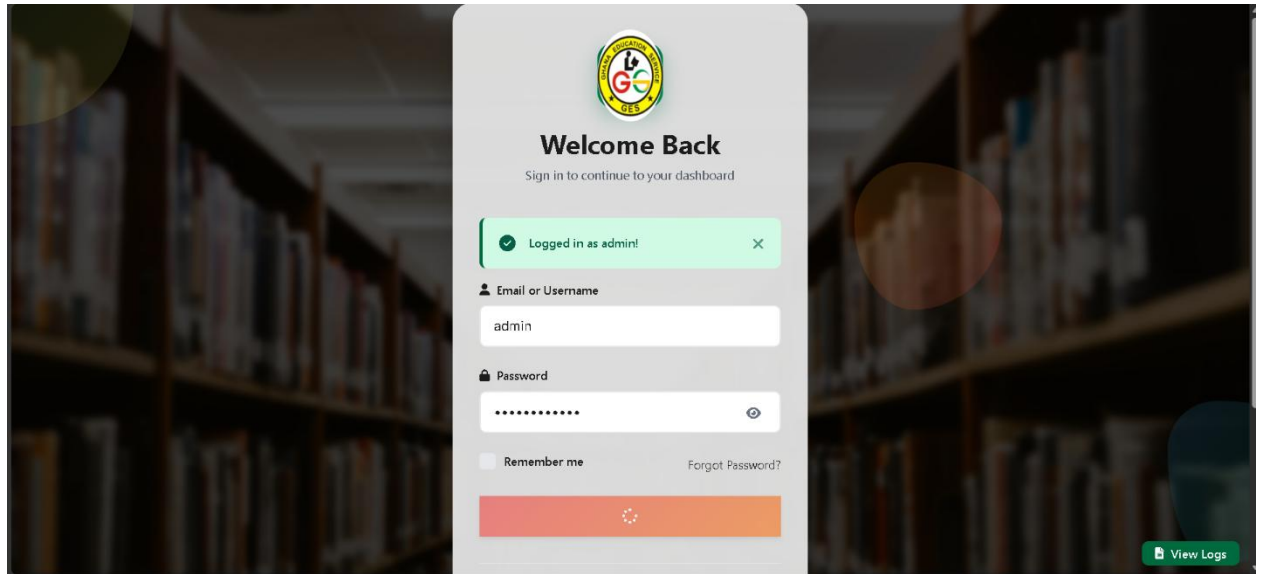


4.8.4 System Messages and API Responses

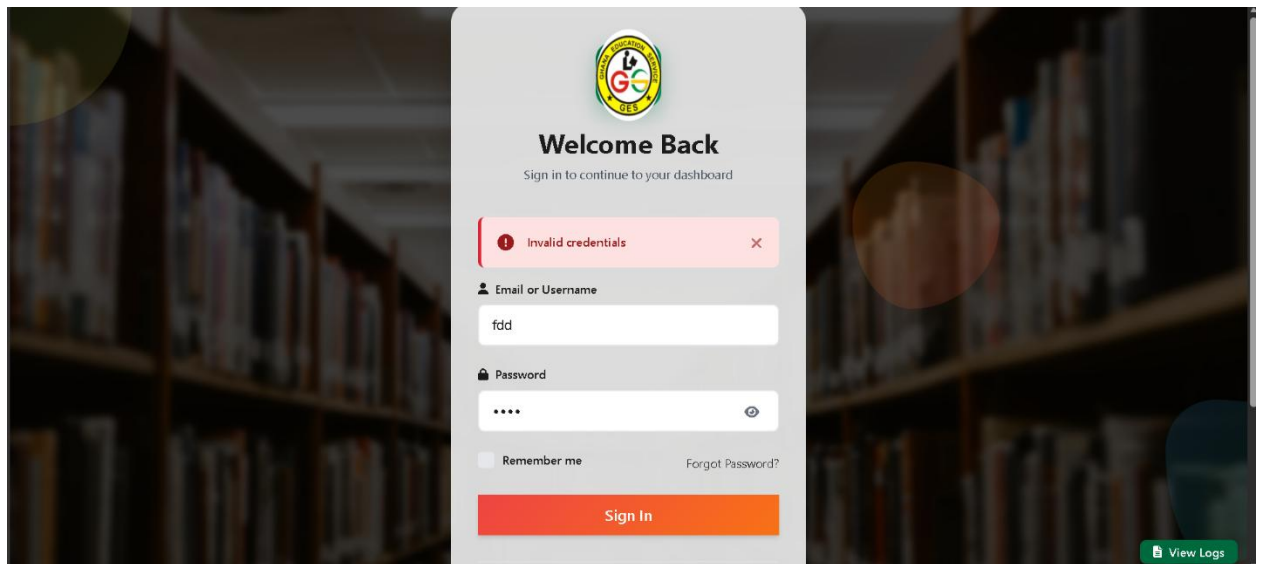
In addition to dashboards and reports, the system generates immediate responses following user actions. These responses inform users whether an operation was successful or whether an error occurred.

Examples include:

- Successful login messages.



- Invalid login notifications.



- Successful record creation messages.

LMS Admin

Overview

Students

Promote Students

Teachers

Users

Parents

Subjects

My Institution

Classes

Programs

Logout

Users

Search...

User created successfully

User Management

Manage platform users, roles, and account state across your institution

Import CSV

Export CSV

Export PDF

Add User

114 Total Users

113 Active Users

4 Admins

1 Inactive Users

Search by name, email, or username...

All Roles

All Statuses

114 of 114 users

USER	ROLE	USERNAME	DATE JOINED	STATUS	ACTIONS
AG Afuaamonu Gyamfua afuaamonu.gyamfua@gmail.com	Student	afuaamonu.gyamfua	Aug 11, 2026	ACTIVE	
UE University Education, Winneba odioriscilla22@gmail.com	Teacher	university.education	May 6, 2026	ACTIVE	
JS jdsjhjsdhsjdhs sdsdsdsds testapowithn3thunt3r@gmail.com	Teacher	jdsjhjsdhsjdhsjdhs.ds.d174	Mar 15, 2026	ACTIVE	

- Validation error messages.

LMS Admin

Overview

Students

Promote Students

Teachers

Users

Parents

Subjects

My Institution

Classes

Programs

Logout

Users

Search...

User created successfully

User Management

Manage platform

Import CSV

Export CSV

Export PDF

Add User

114 Total Users

113 Active Users

4 Admins

1 Inactive Users

Search by name, email, or username...

All Roles

All Statuses

113 of 113 users

USER	ROLE	USERNAME	DATE JOINED	STATUS	ACTIONS
UE University Education, Winneba odioriscilla22@gmail.com	Teacher	university.education	May 6, 2026	ACTIVE	
JS jdsjhjsdhsjdhs sdsdsdsds testapowithn3thunt3r@gmail.com	Teacher	jdsjhjsdhsjdhsjdhs.ds.d174	Mar 15, 2026	ACTIVE	
GF gvjht gthvngs				ACTIVE	

Create User

First Name *

Last Name *

First name

Last name

Email *

Username *

user@example.com

username

ACCOUNT SETTINGS

Role

Status

Select role

Active

Password

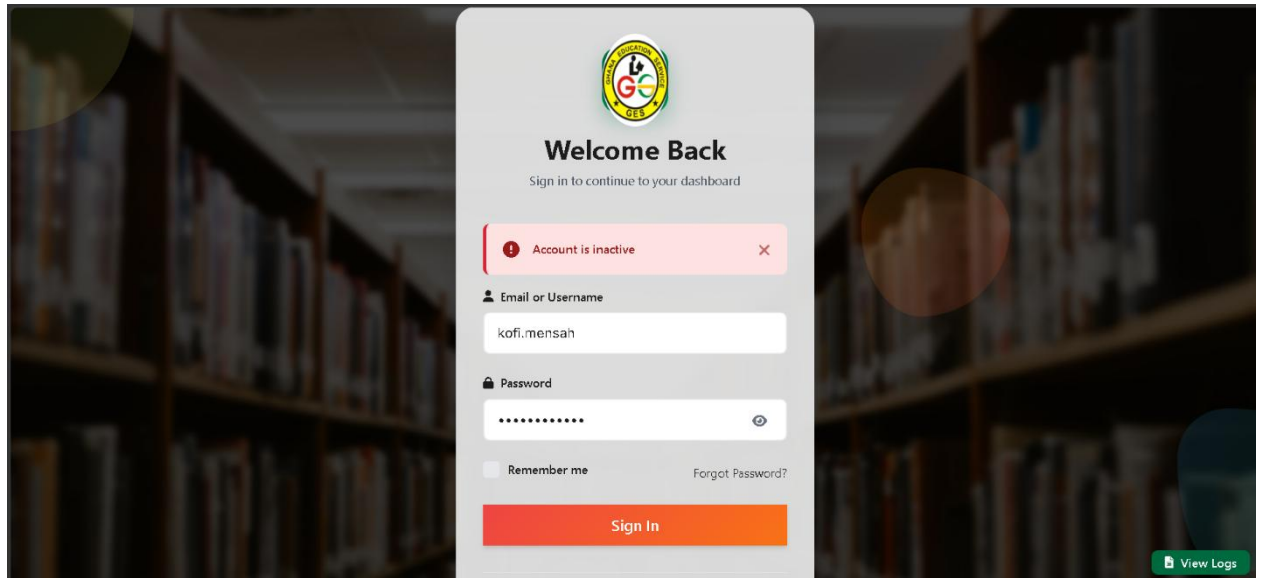
Required for new users; leave blank to keep current when editing

Username and email are required.

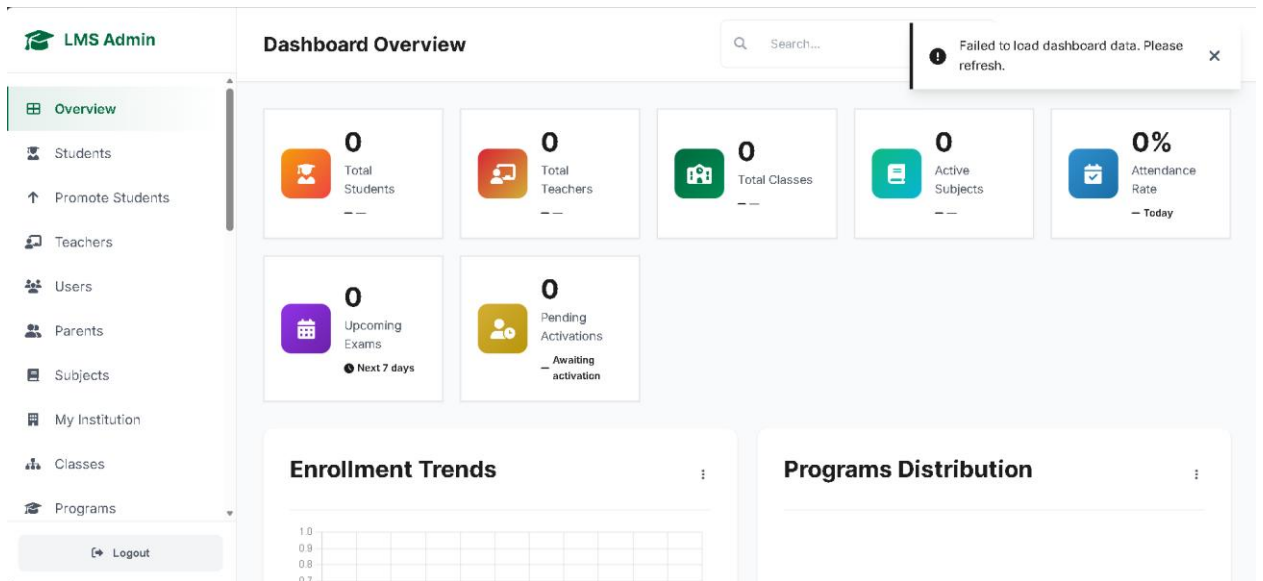
Cancel

Save User

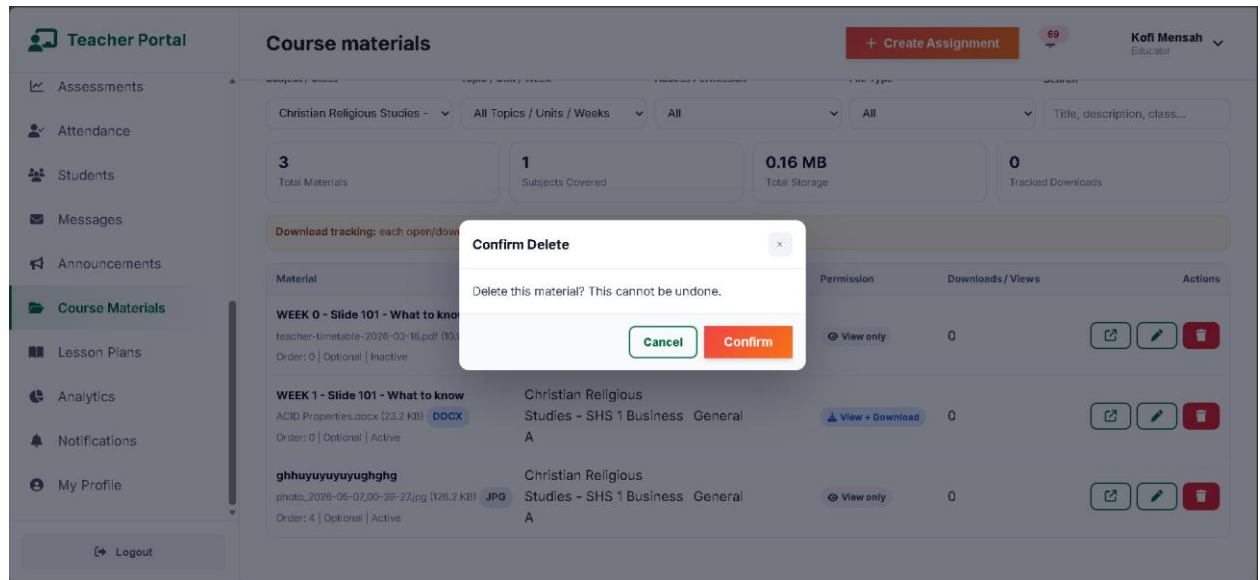
- Successful assignment submission messages.



- Database or server error messages.



- Successful update and deletion confirmations.



The backend returns these responses to the frontend through the RESTful API using **JSON**. The frontend interprets the response and presents an appropriate message or updated information to the user.

4.8.5 Report Generation

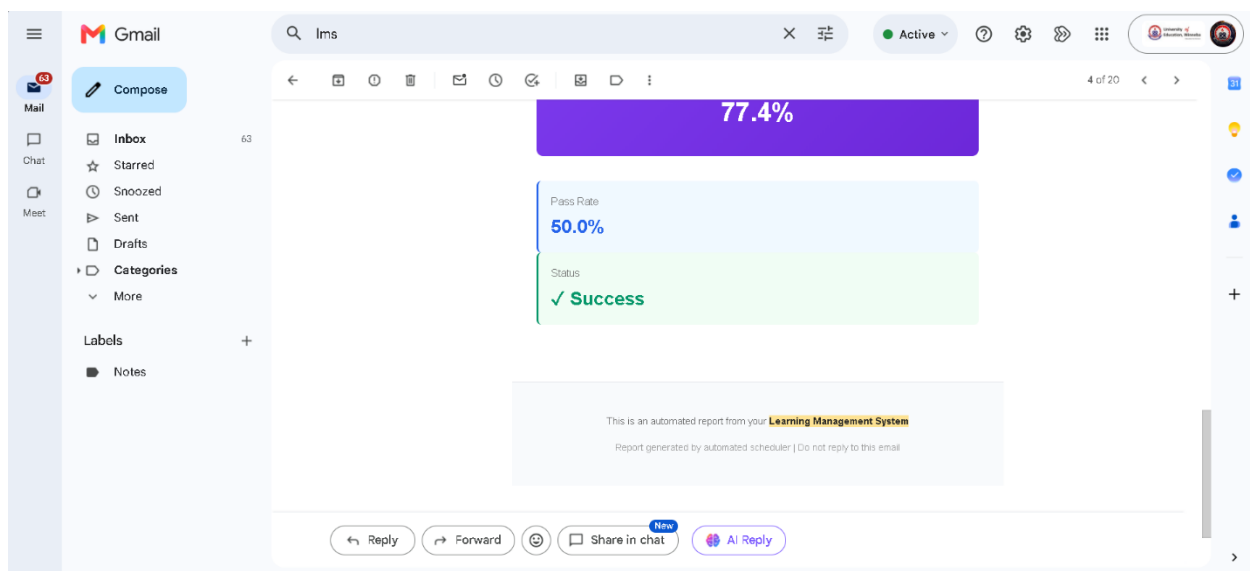
The LMS provides report outputs based on information stored in the centralised database. Reports may be generated according to relevant criteria such as student, course, department, academic year, or date range.

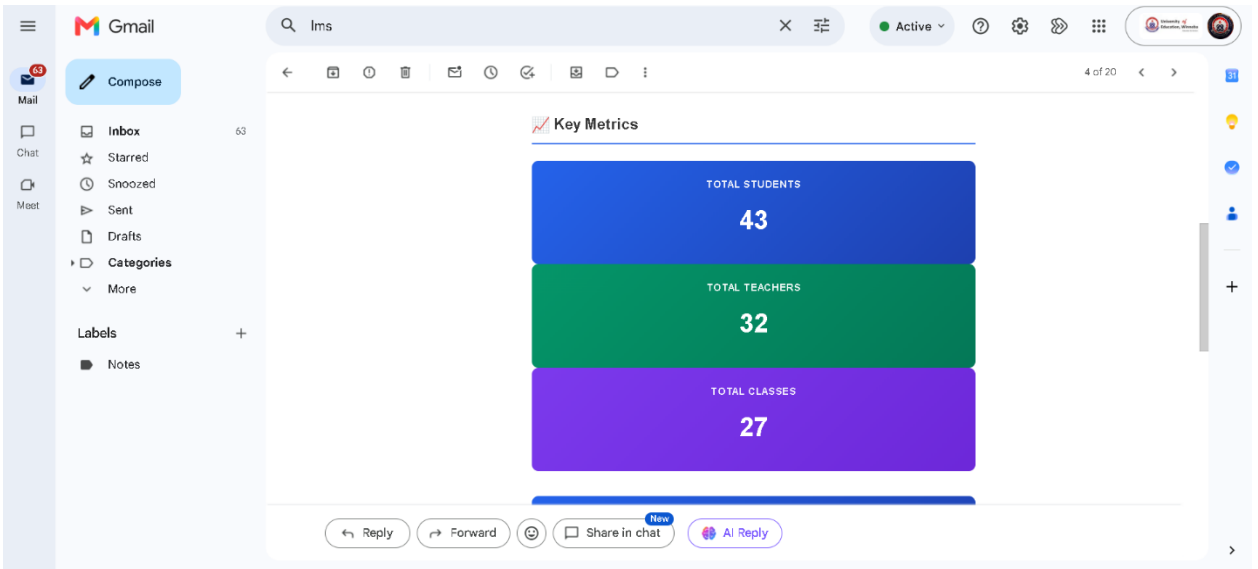
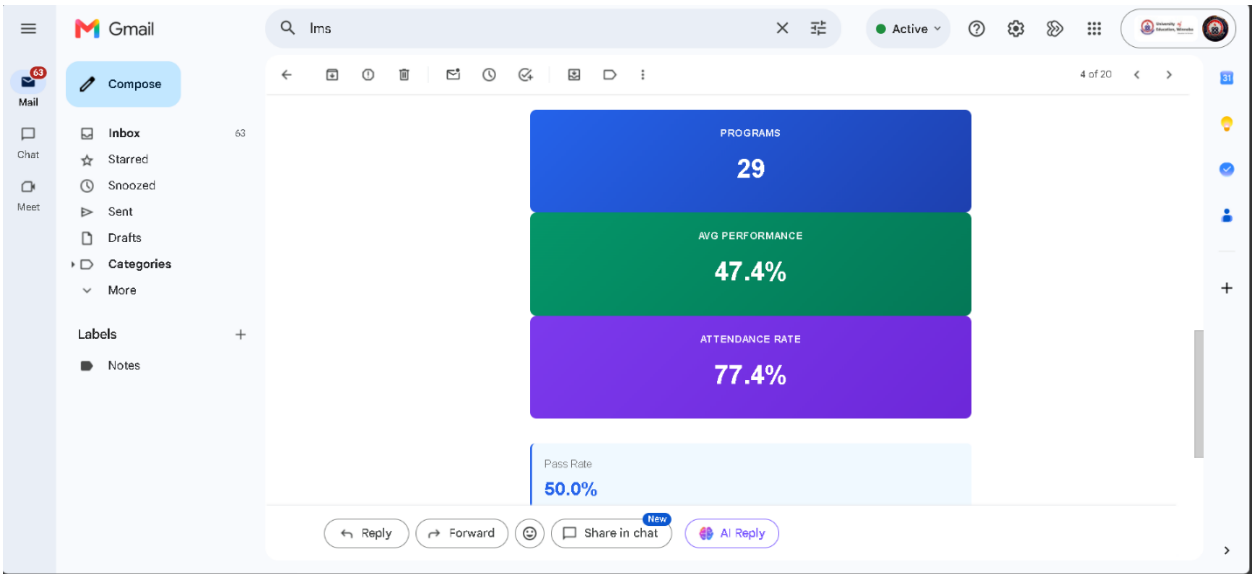
Major report categories include:

Report	Main User	Purpose
Student Performance Report	Administrator, Teacher	Shows student academic performance.
Attendance Report	Administrator, Teacher	Summarises student attendance.

Course Report	Administrator, Teacher	Provides course and enrolment information.
Assignment Report	Teacher, Administrator	Shows assignment submission and grading information.
Assessment Report	Teacher, Administrator	Presents assessment results and performance.
User Report	Administrator	Provides information about system users.
Activity Report	Administrator	Provides information about recorded system activities.

Figure 4.3 Activity Reports on Automated Report Schedules





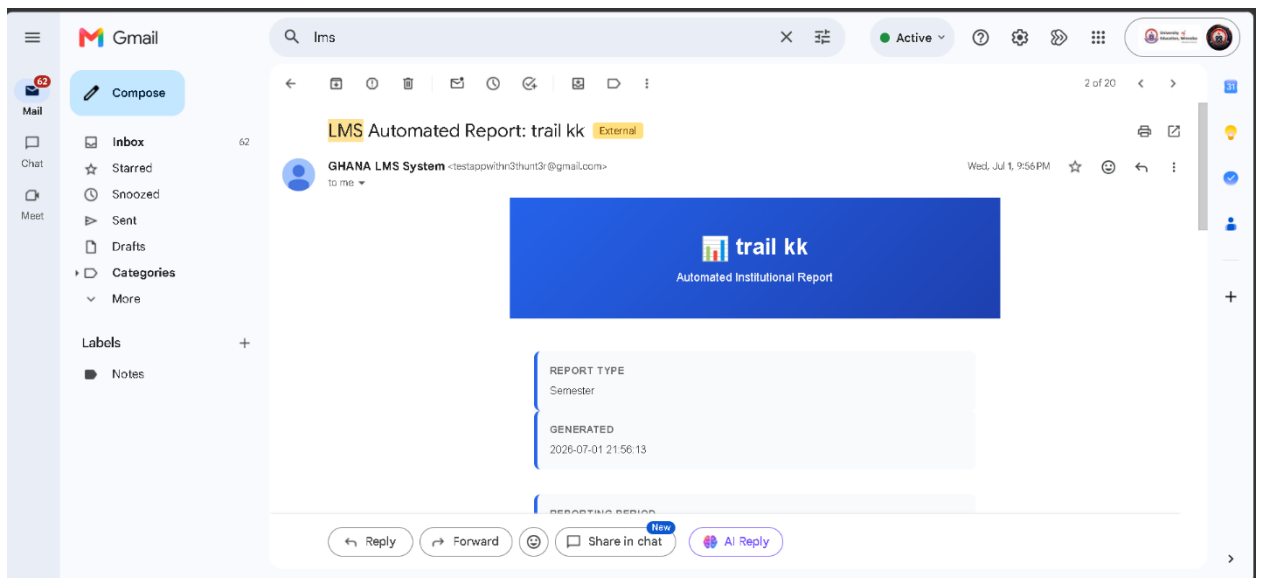
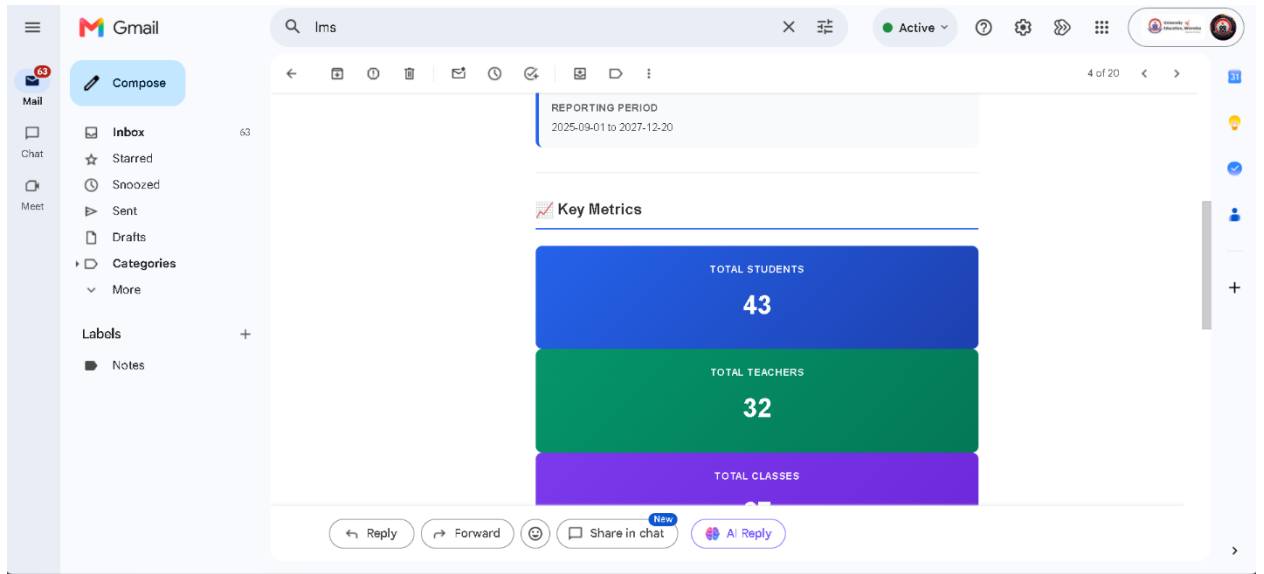


Figure 4.4 Activity Reports on Email sent for Password Reset Request

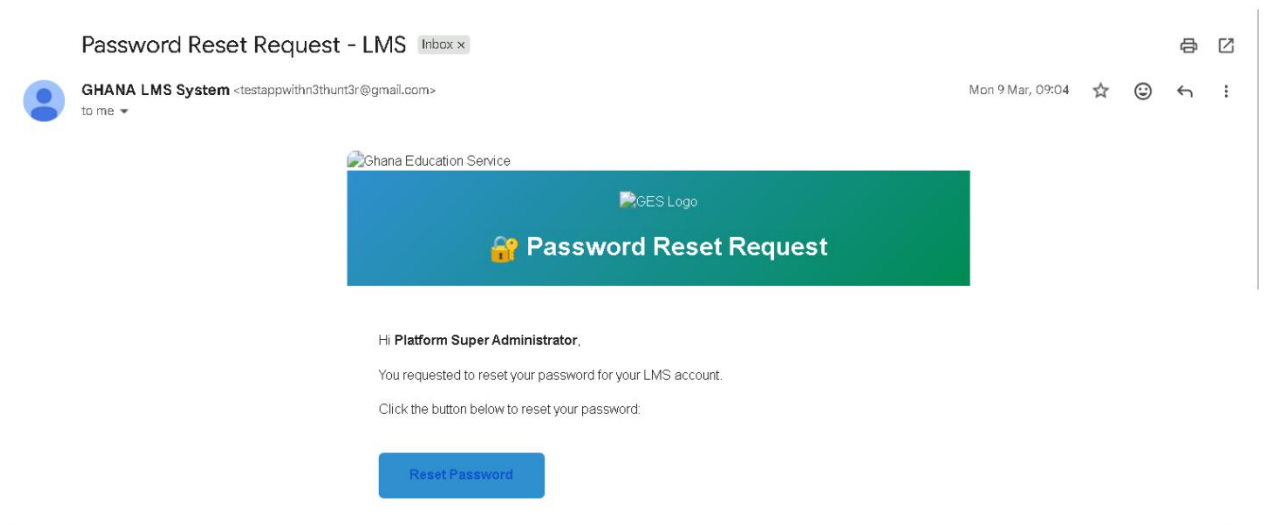
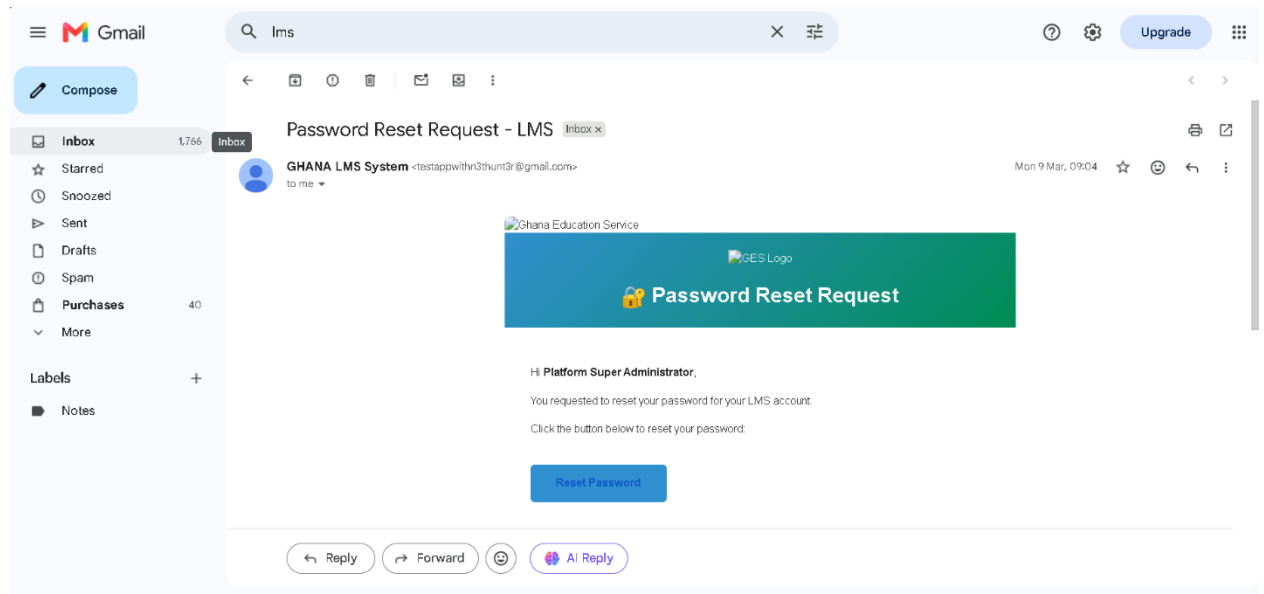


Figure 4.5 Present Report page from Administrator

LMS Admin

Semesters

Quizzes

Grade Review

Grade Scales

Reports

Messages

Announcements

Events

Notifications

System Settings

Logout

Reports

Automated Report Schedules

Configure recurring report jobs and delivery recipients.

SCHEDULE NAME

REPORT TYPE

FREQUENCY

RECIPIENT EMAIL

Monthly Admin Pack

Semester Reports

Weekly

admin@school.edu.gh

+ Add Schedule

trail kk

Type: Semester | Frequency: weekly

Recipient: 5231570098@st.uew.edu.gh

Status: Active

Pause

Delete

Weekly Class Reports

Type: Class | Frequency: weekly

Recipient: benedictamankwa9@gmail.com

Status: Active

Pause

Delete

Monthly Program Summary

Type: Program | Frequency: monthly

Recipient: benedictamankwa9@gmail.com

Status: Paused

Resume

Delete

Weekly Class Reports

Semester Attendance Report

Type: Attendance | Frequency: term_end

Recipient: benedictamankwa9@gmail.com

Status: Active

Pause

Delete

Monthly Program Summary

Figure 4.6 Present My Profile page showing User Activity

The image displays two versions of the 'Myprofile' page in an LMS Admin interface. Both versions share a common left sidebar with navigation links: Overview, Students, Promote Students, Teachers, Users, Parents, Subjects, My Institution, Classes, and Programs. The top right of both pages shows a search bar and the user's name 'Justice Mensah' with a dropdown arrow.

Top Screenshot: The profile card for 'Prof Justice Mensah' (Admin) is shown. It includes a profile picture, email 'admin@accrashes.edu.gh', and statistics: 43 STUDENTS, 32 TEACHERS, and 27 CLASSES. Below these are links for Member since 4 Dec 2025, Last login: —, 2FA Enabled, Personal Info, Account Security, Preferences, and Activity Log. To the right of the profile card is a list of recent activities:

- Addedstudent: University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA student, 6 May
- Updatedstudent: University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA student, 3 May
- Updatedstudent: jbknbuhbyu buybuhbuh student, 3 May
- Updatedstudent: jdsjhsdhsjds sdsdsdsds student, 3 May
- Addedteacher: Jane Smith teacher, 17 Apr
- Teacher activated

Bottom Screenshot: The profile card is identical to the top one. However, the right side features a 'Recent Activity' section with a date range selector (mm/dd/yyyy to mm/dd/yyyy) and a list of activities:

- Addedstudent: University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA student, 6 May
- Updatedstudent: University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA student, 3 May

Figure 4.7 Present User Report from Administrator

 Users Report Exported: August 11, 2026 Total: 113 users							
#	NAME	EMAIL	USERNAME	ROLE	STATUS	PHONE	DATE JOINED
1	University Education, Winneba	odiaprisilla22@gmail.com	university.education	Teacher	active	0594500785	May 6, 2026
2	jdsjhjsdhsjdhs sdsdsdsds	testappwithn3thunt3r@gmail.com	jdsjhjsdhsjdhsjdhs.dsd174	Teacher	active	3232233	Mar 15, 2026
3	gvjhbjbkyiy fcgfgcggj	ghvgvgv@fchgjhvv	loknb	Teacher	active	—	Mar 14, 2026
4	b hg hg hg h hg jg jh uvhvb	gvvgvhgj@vgvh.bubu	guybybbjh	Parent	active	—	Mar 14, 2026
5	jkbkbuhbyu buybuhbuh	vybuybubuybu@vgvhg.buvuyu	buybbbbi	Student	active	—	Mar 14, 2026
6	ghgfytygyy gvttyuihyhiuh	gvgygohh@vviuygiuy.vttvuyt	vgvtyiyuyyib	Admin	active	—	Mar 14, 2026
7	sdsdsds dsdsds	sdsdsd@ewewe.ewe	dssfefwe	Student	active	—	Mar 14, 2026
8	dsdsdsdsd ddfdwfwwf	listopincode@gmail.com	uewoff	Admin	active	—	Mar 14, 2026
9	asasacasda fgsfwefgdsggf	sfsdfsdf@efesf.fdsfsdf	ssdsdfsdfs	Parent	active	—	Mar 14, 2026
10	sdsdsd sdsdsdsd	dsdsds@ewew.wewe	sdsdsd.sdsdsdsd285	Teacher	active	—	Mar 14, 2026
11	Domeewe wewewe	ewewewewe@wewewe.wewe	domeewe.wewewe	Teacher	active	—	Mar 14, 2026
12	Fristt MAndd	wekjwe@wewe.ewew	donne.ewe	Parent	active	ewewe	Mar 14, 2026
13	Benedict Amankwa	hvjvj@gchgchgc.fchgch	mvjhvhvjhvjh	Parent	active	—	Mar 14, 2026
14	Benedict Amankwa	jhhvgcgjhfyfdy@hgchc.uyfu	ghgchgch	Parent	active	0909090988	Mar 14, 2026
15	JOHN bEN	JOHNBEN10@gmail.com	john.ben663	Teacher	active	0244110018	Mar 14, 2026
16	Ama Mensah	ama@example.com	amensah	Admin	active	—	Mar 14, 2026
17	ssdsd Doe	jane.doe@school.com	jane.doe	Parent	inactive	—	Mar 13, 2026

 Users Report Exported: August 11, 2026 Total: 113 users Print / Save PDF							
#	NAME	EMAIL	USERNAME	ROLE	STATUS	PHONE	DATE JOINED
1	University Education, Winneba	odiaprisilla22@gmail.com	university.education	Teacher	active	0594500785	May 6, 2026
2	jdsjhjsdhsjdhs sdsdsdsds	testappwithn3thunt3r@gmail.com	jdsjhjsdhsjdhsjdhs.dsd174	Teacher	active	3232233	Mar 15, 2026
3	gvjhbjbkyiy fcgfgcggj	ghvgvgv@fchgjhvv	loknb	Teacher	active	—	Mar 14, 2026
4	b hg hg hg h hg jg jh uvhvb	gvvgvhgj@vgvh.bubu	guybybbjh	Parent	active	—	Mar 14, 2026
5	jkbkbuhbyu buybuhbuh	vybuybubuybu@vgvhg.buvuyu	buybbbbi	Student	active	—	Mar 14, 2026
6	ghgfytygyy gvttyuihyhiuh	gvgygohh@vviuygiuy.vttvuyt	vgvtyiyuyyib	Admin	active	—	Mar 14, 2026
7	sdsdsds dsdsds	sdsdsd@ewewe.ewe	dssfefwe	Student	active	—	Mar 14, 2026
8	dsdsdsdsd ddfdwfwwf	listopincode@gmail.com	uewoff	Admin	active	—	Mar 14, 2026
9	asasacasda fgsfwefgdsggf	sfsdfsdf@efesf.fdsfsdf	ssdsdfsdfs	Parent	active	—	Mar 14, 2026
10	sdsdsd sdsdsdsd	dsdsds@ewew.wewe	sdsdsd.sdsdsdsd285	Teacher	active	—	Mar 14, 2026
11	Domeewe wewewe	ewewewewe@wewewe.wewe	domeewe.wewewe	Teacher	active	—	Mar 14, 2026
12	Fristt MAndd	wekjwe@wewe.ewew	donne.ewe	Parent	active	ewewe	Mar 14, 2026
13	Benedict Amankwa	hvjvj@gchgchgc.fchgch	mvjhvhvjhvjh	Parent	active	—	Mar 14, 2026
14	Benedict Amankwa	jhhvgcgjhfyfdy@hgchc.uyfu	ghgchgch	Parent	active	0909090988	Mar 14, 2026
15	JOHN bEN	JOHNBEN10@gmail.com	john.ben663	Teacher	active	0244110018	Mar 14, 2026
16	Ama Mensah	ama@example.com	amensah	Admin	active	—	Mar 14, 2026
17	ssdsd Doe	jane.doe@school.com	jane.doe	Parent	inactive	—	Mar 13, 2026
18	Jane Smith	jane.smith@school.com15	jane.sm1th15	Teacher	active	1234567890	Mar 10, 2026
19	Jane Smith	jane.smith@school.com16	jane.sm1th16	Teacher	active	1234567890	Mar 10, 2026
20	Jane Smith	jane.smith@school.com18	jane.sm1th18	Teacher	active	—	Mar 10, 2026

LMS Admin

Overview

Students

Promote Students

Teachers

Users

Parents

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My Institution

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Users

Search...

Justice Mensah
Administrator

User Management

Manage platform users, roles, and account state across your institution

Import CSV

Export CSV

Export PDF

Add User

113

Total Users

112

Active Users

4

Admins

1

Inactive Users

Search by name, email, or username...

All Roles

All Statuses

113 of 113 users

USER	ROLE	USERNAME	DATE JOINED	STATUS	ACTIONS
UE University Education, Winneba pdiaorisolln22@gmail.com	Teacher	university.education	May 6, 2026	ACTIVE	
JS jdsjhjsdhjsdhs sdsdsdsds testapowthn3thunk3r@gmail.com	Teacher	jdsjhjsdhjsdhsjdh.dsdl74	Mar 15, 2026	ACTIVE	
GF gvjhbjhbkyy fcgrfgcggi ghvgvqv@fchgjhvv	Teacher	1okhb	Mar 14, 2026	ACTIVE	

Admin Reports

Academic Year: 2025-2027 1 | Semester: First Term

Type	Name	Group	Average %	Pass Rate	Count	Notes
Enrollment	Class	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Business A	Class	0.00%	0.00%	4	Current enrollment count by class.
Enrollment	SHS 1 General Arts A	Class	0.00%	0.00%	4	Current enrollment count by class.
Enrollment	SHS 1 General Arts B	Class	0.00%	0.00%	3	Current enrollment count by class.
Enrollment	SHS 1 Science 1	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science W	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science E	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science F	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science G	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science H	Class	0.00%	0.00%	1	Current enrollment count by class.
Enrollment	SHS 1 Science 2	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science 5	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science 8	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 General Science A	Class	0.00%	0.00%	6	Current enrollment count by class.
Enrollment	SHS 1 General Science B	Class	0.00%	0.00%	3	Current enrollment count by class.
Enrollment	SHS 1 Science D	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science E	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science F	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science G	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science H	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science S	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science W	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Visual Art 1	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Visual Art 1111	Class	0.00%	0.00%	0	Current enrollment count by class.

Figure 4.8 Present Employment Report from Administrator

LMS Admin

School Calendar

Assessments

Academic Years

Grade Levels

Semesters

Quizzes

Grade Review

Grade Scales

Reports

Messages

Logout

Reports

Search...

Justice Mensah Administrator

Report Output

Includes semester, annual, WASSCE, student progress, teacher, attendance and enrollment views.

TYPE	NAME	GROUP	AVERAGE %	PASS RATE	COUNT	NOTES
Enrollment	Class	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Business A	Class	0.00%	0.00%	4	Current enrollment count by class.
Enrollment	SHS 1 General Arts A	Class	0.00%	0.00%	4	Current enrollment count by class.
Enrollment	SHS 1 General Arts B	Class	0.00%	0.00%	3	Current enrollment count by class.
Enrollment	SHS 1 Science 1	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science W	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science E	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science F	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science G	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science H	Class	0.00%	0.00%	1	Current enrollment count by class.
Enrollment	SHS 1 Science 2	Class	0.00%	0.00%	0	Current enrollment count by class.
Enrollment	SHS 1 Science 5	Class	0.00%	0.00%	0	Current enrollment count by class.

LMS Admin

School Calendar

Assessments

Academic Years

Grade Levels

Semesters

Quizzes

Grade Review

Grade Scales

Reports

Messages

Logout

Reports

Search...

Justice Mensah Administrator

Admin Reports Control Center

Generate school-wide intelligence: performance by class/program, attendance, enrollment, WASSCE readiness, and export-ready leadership reports.

Report Filters & Exports

Choose a report family, then build or export your current analytics view.

ACADEMIC YEAR

SEMESTER

REPORT FAMILY

SEARCH

EXPORT FORMAT

2025-2027 1

First Term

Enrollment Reports

Class, program, student, tea

PDF

Load Data

Export Current

Export All Reports

Admin reports loaded successfully.

43

Total Students

32

Total Teachers

27

Total Classes

29

Total Programs

50.00%

Pass Rate

76.20%

Attendance Rate

Class Performance

Average outcomes by class for current term.

Program Performance

Comparative distribution and trend by program.

Figure 4.9 Present Assessment Page from Administrator

LMS Admin

Classes

Programs

Attendance

Timetable

Course Materials

School Calendar

Assessments

Academic Years

Grade Levels

Semesters

Logout

Assessments

Search...

Justice Mensah
Administrator

Quiz

15

Short quizzes and tests

Save Weights

Reset Defaults

Total: 100%

Assessment Schedule Overview

Institution-wide read-only schedule pulled from teacher-created assessments.

Refresh

ASSESSMENT	CLASS / COURSE	TYPE	DUE DATE	WEIGHT	STATUS
No assessments found yet.					

Assessment Calendar (Read-only)

Chronological list for tracking deadlines and exam load.

No calendar entries available.

LMS Admin

Classes

Programs

Attendance

Timetable

Course Materials

School Calendar

Assessments

Academic Years

Grade Levels

Semesters

Logout

Assessments

Search...

Justice Mensah
Administrator

Admin Scope: This page is configuration and monitoring only. Admins cannot create or assign assessments; only teachers can.

Total Assessments

0

Upcoming (7 days)

0

Published

0

Classes With Assessments

0

Assessment Categories & Weights

School-wide configuration used for reporting and policy checks.

New category name (e.g., Mid-Term Exam)

Weight %

Optional category description

+ Add Category

Assignment

20

Homework and take-home assignments

Upcoming Assessments by Class

Read-only view of what teachers already scheduled.

No upcoming assessments in the next 14 days.

Figure 4.10 Present Assessment Report from Teacher

Teacher Assessment Scores

SHS 1 General Science A - Integrated Science

student_name	student_id_number	Assignment (20.00%) score	Assignment (20.00%) max_score	Class Work (10.00%) score	Class Work (10.00%) max_score	Final (30.00%) score	Final (30.00%) max_score	Midterm (25.00%) score	Midterm (25.00%) max_score	Quiz (15.00%) score	Quiz (15.00%) max_score
Akosua Atta	ASHS-2024-0018	78	100	60	100	50	100	60	100	36	100
Kofi Addo	ASHS-2024-0003	59	100	65	100	30	100	12	100	10	100
Kwame Frimpong	ASHS-2024-0013	100	100	10	100	10	100	10	100	10	100
Kwame Osei	ASHS-2024-0001	100	100	100	100	100	100	100	100	100	100
Kwesi Boateng	ASHS-2024-0009	10	100	10	100	10	100	10	100	10	100
University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA	ASHS-2026-0001	34	100	34	100	92	100	23	100	93	100

Figure 4.11 Present Assessment Page from Teacher

Teacher Portal

Assessments

+ Create Assignment

	100	100	100	100	100
Akosua Atta (ID: ASHS-2024-0018)	78	60	50	60	36
Kofi Addo (ID: ASHS-2024-0003)	59	65	30	12	10
Kwame Frimpong (ID: ASHS-2024-0013)	100	10	10	10	10
Kwame Osei (ID: ASHS-2024-0001)	100	100	100	100	100
Kwesi Boateng (ID: ASHS-2024-0009)	10	10	10	10	10
University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA (ID: ASHS-2026-0001)	34	34	92	23	93

Save All (Draft)

Publish All

Reset Changes

CSV Template

Import CSV

Export CSV

Export PDF

Teacher Portal

My Courses

Timetable

School Calendar

Assignments

Quizzes

Grading

Reports

Assessments

Attendance

Students

Logout

Assessments

Create Assignment

Student Assessment Scores

SHS 1 General Science A - Integrated Science (6 students)

Search student name...

STUDENT NAME	Assignment (20.00%)	Class Work (10.00%)	Final (30.00%)	Midterm (25.00%)	Quiz (15.00%)
	100	100	100	100	100
Akosua Atta (ID: ASHS-2024-0018)	78	60	50	60	36
Kofi Addo (ID: ASHS-2024-0003)	59	65	30	12	10
Kwame Frimpong (ID: ASHS-2024-0013)	100	10	10	10	10
Kwame Osei (ID: ASHS-2024-0001)	100	100	100	100	100
Kwesi Boateng (ID: ASHS-2024-0009)	10	10	10	10	10
University Education, Winneba (Winneba)	34	34	92	23	93

Teacher Portal

My Courses

Timetable

School Calendar

Assignments

Quizzes

Grading

Reports

Assessments

Attendance

Students

Logout

Assessments

Create Assignment

Assessment Score Management

Manage Category Scores

Select a class, choose categories, and save or publish student scores.

CLASS / SUBJECT

Integrated Science

Load Students & Scores

Clear Selection

ASSESSMENT CATEGORIES

Assignment (20.00%), Class Work (10.00%), Final (30.00%), Midterm (25.00%), Quiz (15.00%)

Current Semester: First Term | Academic Year: 2025-2027 1

Student Assessment Scores

SHS 1 General Science A - Integrated Science (6 students)

Search student name...

STUDENT NAME	Assignment (20.00%)	Class Work (10.00%)	Final (30.00%)	Midterm (25.00%)	Quiz (15.00%)
	100	100	100	100	100
Akosua Atta (ID: ASHS-					

Figure 4.12 Present Subject/Course Page from Administrator


Subjects Report
Exported: August 11, 2026

Total: 26 subjects

#	SUBJECT	CREDITS	DESCRIPTION	TYPE	CLASSES
1	English Language ENG	3	Core English Language	Core	5
2	Integrated Science INT-SCI	3	Integrated Science	Core	2
3	Mathematics (Core) MATH-C	3	Core Mathematics	Core	6
4	Core Mathematics1 MATH-CORE1	3	Fundamental math subject	Core	0
5	Core Mathematics2 MATH-CORE2	3	Fundamental math subject	Core	0
6	Core Mathematics3 MATH-CORE3	3	Fundamental math subject	Core	2
7	Core Mathematics4 MATH-CORE4	3	Fundamental math subject	Core	0
8	Core Mathematics5 MATH-CORE5	3	Fundamental math subject	Core	0
9	Core Mathematics6 MATH-CORE6	3	Fundamental math subject	Core	1
10	Core Mathematics7 MATH-CORE7	3	Fundamental math subject	Core	1
11	Social Studies SOC-ST	3	Social Studies	Core	0
12	Accounting	2	Accounting	Elective	0

Subjects Report

Total: 26 subjects

Exported: August 11, 2026

Print / Save PDF

#	SUBJECT	CREDITS	DESCRIPTION	TYPE	CLASSES
1	English Language ENG	3	Core English Language	Core	5
2	Integrated Science INT-SCI	3	Integrated Science	Core	2
3	Mathematics (Core) MATH-C	3	Core Mathematics	Core	6
4	Core Mathematics1 MATH-CORE1	3	Fundamental math subject	Core	0
5	Core Mathematics2 MATH-CORE2	3	Fundamental math subject	Core	0
6	Core Mathematics3 MATH-CORE3	3	Fundamental math subject	Core	2
7	Core Mathematics4 MATH-CORE4	3	Fundamental math subject	Core	0
8	Core Mathematics5 MATH-CORE5	3	Fundamental math subject	Core	0
9	Core Mathematics6 MATH-CORE6	3	Fundamental math subject	Core	1
10	Core Mathematics7 MATH-CORE7	3	Fundamental math subject	Core	1
11	Social Studies SOC-ST	3	Social Studies	Core	0
12	Accounting ACCT	3	Accounting	Elective	0
13	Biology BIO	3	Biology	Elective	3
14	Business Management BUS-MGT	3	Business Management	Elective	0

LMS Admin

Students

Promote Students

Teachers

Users

Parents

Subjects

My Institution

Classes

Programs

Attendance

Time Table

Logout

Subjects

Search...

36

Justice Mensah
Administrator

Subjects

Manage the subjects offered across all programs at your institution

Import CSVExport CSVExport PDFAdd Subject

26
Total Subjects

11
Core Subjects

9
Elective Subjects

26
Assigned to Classes

Search subjects...

All Types

SUBJECT	CREDITS	DESCRIPTION	TYPE	CLASSES	ACTIONS
English Language ENG	3	Core English Language	Core	5	
Integrated Science INT-SCI	3	Integrated Science	Core	2	
Mathematics (Core) MATH-C	3	Core Mathematics	Core	6	
Core Mathematics1 MATH-CORE1	3	Fundamental math subject	Core	0	

Figure 4.13 Present Attendance Page from Administrator

LMS Admin

Users

Parents

Subjects

My Institution

Classes

Programs

Attendance

Timetable

Course Materials

School Calendar

Logout

Attendance-details

Search...

Justice Mensah
Administrator

08/11/2026

All Status

Search by student name or ID

No unsaved changes

Mark All Present

Discard Changes

Save Changes

4
Total Records

2
Present

1
Absent

1
Late

STUDENT	ID NUMBER	STATUS	MARKED AT	REMARK
Yaw Boadu	ASHS-2024-0017	Present	8/11/2026, 5:30:06 AM	Optional remark
Afua Darko	ASHS-2024-0012	Absent	8/11/2026, 5:30:06 AM	Optional remark
Kobbina Donkor	ASHS-2026-0018	Present	8/11/2026, 5:30:06 AM	Optional remark
Yaw Owusu	ASHS-2024-0007	Late	8/11/2026, 5:30:06 AM	Optional remark

localhost:8282/congEasyLearning-managt-sys-by-php/ms-front-end/ms-front-end/admin/dashboard.html#

LMS Admin

Users

Parents

Subjects

My Institution

Classes

Programs

Attendance

Timetable

Course Materials

School Calendar

Logout

Attendance-details

Search...

Justice Mensah
Administrator

← Back to Attendance

Attendance Details

SHS 1 Business A (SHS1-BUS-A) • Business • Jane Smith • Mode: Edit • Date: 2026-08-11

Edit

FILTERS

08/11/2026

All Status

Search by student name or ID

No unsaved changes

Mark All Present

Discard Changes

Save Changes

4
Total Records

2
Present

1
Absent

1
Late

STUDENT	ID NUMBER	STATUS	MARKED AT	REMARK
Yaw Boadu	ASHS-2024-0017	Present	8/11/2026, 5:30:06 AM	Optional remark
Afua Darko	ASHS-2024-0012	Absent	8/11/2026, 5:30:06 AM	Optional remark

LMS Admin

Users

Parents

Subjects

My Institution

Classes

Programs

Attendance

Timetable

Course Materials

School Calendar

Logout

Attendance-details

Search...

Justice Mensah
Administrator

← Back to Attendance

Attendance Details

SHS 1 Business A (SHS1-BUS-A) • Business • Jane Smith • Mode: View • Date: 2026-08-11

View

FILTERS

08/11/2026

All Status

Search by student name or ID

4
Total Records

2
Present

1
Absent

1
Late

STUDENT	ID NUMBER	STATUS	MARKED AT	REMARK
Yaw Boadu	ASHS-2024-0017	Present	8/11/2026, 5:30:06 AM	—
Afua Darko	ASHS-2024-0012	Absent	8/11/2026, 5:30:06 AM	—
Kobbina Donkor	ASHS-2026-0018	Present	8/11/2026, 5:30:06 AM	—
Yaw Owusu	ASHS-2024-0007	Late	8/11/2026, 5:30:06 AM	—

LMS Admin

Users

Parents

Subjects

My Institution

Classes

Programs

Attendance

Timetable

Course Materials

School Calendar

Logout

Attendance

Search...

Justice Mensah
Administrator

CLASS	PROGRAM	SUBJECT	TEACHER	TOTAL STUDENTS	PRESENT	ABSENT	LATE	RATE	STATUS	ACTIONS
SHS 1 Business A	Business		Jane Smith	4	2	1	1	50.0%	MARKED	
SHS 1 Business A	Business		Jane Smith	4	4	0	0	100.0%	MARKED	
SHS 1 Business A	Business	University Education, Winneba (Winneba)	BENEDICT OSEI AMANKWA	4	3	0	1	75.0%	MARKED	
SHS 1 Business A	Business		Kofi Mensah	4	0	0	0	-	NOT MARKED	

LMS Admin

Users

Parents

Subjects

My Institution

Classes

Programs

Attendance

Timetable

Course Materials

School Calendar

Logout

Attendance

Search...

36

Justice Mensah
Administrator

Attendance Trends2026-08-11 to 2026-08-11

Absent Students0

Late Arrivals0

CLASS	PROGRAM	SUBJECT	TEACHER	TOTAL STUDENTS	PRESENT	ABSENT	LATE	RATE	STATUS	ACTIONS
SHS 1 Business A	Business	Jane Smith	4	2	1	1	50.0%	MARKED		
SHS 1 Business A	Business	Jane Smith	4	4	0	0	100.0%	MARKED		

LMS Admin

Users

Parents

Subjects

My Institution

Classes

Programs

Attendance

Timetable

Course Materials

School Calendar

Logout

Attendance

Search...

36

Justice Mensah
Administrator

Attendance Management

ManualAuto

Export CSV

Export PDF

Notify Parents

10.3%
9 / 87 students

2
22.2% of present students

1
1.1% absence rate

20 / 20
100.0% completion

TODAY
75.0%
3 marked class-day records

THIS WEEK
75.0%
3 marked class-day records

THIS MONTH
75.0%
3 marked class-day records

08/11/2026

08/11/2026

All Programs

All Grade Levels

All Classes

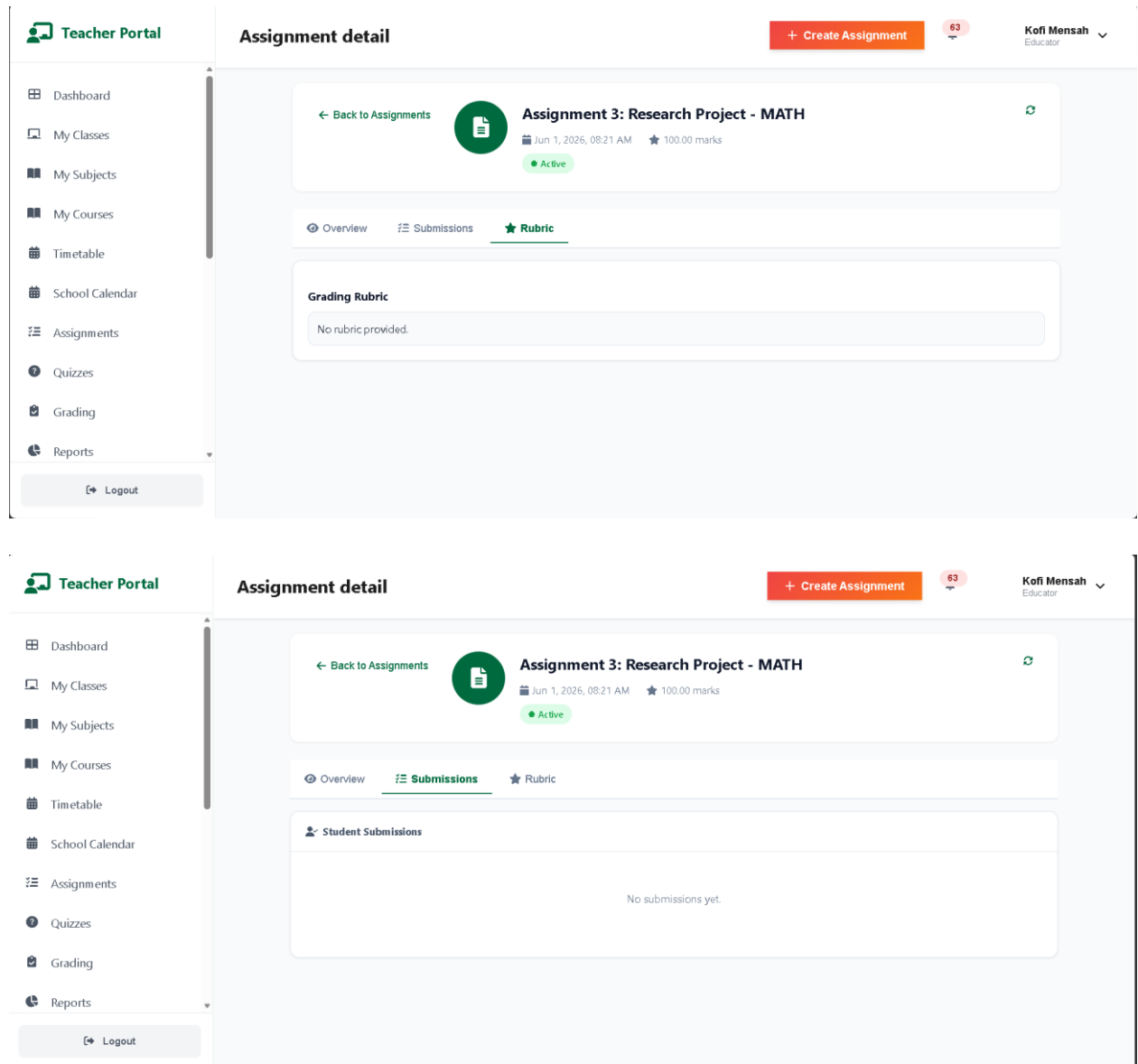
All Status

Attendance Trends2026-08-11 to 2026-08-11

Absent Students0

Late Arrivals0

Figure 4.15 Present Assignment Details Page from Teacher



Teacher Portal

Dashboard

My Classes

My Subjects

My Courses

Timetable

School Calendar

Assignments

Quizzes

Grading

Reports

Logout

Assignment detail

Create Assignment

Kofi Mensah
Educator

STATUS

Active

DUE DATE

Jun 1, 2026, 08:21 AM

TOTAL MARKS

100.00

PASSING MARK

60.00

SUBMISSION TYPE

FILE

SECTION

—

Instructions

Instructions Text

Research current topics in Core Mathematics7

Instructions File

No instruction file uploaded.

Teacher Portal

Dashboard

My Classes

My Subjects

My Courses

Timetable

School Calendar

Assignments

Quizzes

Grading

Reports

Logout

Assignment detail

Create Assignment

Kofi Mensah
Educator

Back to Assignments

Assignment 3: Research Project - MATH

Jun 1, 2026, 08:21 AM

100.00 marks

Active

Refresh

Overview

Submissions

Rubric

Assignment Details

STATUS

Active

DUE DATE

Jun 1, 2026, 08:21 AM

TOTAL MARKS

100.00

PASSING MARK

60.00

SUBMISSION TYPE

FILE

SECTION

—

Instructions

Figure 4.16 Present Assignment Edit Page from Teacher

The screenshot shows the 'Edit Assignment' modal in the Teacher Portal. The modal is titled 'Edit Assignment' and contains the following fields:

- Assignment Title ***: A text input field containing 'Assignment 3: Research Project - MATH'.
- Class/Subject ***: A dropdown menu showing 'Core Mathematics7 (SHS 1 Generi'.
- Section (Optional)**: A dropdown menu showing 'No Section (Optional)'.
- Instructions (Optional)**: A text area containing 'Research current topics in Core Mathematics7'.
- Rubric (Optional)**: A text area containing 'Assessment criteria and grading rubric'.
- Due Date ***: A date input field.
- Total Marks ***: A number input field.

At the bottom of the modal are two buttons: 'Cancel' and 'Save Assignment'.

Figure 4.17 Present Assignment Create Page from Teacher

The screenshot shows the 'Create Assignment' modal in the Teacher Portal. The modal is titled 'Create Assignment' and contains the following fields:

- Assignment Title ***: A text input field.
- Class/Subject ***: A dropdown menu showing 'Select Class/Subject'.
- Section (Optional)**: A dropdown menu showing 'Select Class/Subject first'.
- Instructions (Optional)**: A text area containing 'Type assignment instructions for students'.
- Rubric (Optional)**: A text area containing 'Assessment criteria and grading rubric'.
- Due Date ***: A date input field.
- Total Marks ***: A number input field.

At the bottom of the modal are two buttons: 'Cancel' and 'Save Assignment'.

Figure 4.18 Present Assignment Page from Teacher

Teacher Portal

Dashboard
My Classes
My Subjects
My Courses
Timetable
School Calendar
Assignments
Quizzes
Grading
Reports
Logout

Assignments

Create Assignment

63

Kofi Mensah Educator

Assignments

Create and manage assignments, deadlines, and resources.

Create Assignment

Class/Subject

Status

Search

All Classes

All Status

Search assignments...

Filter

Reset

37
Total Assignments

37
Active Assignments

0
Draft Assignments

1
Pending to Grade

Title	Class/Subject	Due Date	Total Marks	Status	Actions
Assignment 3: Research Project - MATH Core Mathematics7 (SHS 1 General Science A)	Core Mathematics7	Jun 1, 2026, 08:21 AM	100.00	Active (Overdue)	View Edit Delete
Assignment 3: Research Project - MATH Core Mathematics3 (SHS 1 General Science A)	Core Mathematics3	Jun 1, 2026, 08:21 AM	100.00	Active (Overdue)	View Edit Delete
Assignment 3: Research Project - INT Integrated Science (SHS 1 General Science A)	Integrated Science	Jun 1, 2026, 08:21 AM	100.00	Active (Overdue)	View Edit Delete

Figure 4.19 Present Assignment Submission Page from Teacher

Teacher Portal

My Classes
My Subjects
My Courses
Timetable
School Calendar
Assignments
Quizzes
Grading
Reports
Assessments
Logout

Submissions

Create Assignment

63

Kofi Mensah Educator

← Back Student Submissions

TOTAL STUDENTS

0

SUBMITTED

0/0 (0%)

GRADED

0/0 (0%)

Submissions

No submissions yet.

Figure 4.20 Present Subject/Course Page from Teacher

Teacher Portal

Dashboard

My Classes

My Subjects

My Courses

Timetable

School Calendar

Assignments

Quizzes

Grading

Reports

Logout

My subjects

Create Assignment

63

Kofi Mensah
Educator

My Subjects

Refresh

Search subject or code...

5
Subjects Teaching

Christian Religious Studies

CBS

Active

Classes Assigned: 5

Total Students: 11

Active Courses: 5 / 5

SHS 1 General Arts B SHS 1 General Arts A

SHS 1 Science W SHS 1 Science 1 +1 more

Assigned Classes

Subject Schedule

11
Classes Assigned

Core Mathematics3

MATH-COE3

Active

Classes Assigned: 1

Total Students: 6

Active Courses: 1 / 1

SHS 1 General Science A

Assigned Classes

Subject Schedule

32
Students Across Subjects

Core Mathematics7

MATH-COE7

Active

Classes Assigned: 1

Total Students: 6

Active Courses: 1 / 1

SHS 1 General Science A

Assigned Classes

Subject Schedule

Teacher Portal

Dashboard

My Classes

My Subjects

My Courses

Timetable

School Calendar

Assignments

Quizzes

Grading

Reports

Logout

My subjects

Create Assignment

63

Kofi Mensah
Educator

My Subjects

Refresh

Search subject or code...

5
Subjects Teaching

Christian Religious Studies

CBS

Active

Classes Assigned: 5

Total Students: 11

Active Courses: 5 / 5

SHS 1 General Arts B SHS 1 General Arts A

SHS 1 Science W SHS 1 Science 1 +1 more

Assigned Classes

Subject Schedule

11
Classes Assigned

Core Mathematics3

MATH-COE3

Active

Classes Assigned: 1

Total Students: 6

Active Courses: 1 / 1

SHS 1 General Science A

Assigned Classes

Subject Schedule

32
Students Across Subjects

Core Mathematics7

MATH-COE7

Active

Classes Assigned: 1

Total Students: 6

Active Courses: 1 / 1

SHS 1 General Science A

Assigned Classes

Subject Schedule

Subject Schedule — Christian Religious Studies

SHS 1 Science W

1:53 PM - 11:53 PM Monday

Room: TBA

SHS 1 Business A

5:56 PM - 11:52 PM Monday

Room: TBA

Close

Teacher Portal

Dashboard

My Classes

My Subjects

My Courses

Timetable

School Calendar

Assignments

Quizzes

Grading

Reports

Logout

My subjects

Create Assignment

63

Kofi Mensah
Educator

Refresh

My Subjects

Assigned Classes — Christian Religious Studies

#	Class	Program	Students	Status
1	SHS 1 Business A	Business	4	Active
2	SHS 1 General Arts A	General Arts	4	Active
3	SHS 1 General Arts B	General Arts	3	Active
4	SHS 1 Science 1	General Science	0	Active
5	SHS 1 Science W	General Science	0	Active

Close

SHS 1 General Arts B

SHS 1 General Arts A

SHS 1 Science W

SHS 1 Science 1

+1 more

SHS 1 General Science A

SHS 1 General Science A

Assigned Classes

Subject Schedule

Assigned Classes

Subject Schedule

Assigned Classes

Subject Schedule

Figure 4.21 Present Report Page filtering Academic by Program from Administrator

LMS Admin

Grade Review

Grade Scales

Reports

Messages

Announcements

Events

Notifications

System Settings

Institution Settings

Analytics

Logout

Reports

Search...

36

Justice Mensah
Administrator

50

40

30

20

10

0

SHS 1 General Sci...

General Science

Custom Report Builder

Create your own view for leadership meetings and audits.

SOURCE

METRIC

TOP ROWS

Class Performance

Average Score

Top 5

Build Custom Report

Report Output

Includes semester, annual, WASSCE, student progress, teacher, attendance and enrollment views.

TYPE	NAME	GROUP	AVERAGE %	PASS RATE	COUNT	NOTES
Program	General Science	Program	47.38%	50.00%	6	Academic performance by program

Automated Report Schedules

LMS Admin

Grade Review

Grade Scales

Reports

Messages

Announcements

Events

Notifications

System Settings

Institution Settings

Analytics

Logout

Reports

Search...

36

Justice Mensah
Administrator

Class Performance

Average outcomes by class for current term.

100

90

80

70

60

50

40

30

20

10

0

SHS 1 General Sci...

Program Performance

Comparative distribution and trend by program.

General Science

Custom Report Builder

Create your own view for leadership meetings and audits.

SOURCE

METRIC

TOP ROWS

Class Performance

Average Score

Top 5

Build Custom Report

LMS Admin

Grade Review

Grade Scales

Reports

Messages

Announcements

Events

Notifications

System Settings

Institution Settings

Analytics

Logout

Reports

Search...

Justice Mensah
Administrator

Admin Reports Control Center

Generate school-wide intelligence: performance by class/program, attendance, enrollment, WASSCE readiness, and export-ready leadership reports.

Report Filters & Exports

Choose a report family, then build or export your current analytics view.

ACADEMIC YEAR

SEMESTER

REPORT FAMILY

SEARCH

EXPORT FORMAT

2025-2027 1

First Term

Academic by Program

Class, program, student, tea

PDF

Load Data

Export Current

Export All Reports

Admin reports loaded successfully.

43
Total Students

32
Total Teachers

27
Total Classes

29
Total Programs

50.00%
Pass Rate

77.80%
Attendance Rate

Class Performance

Average outcomes by class for current term.

Program Performance

Comparative distribution and trend by program.

Figure 4.22 Present Attendance Report from Administrator

Admin Reports

Academic Year: 2025-2027 1 | Semester: First Term

Type	Name	Group	Average %	Pass Rate	Count	Notes
Attendance	School Attendance	All Classes	77.80%	77.80%	27	Attendance summary for active term.

LMS Admin

Quizzes

Grade Review

Grade Scales

Reports

Messages

Announcements

Events

Notifications

System Settings

Institution Settings

Logout

Reports

Search...

Justice Mensah Administrator

Custom Report Builder

Create your own view for leadership meetings and audits.

SOURCE

METRIC

TOP ROWS

Class Performance

Average Score

Top 5

Build Custom Report

Report Output

Includes semester, annual, WASSCE, student progress, teacher, attendance and enrollment views.

TYPE	NAME	GROUP	AVERAGE %	PASS RATE	COUNT	NOTES
Attendance	School Attendance	All Classes	77.80%	77.80%	27	Attendance summary for active term.

Automated Report Schedules

Configure recurring report jobs and delivery recipients.

SCHEDULE NAME

REPORT TYPE

FREQUENCY

RECIPIENT EMAIL

Monthly Admin Pack

Semester Reports

Weekly

admin@school.edu.gh

Add Schedule

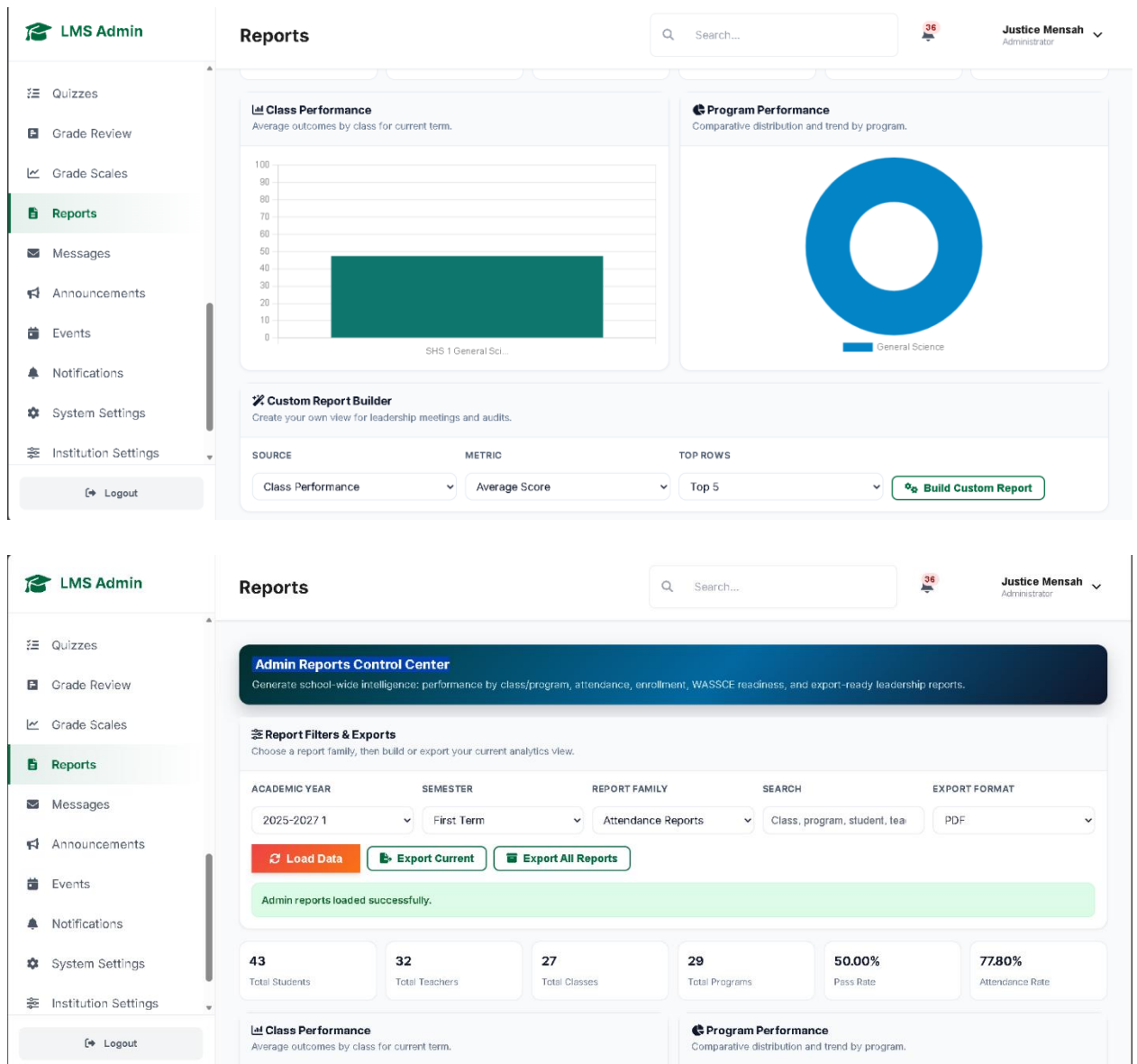


Figure 4.23 Present Attendance Report from Teacher

Attendance Report — 2026-03-01

Date	Class	Subject	Student	Status	Remarks
2026-03-01	SHS 1 General Science A	Integrated Science	Kofi Addo	present	
2026-03-01	SHS 1 General Science A	Integrated Science	Akosua Atta	absent	
2026-03-01	SHS 1 General Science A	Integrated Science	Kwesi Boateng	present	
2026-03-01	SHS 1 General Science A	Integrated Science	University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA	present	
2026-03-01	SHS 1 General Science A	Integrated Science	Kwame Frimpong	present	
2026-03-01	SHS 1 General Science A	Integrated Science	Kwamew Osei	present	
2026-03-25	SHS 1 General Science A	Integrated Science	Kofi Addo	present	
2026-03-25	SHS 1 General Science A	Integrated Science	Akosua Atta	present	
2026-03-25	SHS 1 General Science A	Integrated Science	Kwesi Boateng	present	
2026-03-25	SHS 1 General Science A	Integrated Science	Kwame Frimpong	present	
2026-03-25	SHS 1 General Science A	Integrated Science	Kwamew Osei	absent	

Teacher Portal

📅 Timetable

📅 School Calendar

📋 Assignments

📝 Quizzes

📊 Grading

📈 Reports

📄 Assessments

Attendance

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Messages

Logout

Attendance

+ Create Assignment

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Kofi Mensah
Educator

DATE	CLASS	SUBJECT	STUDENT	STATUS	NOTE
2026-03-01	SHS 1 General Science A	Integrated Science	Kofi Addo	Present	—
2026-03-01	SHS 1 General Science A	Integrated Science	Akosua Atta	Absent	—
2026-03-01	SHS 1 General Science A	Integrated Science	Kwesi Boateng	Present	—
2026-03-01	SHS 1 General Science A	Integrated Science	University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA	Present	—
2026-03-01	SHS 1 General Science A	Integrated Science	Kwame Frimpong	Present	—
2026-03-01	SHS 1 General Science A	Integrated Science	Kwamew Osei	Present	—
2026-03-25	SHS 1 General Science A	Integrated Science	Kofi Addo	Present	—
2026-03-25	SHS 1 General Science A	Integrated Science	Akosua Atta	Present	—
2026-03-25	SHS 1 General Science A	Integrated Science	Kwesi Boateng	Present	—
2026-03-25	SHS 1 General Science A	Integrated Science	Kwame Frimpong	Present	—
2026-03-25	SHS 1 General Science A	Integrated Science	Kwamew Osei	Absent	—

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Attendance

Daily Attendance Workspace

Track attendance, import offline sheets, and review attendance history.

Import CSVExport CSVExport PDF

Mark AttendancePast Attendance

CLASSSUBJECTFROMTO

SHS 1 General Science AIntegrated Science03/01/202603/30/2026Search

DATE	CLASS	SUBJECT	STUDENT	STATUS	NOTE
2026-03-01	SHS 1 General Science A	Integrated Science	Kofi Addo	Present	—
2026-03-01	SHS 1 General Science A	Integrated Science	Akosua Atta	Absent	—
2026-03-01	SHS 1 General Science A	Integrated Science	Kwesi Boateng	Present	—
2026-03-01	SHS 1 General Science A	Integrated Science	University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA	Present	—
2026-03-01	SHS 1 General Science A	Integrated Science	Kwame Frimpong	Present	—

Attendance

Create Assignment63Kofi MensahEducator

Attendance

Daily Attendance Workspace

Track attendance, import offline sheets, and review attendance history.

Import CSVExport CSVExport PDF

Mark AttendancePast Attendance

CLASSSUBJECTFROMTO

SHS 1 General Science AIntegrated Science03/01/202603/30/2026Search

#	STUDENT	STUDENT ID	STATUS	NOTE (OPTIONAL)
1	Kofi Addo	ASHS-2024-0003	Present Excused	Optional note
2	Akosua Atta	ASHS-2024-0018	Present Excused	Optional note
3	Kwesi Boateng	ASHS-2024-0009	Present Excused	Optional note
4	University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA	ASHS-2026-0001	Present Excused	Optional note
5	Kwame Frimpong	ASHS-2024-0013	Present Excused	Optional note
6	Kwamew Osei	ASHS-2024-0001	Present Excused	Optional note

student(s) not yet marked

Submit Attendance

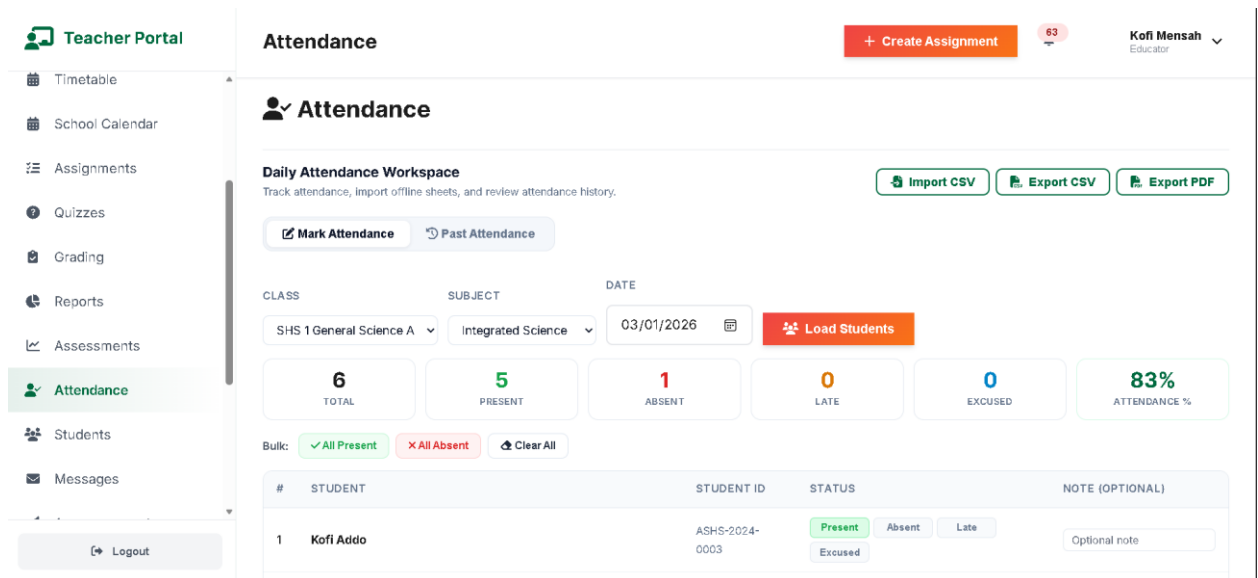


Figure 4.24 Present Student Report from Teacher

Student Report Card

SHS 1 General Science A - Integrated Science

Student: Akosua Atta (ASHS-2024-0018)

Average: 57.00% | Grade: C5 | Status: Published

Subject	Total Score	Percentage	Grade
Integrated Science	284.00	57.00%	C5

Generated: 8/11/2026, 5:05:06 AM

Figure 4.25 Present Student Report from Student

ACCRA SENIOR HIGH SCHOOL

Excellence Through Knowledge with fuck you

Website: <https://accrashs.edu.gh> | Email: info@accrashs.edu.gh

Phone: +233 30 222 1111 | Postal: N/A

STUDENT REPORT CARD

Academic Year: 2025-2027 1 | Semester: First Term

Field	Value	Field	Value
Student Name	MR. OSEI KWAMEW	Student Index	ASHS-2024-0001
Gender	MALE	Class	SHS 1 General Science A
Programme	General Science	Level	SHS 1
Email	kwame.osei@student.accrashs.edu.gh	Phone	+233 55 111 2001
Address	Tema, Accra, western	No. on Roll	6

Summary Metric	Value	Summary Metric	Value
Student's Average Mark	690.00	Position in Class	1st
Last Semester Avg	N/A	Last Semester Position	N/A
Total Subjects Score	690.00	Best Six Aggregate	1.00
Attendance	6 out of 11 (54.55%)		

Subject	Total	Percentage	Class Avg	Position	Grade	Remarks
CORE SUBJECTS						
Integrated Science	690.00	100%	47.38	1st	A1	Excellent

Grade Scale (WAEC)	Interpretation
80 - 100 (A1)	Excellent
70 - 79.99 (B2)	Very Good

Figure 4.26 Present Semster Report from Administrator

Teacher Portal

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Kofi Mensah
Educator

60

40

30

20

10

0

Kwamew Osei

University Ed...

Akosua Atta

Kofi Addo

Kwame Frimpong

Kwesi Boateng

Pending

Approved

Published

Needs correction

Student Report Table

Performance details and report workflow state per student.

WAEC

STUDENT	STUDENT ID	AVERAGE %	GRADE	STATUS	GENERATED	COMMENT
Akosua Atta	ASHS-2024-0018	57.00%	C5	Published	5/7/2026, 7:16:03 AM	-
Kofi Addo	ASHS-2024-0003	31.80%	F9	Published	5/7/2026, 7:16:03 AM	-
Kwame Frimpong	ASHS-2024-0013	28.00%	F9	Published	5/7/2026, 7:16:03 AM	-
Kwamew Osei	ASHS-2024-0001	100.00%	A1	Published	5/7/2026, 7:16:03 AM	-
Kwesi Boateng	ASHS-2024-0009	10.00%	F9	Published	5/7/2026, 7:16:03 AM	-
University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA	ASHS-2026-0001	57.50%	C5	Published	5/7/2026, 7:16:03 AM	-

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Kofi Mensah
Educator

Student Progress Distribution

Top student percentages for the selected class-subject.

100

90

80

70

60

50

40

30

20

10

0

Kwamew Osei

University Ed...

Akosua Atta

Kofi Addo

Kwame Frimpong

Kwesi Boateng

Pending

Approved

Published

Needs correction

Workflow Mix

Pending, approved, published and correction-needed counts.

Pending

Approved

Published

Needs correction

Student Report Table

Performance details and report workflow state per student.

WAEC

STUDENT	STUDENT ID	AVERAGE %	GRADE	STATUS	GENERATED	COMMENT
Akosua Atta	ASHS-2024-0018	57.00%	C5	Published	5/7/2026, 7:16:03 AM	-

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Kofi Mensah
Educator

Teacher Reports Workspace

Build class-subject analysis, student progress reports, and printable report cards for the active term.

Report Filters

Choose class-subject and report mode before generating outputs.

ACADEMIC YEAR

SEMESTER

CLASS / SUBJECT

REPORT TYPE

STUDENT (REPORT CARD)

SEARCH STUDENT

2025-2027 1

First Term

SHS 1 General Science

Semester

Select student

Type name or ID

Load Reports

Generate Report Card

Export PDF

Print View

Selected semester: First Term

Academic Year: 2025-2027 1

Reports loaded successfully.

47.38%
Average Score

100.00%
Highest Score

10.00%
Lowest Score

50.00%
Pass Rate (>=50%)

6
Students with Reports

Student Progress Distribution

Top student percentages for the selected class-subject.

Workflow Mix

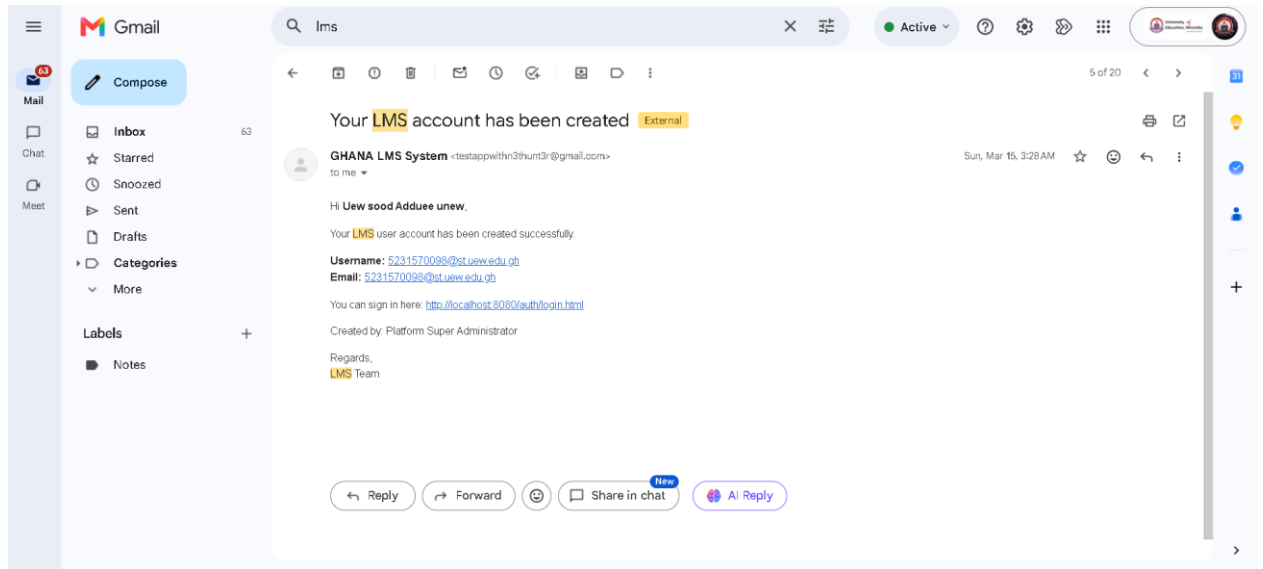
Pending, approved, published and correction-needed counts.

Admin Reports

Academic Year: 2025-2027 1 | Semester: First Term

Type	Name	Group	Average %	Pass Rate	Count	Notes
Progress	Kwame Osei	SHS 1 General Science A	100.00%	100.00%	1	Top student progress profile.
Progress	University Education, Winneba (Winneba) BENEDICT OSEI AMANKWA	SHS 1 General Science A	57.50%	100.00%	1	Top student progress profile.
Progress	Akosua Atta	SHS 1 General Science A	57.00%	100.00%	1	Top student progress profile.
Progress	Kofi Addo	SHS 1 General Science A	31.80%	0.00%	1	Top student progress profile.
Progress	Kwame Frimpong	SHS 1 General Science A	28.00%	0.00%	1	Top student progress profile.
Progress	Kwesi Boateng	SHS 1 General Science A	10.00%	0.00%	1	Top student progress profile.

Figure 4.27 Activity Reports on Email sent when New Account is Created



4.8.6 Output Design Principles

The outputs of the system are designed according to the following principles:

- **Accuracy:** Information displayed to users should reflect the data stored and processed by the system.
- **Clarity:** Outputs should be presented using clear labels, tables, cards, and appropriate visual elements.
- **Relevance:** Users should receive information relevant to their roles and activities.
- **Security:** Sensitive academic information should only be accessible to authorised users.
- **Consistency:** Reports, dashboards, messages, and other outputs should follow a consistent interface design.
- **Timeliness:** System responses and updated information should be provided promptly after successful processing.

- **Usability:** Outputs should be easy to read, understand, and navigate.

The output design therefore ensures that information generated by the LMS is presented in a structured and user-friendly manner. By providing role-specific dashboards, reports, academic results, notifications, and system responses, the proposed system enables administrators, teachers, and students to access relevant information efficiently.

4.9 User Interface (UI) Design

The User Interface (UI) design of the proposed School Learning Management System focuses on providing a simple, consistent, responsive, and user-friendly environment through which administrators, teachers, and students can interact with the system. The interface is designed according to the roles and responsibilities of users, ensuring that each user is presented with the functions and information relevant to their activities.

The UI is implemented using **HTML5, CSS3, JavaScript, and Bootstrap**. The use of responsive design principles enables the system interface to adapt to different screen sizes and devices, including desktop computers, laptops, tablets, and mobile devices.

4.9.1 Administrator Interface

The Administrator interface provides access to the major administrative functions of the LMS. The administrator dashboard presents an overview of important system information and provides navigation to the available management modules.

The major administrator interface components include:

- **Dashboard:** displays summaries of users, courses, departments, academic activities, and system statistics.

- **User Management:** provides interfaces for creating, viewing, updating, and managing user accounts.
- **Department Management:** provides forms for creating and managing academic departments.
- **Course Management:** allows administrators to create, update, view, and manage courses.
- **Academic Session Management:** provides interfaces for managing academic years and sessions.
- **Reports:** provides access to academic, attendance, course, and system reports.
- **Audit Logs** allows administrators to review recorded system activities.
- **System Settings:** provides access to permitted system configuration options.

4.9.2 Teacher Interface

The Teacher interface is designed to support teaching, learning, assessment, and student management activities. The teacher dashboard provides access to assigned courses and the functions required to manage academic activities.

The major teacher interface components include:

- **Dashboard:** provides an overview of courses, students, assignments, assessments, and notifications.
- **Course Management:** allows teachers to access and manage assigned courses.

- **Learning Materials** provides interfaces for uploading, viewing, updating, and managing learning materials.
- **Assignments:** allows teachers to create assignments, specify deadlines, view submissions, and provide grades and feedback.
- **Quizzes and Assessments** provides interfaces for creating and managing online assessments.
- **Attendance:** allows teachers to record and view student attendance.
- **Grades:** provides interfaces for entering and viewing student grades and performance.
- **Announcements:** allows teachers to publish relevant academic information and notifications.

4.9.3 Student Interface

The student interface provides students with access to learning resources and academic activities. The dashboard presents relevant information such as enrolled courses, assignments, assessments, grades, attendance, and notifications.

The major student interface components include:

- **Dashboard:** displays an overview of the student's academic activities.
- **Courses:** provides access to enrolled courses.
- **Learning Materials** allows students to view or download available course materials.

- **Assignments:** displays assignment details, deadlines, submission status, grades, and feedback.
- **Quizzes:** provides access to available online quizzes and assessments.
- **Attendance:** allows students to view available attendance information.
- **Grades:** provides access to academic results and performance information.
- **Notifications:** displays announcements, reminders, and other relevant academic notifications.

4.9.4 Navigation Design

The navigation structure is designed to provide users with direct access to the functions available within their roles. The interface uses consistent navigation elements such as menus, dashboard cards, buttons, tables, forms, and navigation links.

Role-based navigation ensures that users do not receive menu options for functions that they are not authorised to perform. For example, administrative functions are presented to administrators, teaching and assessment functions are provided to teachers, and learning-related functions are provided to students.

4.9.5 Form Design

Forms are used throughout the system to capture and update information. The forms are designed with clear labels, appropriate input controls, validation messages, and action buttons.

Examples include:

- Login forms.
- User registration and management forms.
- Course forms.
- Learning material upload forms.
- Assignment creation forms.
- Assignment submission forms.
- Quiz and assessment forms.
- Attendance forms.
- Grading forms.
- Announcement forms.

Input validation provides immediate feedback when required information is missing or incorrectly entered.

4.9.6 Table and Data Display Design

Tables are used to display structured information such as users, courses, students, assignments, submissions, attendance records, grades, notifications, and reports.

The table design provides:

- Clear column headings.
- Appropriate data formatting.
- Search and filtering where applicable.

- Pagination for large datasets where implemented.
- Action controls based on user permissions.
- Responsive presentation on smaller screens.

This design enables users to locate and manage records efficiently.

4.9.7 Responsive Design

The interface uses responsive design principles to support different screen sizes. Layouts, forms, navigation elements, tables, and dashboard components are designed to adapt to the available screen dimensions.

The responsive interface allows users to access the LMS through supported desktop and mobile browsers without requiring a separate application for each device type.

4.9.8 UI Feedback and Error Handling

The interface provides feedback after users perform system operations. Success messages are displayed when an operation is completed successfully, while validation and error messages are displayed when an operation cannot be completed.

Examples include:

- Successful login confirmation.
- Invalid login notification.
- Successful record creation message.
- Validation error message.

- Successful file upload notification.
- Assignment submission confirmation.
- Access-denied message.
- Server or API error notification.

This feedback helps users understand the outcome of their actions and reduces uncertainty when interacting with the system.

4.9.9 UI Design Principles

The proposed LMS interface follows the following principles:

- **Consistency:** similar functions use consistent layouts, controls, labels, and navigation.
- **Simplicity:** interfaces avoid unnecessary elements and present only relevant information.
- **Usability:** users can perform common activities with minimal steps.
- **Responsiveness:** interfaces adapt to different screen sizes.
- **Accessibility:** text, controls, and navigation are presented in a clear and understandable manner.
- **Security:** restricted functions and sensitive information are displayed only to authorised users.
- **Feedback:** the system provides clear responses to user actions.

- **Role-based presentation:** each user receives an interface appropriate to their assigned role.

The UI design therefore provides a consistent interaction environment for administrators, teachers, and students. The combination of role-based navigation, responsive layouts, structured forms, data tables, dashboards, and feedback mechanisms supports efficient interaction with the proposed Learning Management System.

Figure 4.28 presents representative interfaces for the Login page.

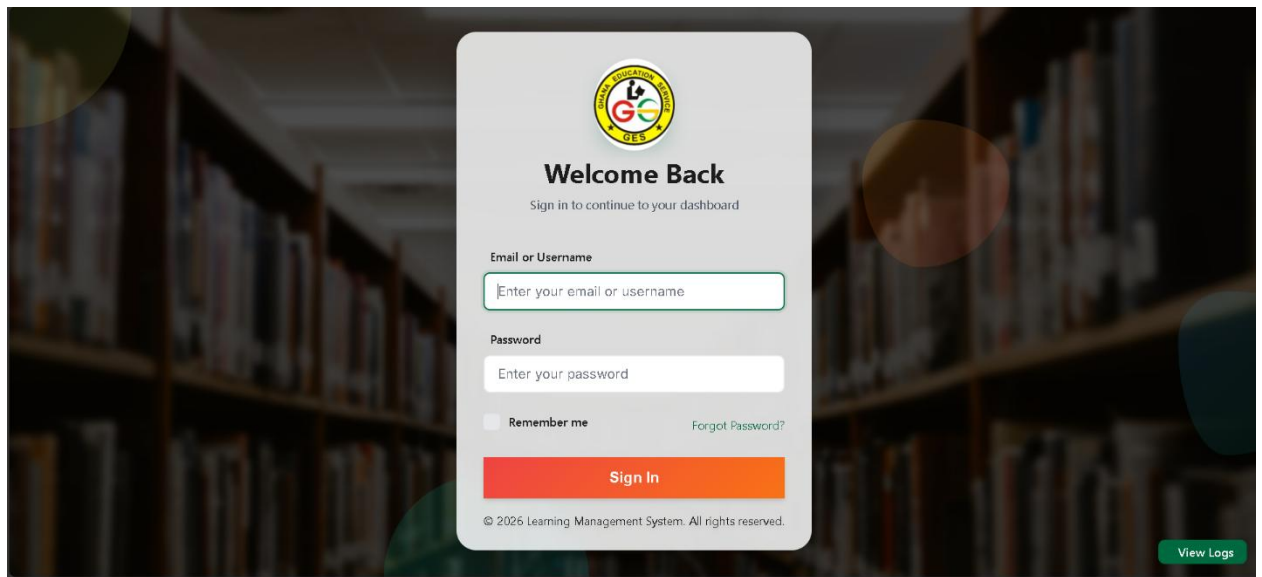


Figure 4.29 presents representative interfaces for the Forgot Password page.

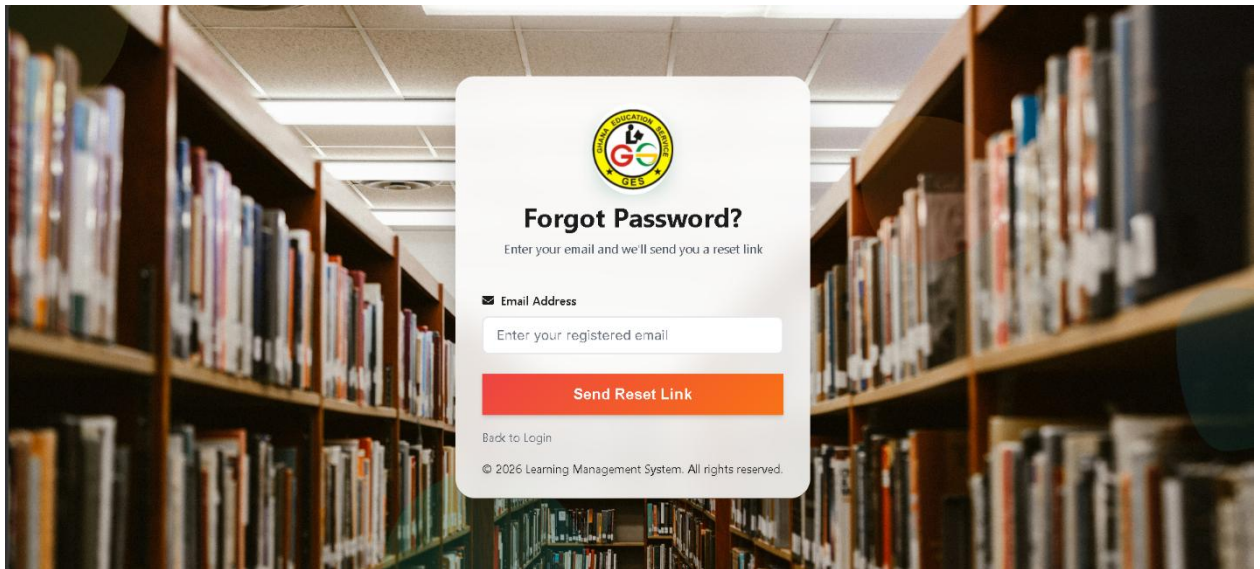


Figure 4.30 presents representative interfaces for the Administrator dashboard.

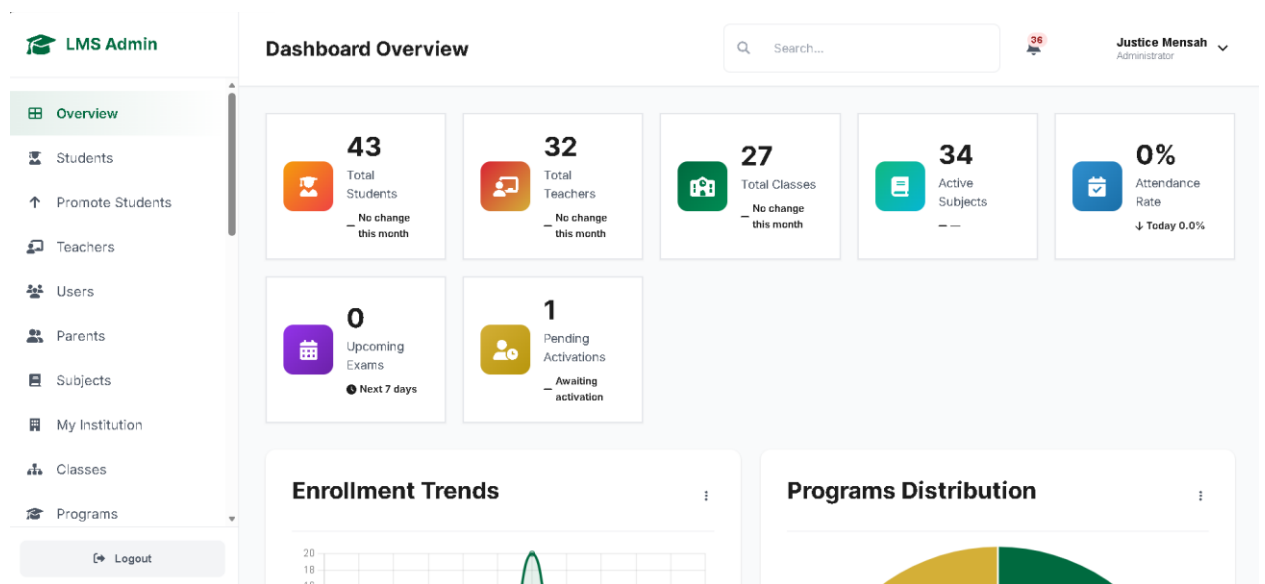


Figure 4.31 presents representative interfaces for the Teacher dashboard.

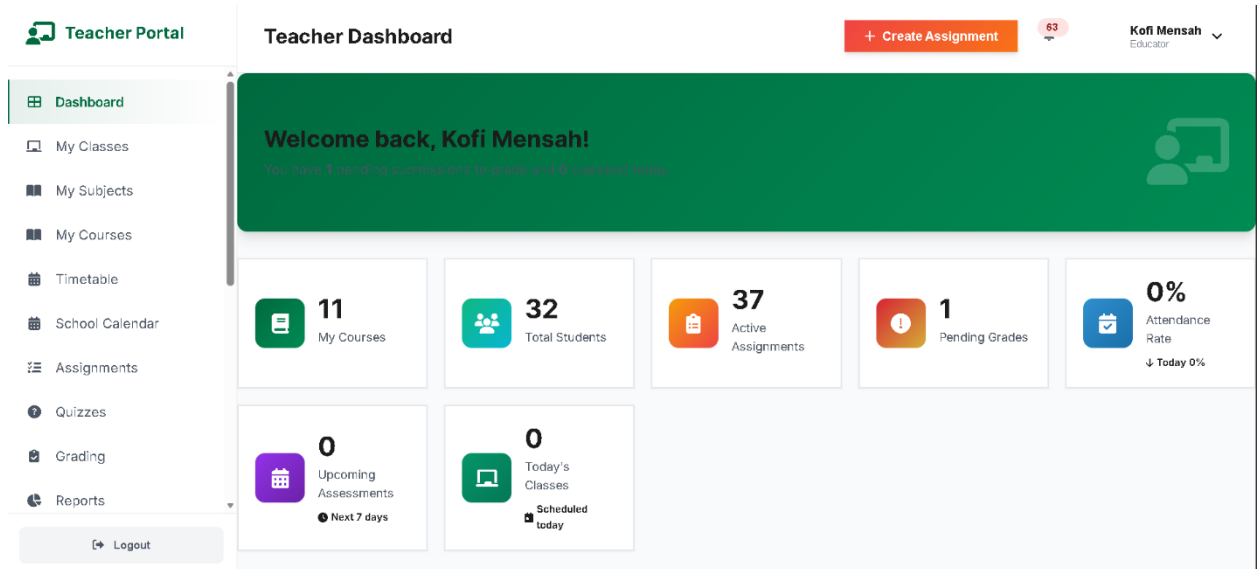
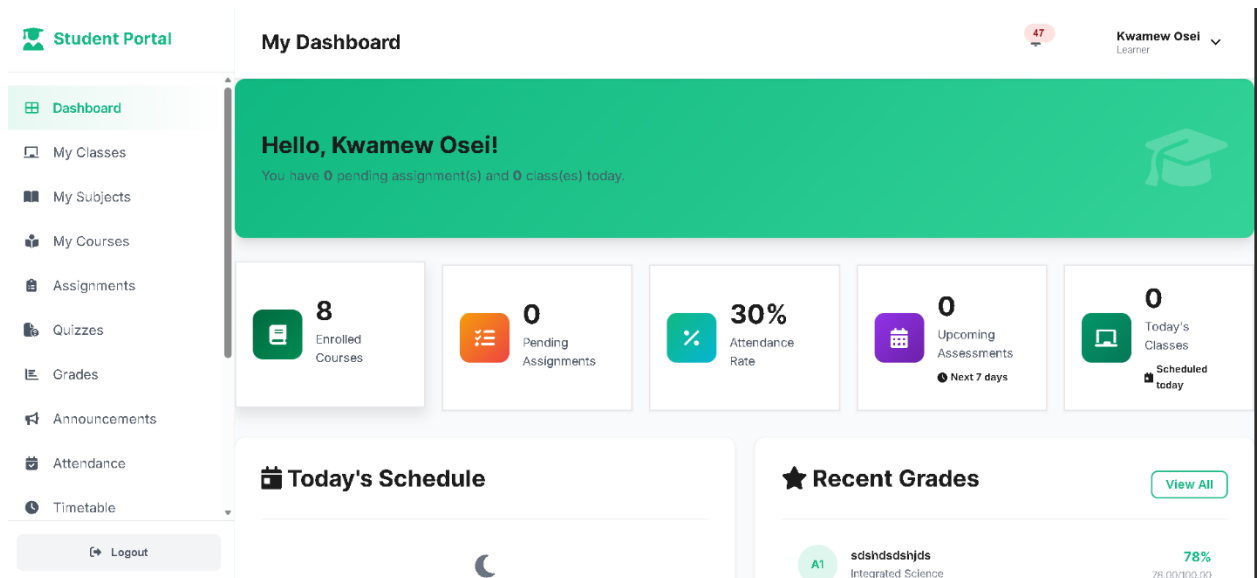


Figure 4.32 presents representative interfaces for the student dashboard.



4.10 System Modules

The proposed School Learning Management System is organised into several interconnected modules. Each module is responsible for a specific set of academic or

administrative functions while communicating with other modules through the backend RESTful API. This modular structure improves system organisation, maintainability, and separation of responsibilities.

The major modules implemented in the proposed system are described below.

4.10.1 Authentication and Security Module

The Authentication and Security Module controls access to the Learning Management System. It ensures that only authenticated and authorised users can access protected system functions.

The module is responsible for:

- User login and logout.
- Authentication of user credentials.
- Password management.
- Session management.
- Role-based access control.
- Authorisation of protected operations.
- Access restriction based on user roles.
- Recording relevant authentication and security activities.

The module ensures that administrators, teachers, and students are provided with access appropriate to their assigned roles.

4.10.2 User Management Module

The User Management Module allows administrators to manage accounts used to access the system. It provides centralised management of user information and account status.

The module supports:

- Creating user accounts.
- Viewing user information.
- Updating user accounts.
- Deactivating or managing account status.
- Assigning user roles.
- Managing permitted user information.
- Searching and filtering user records where implemented.

The module helps maintain consistent user information within the centralised database.

4.10.3 Academic Management Module

The Academic Management Module manages the academic structure of the institution. It provides administrators with tools for managing departments, courses, academic years, sessions, and related academic information.

The module supports:

- Department management.
- Course creation and management.

- Course code and course information management.
- Teacher-course assignment.
- Academic year management.
- Academic session management.
- Management of relevant academic records.

This module provides the foundation for organising the academic activities handled by the LMS.

4.10.4 Enrolment Management Module

The Enrolment Management Module manages the relationship between students and courses. It records the courses in which students are enrolled and provides information required for learning and assessment activities.

The module supports:

- Student enrolment.
- Course enrolment records.
- Viewing enrolled students.
- Viewing student course information.
- Maintaining relationships between students and courses.

The enrolment information is used by other modules such as assignments, attendance, assessments, and results.

4.10.5 Learning Material Module

The Learning Material Module provides a centralised platform for managing digital learning resources. Teachers can upload materials associated with their courses, while authorised students can access available resources.

The module supports:

- Uploading learning materials.
- Associating materials with courses.
- Viewing available materials.
- Downloading permitted materials.
- Updating material information.
- Managing material records.

Centralising learning materials reduces dependence on separate file-sharing platforms and makes course resources easier to access.

4.10.6 Assignment Management Module

The Assignment Management Module supports the creation, distribution, submission, and grading of assignments.

Teachers can:

- Create assignments.
- Provide assignment descriptions and instructions.

- Set submission deadlines.
- Specify marks or grading information.
- View student submissions.
- Grade submitted assignments.
- Provide feedback.

Students can:

- View available assignments.
- View assignment instructions and deadlines.
- Submit completed assignments.
- View submission status.
- View grades and feedback where available.

4.10.7 Quiz and Assessment Module

The Quiz and Assessment Module supports online assessments and quizzes. It enables teachers to create assessments and students to complete them through the LMS.

The module supports:

- Creating quizzes.
- Creating quiz questions.
- Defining available answers and correct responses where applicable.
- Setting assessment parameters.

- Publishing assessments.
- Student quiz attempts.
- Recording assessment responses.
- Calculating or recording scores.
- Viewing assessment results.

The module provides a structured approach to conducting online assessments.

4.10.8 Attendance Management Module

The Attendance Management Module enables teachers to record and manage student attendance for courses and academic activities.

The module supports:

- Recording attendance.
- Selecting courses and relevant dates.
- Recording student attendance status.
- Viewing attendance records.
- Monitoring attendance information.
- Generating attendance-related information and reports.

The centralised attendance records make it easier to retrieve and monitor student participation.

4.10.9 Results and Grading Module

The Results and Grading Module manages student academic performance information. It provides functions for recording, processing, and presenting grades and results.

The module supports:

- Recording student marks.
- Recording grades.
- Associating results with students and courses.
- Viewing student performance.
- Providing academic result information.
- Supporting generation of performance reports.

The module ensures that academic performance information is stored centrally and can be retrieved by authorised users.

4.10.10 Communication and Notification Module

The Communication and Notification Module supports communication between authorised users of the LMS. It provides a centralised mechanism for distributing academic announcements and system notifications.

The module supports:

- Creating announcements.
- Publishing academic information.
- Sending notifications.

- Displaying notifications to users.
- Tracking notification status where implemented.
- Supporting relevant user communication.

The module reduces dependence on fragmented communication channels and helps ensure that important academic information reaches the intended users.

4.10.11 Reporting Module

The Reporting Module converts stored academic and administrative data into useful reports for authorised users.

The module supports reports relating to:

- Student performance.
- Attendance.
- Courses.
- Assignments.
- Assessments.
- Users.
- Academic activities.
- Other available system information.

Administrators and authorised teachers can use reports to support monitoring, decision-making, and academic management.

4.10.12 Audit and Activity Monitoring Module

The Audit and Activity Monitoring Module records relevant activities performed within the system. It supports monitoring, accountability, and investigation of system activities.

The module can record information such as:

- User performing an activity.
- Action performed.
- Date and time of the activity.
- Relevant system operation.
- IP address where applicable.

Administrators can use audit information to monitor system usage and identify potentially unauthorised or unusual activities.

4.10.13 Module Integration

Although the LMS is divided into separate modules, the modules operate as an integrated system. For example, the Authentication Module determines whether a user can access the system, while the User Management Module determines the user's role and account information. The Enrolment Module provides course relationships used by the Learning Material, Assignment, Assessment, Attendance, and Results modules.

Communication between the frontend and these modules is handled through the RESTful API. The API processes requests, applies authentication and authorisation rules, performs

the required business logic, communicates with the MySQL database, and returns JSON responses to the frontend.

The overall relationship between the major modules can therefore be represented as:

**User → Authentication → Role-Based Dashboard → Relevant Module → REST API
→ MySQL Database → JSON Response → User Interface**

The modular design enables the proposed Learning Management System to provide integrated academic and administrative services while maintaining clear separation of responsibilities among the different functional components.

4.11 Technologies

The development of the proposed School Learning Management System involved a combination of web development, server-side, database, API, and development tools. The selected technologies were chosen to support the requirements of the system, including maintainability, usability, security, data management, and communication between the frontend and backend.

4.11.1 HTML5

HTML5 was used to structure the content and components of the LMS user interface. It provides the basic structure for pages, forms, tables, navigation elements, dashboards, and other interface components.

HTML5 supports the organisation of information into meaningful elements, making the interface easier to develop and maintain.

4.11.2 CSS3

CSS3 was used to control the visual presentation and layout of the system. It was applied to elements such as colours, typography, spacing, buttons, forms, tables, navigation menus, and dashboard components.

CSS3 also supports responsive layouts, allowing the interface to adapt to different screen sizes and devices.

4.11.3 JavaScript

JavaScript was used to provide client-side functionality and dynamic interaction within the LMS. It enables the frontend to respond to user actions, validate information, manipulate interface elements, and communicate with backend services.

JavaScript also supports asynchronous communication with the REST API, allowing data to be retrieved or submitted without requiring complete page reloads.

4.11.4 Bootstrap

Bootstrap was used as a frontend framework to support responsive and consistent interface development. It provides reusable components and layout utilities for elements such as navigation bars, forms, buttons, cards, tables, alerts, and responsive grids.

The use of Bootstrap helps maintain consistency across the different user interfaces of the LMS.

4.11.5 PHP 8

PHP 8 was used for the server-side implementation of the system. It processes requests received from the frontend, performs business logic, validates data, communicates with the database, and generates API responses.

PHP was selected because it provides suitable support for web application development and integrates effectively with MySQL and Apache.

4.11.6 RESTful API

The system uses a **RESTful Application Programming Interface (API)** to facilitate communication between the frontend and backend. The API provides endpoints through which the frontend can request, create, update, or delete system information.

The API architecture separates the presentation layer from the backend services and allows the frontend to communicate with the application using standard HTTP methods such as:

- **GET** – retrieves information.
- **POST** – creates or submits information.
- **PUT/PATCH** – updates existing information.
- **DELETE** – removes information where authorised.

This architecture supports the API-first design of the proposed LMS.

4.11.7 JSON

JavaScript Object Notation (JSON) is used as the primary data-interchange format between the frontend and backend API.

The API returns information and operation results in JSON format, which JavaScript can process and use to update the user interface. JSON provides a lightweight and structured format for exchanging information between the frontend and backend.

4.11.8 Fetch API

The **Fetch API** is used within the JavaScript frontend to send HTTP requests to the RESTful API and process responses.

For example, when a user submits a form, JavaScript can use the Fetch API to send the information to the appropriate PHP API endpoint. The API processes the request and returns a JSON response, which the frontend uses to provide feedback or update displayed information.

The communication process can be represented as:

Frontend → Fetch API → REST API → PHP → MySQL → JSON Response → Frontend

4.11.9 MySQL

MySQL was used as the relational database management system for the LMS. It provides centralised storage for user, academic, learning, assessment, attendance, results, communication, and system activity information.

MySQL supports relational database concepts such as primary keys, foreign keys, constraints, and relationships, which are important for maintaining the integrity and consistency of the LMS data.

4.11.10 Apache

Apache HTTP Server provides the web server environment required to run the PHP-based backend of the system. It receives web requests and allows the PHP application and REST API services to be accessed through the configured server environment.

4.11.11 XAMPP

XAMPP was used as the local development environment for the system. It provides an integrated environment containing components such as Apache and MySQL, allowing the LMS to be developed and tested locally.

XAMPP simplifies the configuration of the development environment and provides the infrastructure required to run the PHP application and database during development.

4.11.12 Visual Studio Code

Visual Studio Code (VS Code) was used as the primary source-code editor during system development. It provided an environment for creating and managing the HTML, CSS, JavaScript, PHP, and other project files.

Its development features support code editing, project organisation, debugging, and integration with version-control tools.

4.11.13 Git

Git was used as the version-control system for managing changes made to the source code. It allows different versions of the project to be tracked and provides a mechanism for maintaining the development history.

Git also makes it possible to revert changes when necessary and supports structured software development.

4.11.14 GitHub

GitHub was used as a remote repository for storing and managing the project's source code. It provides version-control collaboration capabilities and serves as a backup location for the project repository.

The repository also supports tracking changes and managing different development versions of the LMS.

4.11.15 Postman

Postman was used to test and validate the RESTful API endpoints during development. API requests could be sent directly to individual endpoints to verify authentication, data processing, database operations, validation, and JSON responses.

Postman was particularly useful for testing backend services independently from the frontend.

4.11.16 Summary of Technologies

Technology/Tool	Purpose in the System
HTML5	Structures the web interface.
CSS3	Provides styling, layout, and responsive presentation.
JavaScript	Provides client-side interaction and dynamic functionality.
Bootstrap	Supports responsive layouts and reusable interface components.

PHP 8	Implements server-side processing and REST API services.
REST API	Provides communication between the frontend and backend.
JSON	Exchanges structured data between the frontend and API.
Fetch API	Sends asynchronous HTTP requests from the frontend.
MySQL	Stores and manages system data.
Apache	Provides the web server environment for the PHP application.
XAMPP	Provides the local development and testing environment.
Visual Studio Code	Provides the development and source-code editing environment.
Git	Tracks and manages source-code changes.
GitHub	Hosts and manages the project's remote repository.
Postman	Tests and validates REST API endpoints.

The combination of these technologies provides the technical foundation for implementing the proposed School Learning Management System. The frontend technologies provide the user interface, PHP and the RESTful API provide application

services, MySQL provides persistent data storage, and the supporting development tools facilitate development, testing, and version control.

4.12 System Testing

System testing was conducted to verify that the proposed School Learning Management System functions according to the specified requirements and that the major components operate correctly as an integrated system. Testing was performed after the development of the major system modules and focused on functionality, integration, security, usability, and reliability.

The testing process involved checking individual system components, interactions between the frontend, RESTful API and database, complete system workflows, and the ability of users to perform their assigned activities. The major testing approaches applied were **unit testing, integration testing, system testing, and user acceptance testing**.

4.12.1 Unit Testing

Unit testing was used to verify individual functions and components of the system independently. The purpose was to identify errors within specific functions before they were integrated with other components.

Examples of components tested include:

- User authentication functions.
- User registration and validation.
- Course creation and updating.
- Learning material upload.

- Assignment creation.
- Assignment submission.
- Attendance recording.
- Grade processing.
- Notification processing.
- API request validation.

Each function was tested using valid and invalid inputs to determine whether it produced the expected result.

4.12.2 Integration Testing

Integration testing was performed to verify communication between different components of the LMS. Attention was given to the interaction between the frontend, RESTful API, authentication and authorisation mechanisms, business logic, and MySQL database.

The following interactions were tested:

Frontend → REST API

The system was tested to ensure that requests submitted through the frontend were correctly sent to the appropriate API endpoints.

REST API → Database

API operations were tested to ensure that data could be correctly inserted, retrieved, updated, and deleted from the MySQL database where authorised.

Database → REST API → Frontend

Responses returned from the database through the API were tested to ensure that the correct information was converted into JSON and displayed appropriately on the frontend.

Integration testing also verified that authentication and role-based permissions were correctly applied when users attempted to access protected resources.

4.12.3 System Testing

System testing evaluated the LMS as a complete integrated application. Rather than testing individual components separately, the entire system was tested according to typical user workflows.

The major system workflows tested included:

- 1. Authentication Workflow**

A user enters login credentials, the system validates the credentials, determines the user's role, and redirects the user to the appropriate dashboard.

- 2. Course Management Workflow**

An authorised administrator or teacher creates or updates course information, the API processes the request, and the information is stored in the MySQL database.

- 3. Learning Material Workflow**

A teacher uploads learning material, the system validates the request and file information, stores the relevant record, and makes the material available to authorised students.

4. Assignment Workflow

A teacher creates an assignment, students access the assignment and submit their work, and the teacher reviews, grades, and provides feedback on the submission.

5. Attendance Workflow

A teacher selects a course and records student attendance. The system validates and stores the attendance information and makes appropriate records available to authorised users.

6. Assessment Workflow

A teacher creates an assessment or quiz, students complete the assessment, and the system records the resulting information.

7. Results Workflow

Student marks and grades are processed and stored, after which authorised, users can view the relevant academic performance information.

8. Communication Workflow

Administrators or teachers publish announcements or notifications, which are then made available to the intended users.

System testing therefore verified that the major modules operate together as a single integrated LMS.

4.12.4 Security Testing

Security-related tests were performed to verify that access to protected functions and information was appropriately restricted.

The testing included:

- Testing valid and invalid login credentials.
- Verifying role-based access control.
- Attempting to access restricted functions using an unauthorised role.
- Testing input validation.
- Testing protected API endpoints.
- Verifying that users cannot directly access functions outside their permissions.
- Testing database operations using controlled inputs.
- Checking appropriate handling of authentication and session information.

The purpose of security testing was to ensure that sensitive academic and user information was protected from unauthorised access and inappropriate system operations.

4.12.5 User Acceptance Testing

User Acceptance Testing (UAT) was used to determine whether the LMS satisfies the functional expectations of its intended users. The testing focused on common tasks performed by administrators, teachers, and students.

Administrators were expected to test functions such as:

- Logging into the system.
- Managing users.
- Managing departments and courses.

- Managing academic sessions.
- Generating reports.
- Reviewing system activities.

Teachers were expected to test functions such as:

- Accessing assigned courses.
- Uploading learning materials.
- Creating assignments and assessments.
- Recording attendance.
- Grading student work.
- Publishing announcements.

Students were expected to test functions such as:

- Logging into the system.
- Accessing enrolled courses.
- Viewing learning materials.
- Submitting assignments.
- Taking available assessments.
- Viewing grades and academic information.
- Receiving notifications.

Feedback from these activities was used to identify interface, functionality, and usability issues requiring correction.

4.12.6 Testing Criteria

The system was considered to have passed a test when the actual result matched the expected result, and the operation did not produce an unacceptable error.

The following criteria were considered:

Criterion	Description
Functional correctness	The system performs the requested operation correctly.
Data integrity	Information is stored and retrieved accurately.
Access control	Users can access only authorised functions.
API communication	Frontend requests and backend responses operate correctly.
Error handling	Invalid operations produce appropriate error messages.
Usability	Users can complete common tasks without unnecessary difficulty.
Reliability	The system performs consistently during repeated operations.

Responsiveness	The interface operates appropriately across supported screen sizes.
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The results of the individual test cases are presented in **Section 4.13**, where the test scenarios, expected results, actual results, and test status are documented.

Overall, system testing provided evidence that the major components of the proposed Learning Management System were tested individually and collectively. The testing process helped identify and correct functional, integration, security, and usability issues before the system was considered ready for deployment and user evaluation.

4.13 Test Cases

Test cases were developed to verify that the major functions of the proposed School Learning Management System operate according to the specified requirements. Each test case defines the function being tested, the input or action performed, the expected result, the actual result, and the status of the test.

The test cases cover authentication, user management, course management, learning materials, assignments, assessments, attendance, results, notifications, access control, and API communication.

4.13.1 Authentication Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
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TC-001	Valid Login	Enter valid username and password	User is authenticated and redirected to the appropriate dashboard	User was authenticated and redirected correctly	Pass
TC-002	Invalid Login	Enter incorrect username/password	Error message is displayed and access is denied	Error message was displayed	Pass
TC-003	Empty Login Fields	Submit login form without credentials	Validation message is displayed	Validation message was displayed	Pass
TC-004	Role-Based Access	Login using different user roles	User receives access according to assigned role	Role-based access worked correctly	Pass
TC-005	Logout	Select logout	User session is terminated and login	Session was terminated successfully	Pass

			page is displayed		
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4.13.2 User Management Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
TC-006	Create User	Enter valid user information	New user account is created	User account was created	Pass
TC-007	Duplicate User	Enter an existing username/email	System rejects duplicate information	Duplicate information was rejected	Pass
TC-008	Update User	Modify existing user information	User record is updated	User record was updated	Pass
TC-009	Invalid User Data	Submit incomplete user information	Validation message is displayed	Validation message was displayed	Pass

4.13.3 Course Management Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
TC-010	Create Course	Enter valid course details	Course is created and stored	Course was created successfully	Pass
TC-011	Update Course	Modify course information	Course information is updated	Course was updated	Pass
TC-012	View Course	Select a course	Correct course information is displayed	Course information was displayed	Pass
TC-013	Invalid Course Data	Submit incomplete course information	Validation error is displayed	Validation error was displayed	Pass

4.13.4 Learning Material Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
TC-014	Upload Material	Upload a valid	Material is uploaded	Material was	Pass

		learning material	and associated with the selected course	uploaded successfully	
TC-015	Invalid File	Upload an unsupported file type	System rejects the file	File was rejected	Pass
TC-016	View Material	Student selects available material	Material information is displayed	Material was displayed	Pass
TC-017	Download Material	Student selects download	Authorised material is downloaded	Material was downloaded successfully	Pass

4.13.5 Assignment Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
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TC-018	Create Assignment	Teacher enters assignment details	Assignment is created	Assignment was created successfully	Pass
TC-019	View Assignment	Student opens an assignment	Assignment details and deadline are displayed	Details were displayed	Pass
TC-020	Submit Assignment	Student uploads valid assignment	Submission is recorded	Submission was recorded successfully	Pass
TC-021	Grade Assignment	Teacher enters grade and feedback	Grade and feedback are saved	Grade and feedback were saved	Pass
TC-022	Invalid Submission	Submit an invalid or unsupported file	System rejects the submission	Submission was rejected	Pass

4.13.6 Attendance Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
TC-023	Record Attendance	Teacher records student attendance	Attendance record is stored	Record was stored successfully	Pass
TC-024	View Attendance	Select course/date	Correct attendance records are displayed	Records were displayed	Pass
TC-025	Invalid Attendance Data	Submit incomplete attendance information	Validation message is displayed	Validation message was displayed	Pass

4.13.7 Assessment and Results Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
TC-026	Create Quiz	Teacher enters valid quiz information	Quiz is created	Quiz was created successfully	Pass

TC-027	Attempt Quiz	Student answers quiz questions	Answers are submitted successfully	Answers were submitted successfully	Pass
TC-028	Record Score	Complete assessment	Score is calculated or recorded correctly	Score was recorded correctly	Pass
TC-029	View Grade	Student opens results	Authorised grade information is displayed	Grade was displayed correctly	Pass

4.13.8 Communication and Notification Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
TC-030	Create Announcement	Teacher/admin enters announcement	Announcement is published	Announcement was published	Pass
TC-031	View Notification	User opens notifications	Relevant notification is displayed	Notification was displayed	Pass

TC-032	Mark Notification Read	User opens notification	Notification status is updated	Status was updated	Pass
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4.13.9 Security and Access Control Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
TC-033	Unauthorised Page Access	Student attempts to access an administrator function	Access is denied	Access was denied	Pass
TC-034	Protected API Request	Send request without valid authentication	API rejects the request	Request was rejected	Pass
TC-035	Role Permission	Teacher attempts to perform an administrator- only operation	Operation is denied	Operation was denied	Pass

TC-036	Input Validation	Submit invalid data to an API endpoint	Request is rejected with an appropriate response	Invalid request was rejected	Pass
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4.13.10 API and Database Test Cases

Test ID	Test Case	Test Input/Action	Expected Result	Actual Result	Status
TC-037	GET Request	Request valid course data	API returns the requested data in JSON format	JSON response was returned	Pass
TC-038	POST Request	Submit valid course information	API creates the record and returns a response	Record was created successfully	Pass
TC-039	UPDATE Request	Submit valid updated information	Existing record is updated	Record was updated successfully	Pass

TC-040	DELETE Request	Delete an authorised record	Record is removed or appropriately deactivated	Operation completed successfully	Pass
TC-041	Database Retrieval	Request stored student information	Correct database records are returned	Correct records were returned	Pass

4.13.11 Summary of Test Results

The test cases covered the major functional and technical components of the proposed LMS. The results indicate that the tested functions performed according to their expected behaviour. Authentication, role-based access control, user management, academic management, learning materials, assignments, attendance, assessments, results, communication, API operations, and database interactions were tested.

The testing results provide evidence that the proposed system can support the major academic and administrative activities for which it was designed. Any errors identified during development were corrected and the relevant functions were retested before the system was considered ready for user evaluation.

4.14 Summary

This chapter presented the design and development structure of the proposed School Learning Management System. The system design was developed based on the

requirements and challenges identified during the analysis of the existing academic management processes.

The chapter described the proposed system design, system architecture, use case design, database design, data dictionary, input design, and output design. The database design established the structure for storing and managing users, departments, courses, enrolments, learning materials, assignments, assessments, attendance, results, notifications, and system activities. The input and output designs described how information is entered into the system, validated, processed, stored, and presented to authorised users.

The chapter also described the major system modules, including authentication and security, user management, academic management, enrolment, learning materials, assignments, assessments, attendance, results, communication, reporting, and audit monitoring. The technologies used to implement the system, including HTML5, CSS3, JavaScript, Bootstrap, PHP 8, RESTful API, JSON, Fetch API, MySQL, Apache, XAMPP, Git, GitHub, and Postman, were also discussed.

System testing was presented through different testing approaches, including unit testing, integration testing, system testing, security testing, and user acceptance testing. Test cases were developed to verify major system functions such as authentication, user management, course management, learning materials, assignments, attendance, assessments, results, notifications, API operations, database interactions, and role-based access control. The testing results indicated that the major functions of the proposed system performed according to their expected behaviour.

Overall, the chapter established the technical and functional design of the proposed School Learning Management System and provided the foundation for its implementation and evaluation. The design integrates the major academic and administrative processes into a centralised web-based platform intended to improve efficiency, accessibility, data management, communication, and security within the institution.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary, conclusion, contributions, recommendations, further research directions, and limitations of the study. The chapter reflects on the extent to which the objectives of the study were achieved through the analysis, design, implementation, and testing of the proposed School Learning Management System (LMS). It also highlights the contribution of the developed system to the management of academic activities and identifies areas for future improvement.

5.2 Summary of the Study

The study focused on the design and implementation of a web-based School Learning Management System for educational institutions. The study was motivated by challenges identified in traditional and fragmented academic management practices, including difficulties in managing student and course information, delays in assignment submission and grading, limited access to learning materials, poor communication among stakeholders, difficulties in generating academic reports, inadequate centralisation of educational records, and dependence on physical processes for academic activities.

The analysis of the existing system revealed that academic information was often managed through paper-based records, spreadsheets, separate applications, and disconnected communication channels. These practices resulted in duplication of information, delays,

increased possibility of errors, difficulty retrieving records, and inefficient coordination among administrators, teachers, and students.

Based on the identified challenges, a web-based Learning Management System was designed to centralise major academic activities within a single platform. The proposed system provides role-based functionality for three main categories of users: administrators, teachers, and students.

The administrator functions include user management, department and programme management, course management, academic session management, reporting, system settings, and monitoring of academic activities. Teachers are provided with facilities for managing courses, uploading learning materials, creating assignments and assessments, grading student work, recording attendance, monitoring student performance, and communicating with students. Students can access enrolled courses and learning materials, submit assignments, participate in assessments, view grades, and receive academic notifications.

The system adopts an API-first architecture in which the frontend communicates with a PHP-based RESTful API. The backend processes requests, applies authentication and authorisation, performs business logic, interacts with the MySQL database, and returns responses in JSON format. This architecture separates the presentation layer from the application and data layers and provides a foundation for future system extension.

The system was implemented using HTML5, CSS3, JavaScript, PHP, MySQL, RESTful API technologies, JSON, and related development tools. The implementation was subsequently subjected to functional, security and access-control, API, and database

testing. The test cases documented in Chapter Four covered major system operations, including authentication, role-based access, academic management, assessments, communication, API requests, and database retrieval.

The results demonstrate that the developed system provides a centralised platform for managing several academic and administrative activities and addresses a number of the weaknesses identified in the existing system.

5.3 Achievement of Objectives

The study was guided by seven specific objectives. The achievement of each objective is discussed below.

5.3.1 Objective One: To analyse the challenges associated with traditional academic management systems

This objective was achieved through the analysis of the existing academic management processes. The analysis identified several challenges, including inefficient record management, fragmented information systems, manual attendance recording, delays in assignment processing and feedback, poor communication among stakeholders, difficulty generating academic reports, and inadequate centralisation of academic information.

The existing workflow analysis provided the basis for identifying the functional and non-functional requirements of the proposed Learning Management System. The identified problems demonstrated the need for a centralised platform capable of automating and integrating major academic activities.

5.3.2 Objective Two: To design a user-friendly Learning Management System for educational institutions

This objective was achieved through the design of the proposed School Learning Management System. The system design incorporated a role-based workflow and separate functionality for administrators, teachers, and students.

The design included the system workflow, system architecture, use case model, Data Flow Diagram, database design, API architecture, input design, output design, and user interface design. The proposed interface was designed to provide users with access to functions relevant to their responsibilities while maintaining a simple and responsive interaction environment.

The design therefore provided a structured foundation for the implementation of the system.

5.3.3 Objective Three: To develop a platform for managing students, teachers, courses, and academic records

This objective was addressed through the implementation of the user management, academic management, course management, enrolment, attendance, results, and reporting functionality.

The developed system provides administrators with facilities for managing users, departments, courses, academic sessions, and other institutional information. Teachers can manage course-related activities and student academic information, while students can access their academic information through the student interface.

The centralised database and RESTful API architecture provide a common platform through which academic information can be stored, retrieved, and processed.

5.3.4 Objective Four: To provide facilities for uploading and accessing learning materials online

This objective was achieved through the Learning Materials Module. Teachers can upload learning resources associated with courses, while students can access the available materials through the system.

The implementation reduces dependence on physical distribution of learning resources and provides students with a centralised location from which course materials can be accessed. This addresses the problem identified in the existing system where access to learning resources could be restricted by physical presence or fragmented distribution channels.

5.3.5 Objective Five: To enable assignment creation, submission, and grading electronically

This objective was achieved through the Assignment Management and Assessment functionality.

Teachers can create assignments and manage student submissions, while students can submit their work through the system. The assessment functionality also provides mechanisms for grading and displaying academic results.

The testing results documented in Chapter Four include assignment, assessment, grading, and result-related test cases. The successful execution of these tests demonstrates that the

implemented system supports the major electronic assignment and assessment processes defined by the study.

5.3.6 Objective Six: To facilitate communication between students, teachers, and administrators

This objective was addressed through the Communication and Notification functionality of the system.

The developed system provides facilities for publishing announcements and displaying notifications to users. This provides a centralised communication mechanism for distributing academic information and reduces dependence on disconnected channels such as printed notices and separate messaging platforms.

The testing documented in Chapter Four includes announcement creation, notification viewing, and notification status operations. These results provide evidence that the implemented communication and notification functions support the intended academic communication processes.

5.3.7 Objective Seven: To improve the efficiency and effectiveness of educational administration

The objective was addressed by integrating major academic and administrative activities into a single web-based platform.

The system reduces the need for repeated manual handling of academic information by providing centralised user, course, learning material, assignment, assessment, attendance,

communication, and reporting functions. The RESTful API architecture also provides a structured mechanism for communication between the frontend and backend components.

The testing results presented in Chapter Four indicate that the tested system functions performed according to their expected behaviour. However, the study did not conduct large-scale institutional performance testing; therefore, claims concerning efficiency should be understood in relation to the implemented functionality and testing scope rather than as a measured comparison of institutional productivity.

5.4 Conclusion

The study concludes that a web-based School Learning Management System can provide an integrated platform for addressing several challenges associated with traditional and fragmented academic management processes.

The developed system brings together important academic activities including user management, course management, learning material distribution, assignment submission, assessment, attendance, results, communication, notifications, and reporting. These functions provide administrators, teachers, and students with a centralised environment for accessing and managing academic information.

The implementation of role-based access mechanisms provides different users with access to functions according to their assigned responsibilities. The API-first architecture separates the frontend from the backend services and database operations, providing a structured foundation for maintenance and future extension.

The MySQL relational database provides the central storage mechanism for academic and administrative information, while the RESTful API enables communication between the frontend and backend through HTTP requests and JSON responses.

The testing conducted on the developed system covered major functional areas, security and access-control operations, API requests, and database interactions. The documented test results indicate that the tested functions produced the expected outcomes.

Although the study was subject to limitations such as time, financial resources, internet dependence, and limited testing within selected educational environments, the developed system provides a practical foundation for improving the management and delivery of academic services.

The study therefore concludes that the proposed School Learning Management System addresses the major problems identified during the analysis of the existing system and provides an integrated platform capable of supporting academic administration, teaching, learning, assessment, communication, and access to educational information.

5.5 Contributions of the Study

The study makes contributions in the areas of academic management, educational technology, and software development.

5.5.1 Academic Contribution

The study provides a structured approach to analysing and addressing challenges associated with traditional academic management processes. It demonstrates how academic activities can be modelled and integrated into a centralised information system.

The study also provides a reference for future research involving web-based Learning Management Systems and educational information systems.

5.5.2 Practical Contribution

The developed system provides educational institutions with a centralised platform for managing academic activities.

The system contributes to:

- Improved access to academic information.
- Centralised management of learning materials.
- Electronic assignment submission and grading.
- Centralised attendance and academic records.
- Improved communication through announcements and notifications.
- Easier access to academic reports.
- Reduced dependence on paper-based academic processes.
- Improved coordination between administrators, teachers, and students.

5.5.3 Technical Contribution

The study demonstrates the application of an API-first architecture in the development of a School Learning Management System.

The separation between the frontend, RESTful API, business logic, and MySQL database provides a structured software architecture that can support future enhancement and integration with other client applications.

The project also demonstrates the use of HTML5, CSS3, JavaScript, PHP, MySQL, REST APIs, JSON, and Fetch API technologies in developing an integrated web-based academic management platform.

5.6 Recommendations

Based on the findings, implementation, and testing of the developed system, the following recommendations are made.

5.6.1 Recommendations for Educational Institutions

Educational institutions should consider adopting integrated Learning Management Systems to centralise academic information and reduce dependence on paper-based and fragmented academic management processes.

Institutions should also provide adequate training for administrators, teachers, and students to ensure that users can effectively utilise the available system functions.

Appropriate server infrastructure, network connectivity, storage capacity, and backup procedures should be provided to support reliable operation of the system.

5.6.2 Recommendations for System Administrators

System administrators should regularly review user accounts and access permissions to ensure that users retain only the privileges required for their responsibilities.

Regular database backups should be maintained to reduce the risk of academic data loss. System software and dependencies should also be updated regularly to address security and compatibility issues.

Administrators should monitor system performance and storage usage as the volume of academic records and uploaded learning materials increases.

5.6.3 Recommendations for Teachers

Teachers should make effective use of the learning material, assignment, assessment, attendance, communication, and grading functions provided by the system.

Learning materials and academic information should be updated regularly to ensure that students have access to current and relevant resources.

Teachers should also use the electronic assessment and grading facilities consistently to improve the organisation and accessibility of student academic information.

5.6.4 Recommendations for Students

Students should use the system regularly to access learning materials, submit assignments, participate in assessments, monitor grades, and receive academic announcements.

Students should also comply with institutional policies concerning responsible use of digital learning resources and protect their account credentials from unauthorised access.

5.7 Suggestions for Further Research

Although the developed system addresses the major objectives of the study, several areas can be explored in future research.

5.7.1 Mobile Application Development

Future researchers may develop dedicated Android and iOS applications that communicate with the existing RESTful API. This would provide students and teachers with additional access to academic services through mobile devices.

5.7.2 Artificial Intelligence Integration

Future versions of the system may incorporate artificial intelligence features such as intelligent tutoring, automated assessment support, learning analytics, and predictive student performance analysis.

5.7.3 Advanced Learning Analytics

Future research may develop advanced dashboards for analysing student engagement, attendance, assessment performance, and learning progress.

5.7.4 Virtual Learning Integration

Integration with video conferencing and virtual classroom platforms could extend the system's capabilities for institutions that support online and blended learning.

5.7.5 Biometric Attendance

Future studies may investigate biometric technologies such as fingerprint or facial recognition for automated attendance management, subject to institutional requirements and applicable privacy considerations.

5.7.6 Cloud Deployment

Future researchers may investigate cloud deployment to improve system accessibility, infrastructure flexibility, and capacity for handling increased usage.

5.7.7 Multi-Institution Support

The system could be extended to support multiple schools or educational institutions through a multi-tenant architecture, allowing institutions to manage their academic activities independently within a shared platform.

5.8 Limitations of the Study

The study encountered several limitations that should be considered when interpreting the results.

First, the project was subject to time constraints associated with the development and documentation period. These constraints limited the extent to which additional features and broader evaluations could be conducted.

Second, financial limitations affected the availability of advanced infrastructure and resources for large-scale deployment and testing.

Third, the system is web-based and therefore depends on reliable internet connectivity for remote access. Poor or unstable connectivity may affect activities such as login, file uploads, assignment submission, and access to learning materials.

Fourth, testing was conducted within a limited educational environment. Consequently, the results should not be interpreted as a comprehensive evaluation across all types and sizes of educational institutions.

Finally, the system may require additional infrastructure and optimisation if deployed with substantially larger numbers of users, academic records, and uploaded learning resources.

5.9 Contribution of the System

The developed **Ghana Senior High School Learning Management System (GSLMS)** contributes to the improvement of teaching, learning, academic administration, and communication within Senior High Schools. The system provides an integrated digital platform that brings together activities that are traditionally performed through separate manual processes.

One major contribution of the system is the **centralisation of academic information**.

The system provides a common platform for managing students, teachers, courses, learning materials, assessments, grades, attendance, timetables, announcements, and other academic activities. This reduces the dependence on paper-based records and disconnected information systems.

The system also contributes to **improved access to learning resources**. Teachers can upload and manage course materials, assignments, quizzes, and other resources, while students can access relevant learning materials through the platform. This supports learning beyond the physical classroom and allows students to revisit materials when necessary.

Another important contribution is the **digitisation of assessment and grading processes**.

The system provides functionality for managing assessments, recording scores, calculating academic results, and presenting relevant information to authorised users.

This can reduce errors associated with manual calculations and make academic records easier to retrieve.

The system also improves **communication among educational stakeholders**. Through announcements, notifications, and messaging functionality, administrators, teachers, students, and parents can receive and exchange relevant information. This creates a more connected learning environment and can improve the flow of information within the school.

Furthermore, the system provides **role-based access control**. Different users are provided with access according to their responsibilities. Administrators can manage institutional and user information, teachers can manage teaching and assessment activities, students can access learning resources and academic information, and parents can monitor relevant information concerning their wards.

The system also contributes to **academic monitoring and decision-making** through reports and analytics. School administrators and authorised users can obtain information about academic activities, attendance, assessments, and other aspects of the learning process. Such information can support the identification of areas requiring attention and facilitate more informed decisions.

From a technical perspective, the project demonstrates the application of **web-based software engineering principles** in developing an educational information system. The separation of the frontend from the PHP-based API backend provides a structured architecture that can support future expansion and integration with other applications.

Overall, the system contributes to the digital transformation of Senior High School education by providing a centralised, accessible, and structured platform for managing learning and academic activities.

5.10 Future Enhancements

Although the developed system provides a wide range of functionality, several enhancements can be implemented in future versions to increase its effectiveness, scalability, accessibility, and usefulness.

5.10.1 Mobile Application

A dedicated Android and iOS mobile application could be developed to complement the web-based system. The application would allow students, teachers, parents, and administrators to access important services from mobile devices. Push notifications could also be incorporated to provide users with immediate updates about assignments, announcements, grades, and other academic activities.

5.10.2 Artificial Intelligence Integration

Future versions could incorporate Artificial Intelligence (AI) to provide intelligent educational services. For example, AI could analyse student performance and identify students who may require additional academic support. An AI-powered learning assistant could also help students find relevant learning resources and provide explanations for selected topics.

5.10.3 Online and Live Classes

The system could be enhanced with integrated virtual classroom functionality. Video conferencing, live lessons, screen sharing, chat, and recorded lessons could be incorporated to support remote and blended learning.

5.10.4 Advanced Learning Analytics

More advanced analytics could be introduced to analyse student performance over time. The system could generate performance trends, identify weak subject areas, compare assessment results, and provide early-warning indicators for students who are experiencing academic difficulties.

5.10.5 Offline Learning Support

Since reliable Internet connectivity may not always be available to all students, future versions could provide offline access to selected learning materials. Materials could be downloaded while Internet access is available and accessed later without an active connection.

5.10.6 SMS and Email Integration

The notification system could be expanded to support SMS and email communication. This would allow parents and students to receive important notifications even when they are not actively logged into the platform.

5.10.7 Online Examination Module

A comprehensive online examination module could be implemented to support timed examinations, automatic submission, randomised questions, question banks, automatic marking, examination reports, and enhanced examination security.

5.10.8 Improved Security

Future versions should continuously strengthen security through mechanisms such as multi-factor authentication, improved password policies, enhanced session management, security monitoring, rate limiting, encrypted sensitive information, and regular security audits.

5.10.9 Cloud Deployment and Scalability

The system could be deployed on a reliable cloud infrastructure to support multiple Senior High Schools simultaneously. A multi-school architecture would allow institutions to operate independently while using the same underlying platform.

5.10.10 Parent and Guardian Engagement

Additional parent-oriented features could be introduced, including detailed academic progress reports, attendance alerts, assignment notifications, teacher-parent communication, and notifications concerning important school activities.

5.10.11 Integration with External Educational Systems

The system could eventually integrate with other educational platforms and services. Such integrations could facilitate data exchange and reduce duplicate data entry between the LMS and other school information systems.

5.10.12 Automated Backup and Disaster Recovery

Automated database backups, file backups, backup verification, and disaster recovery procedures could be implemented to minimise the risk of permanent data loss resulting from hardware failure, human error, security incidents, or other unexpected events.

5.11 Final Remark

The School Learning Management System developed in this study provides a centralised web-based platform for managing important academic and administrative activities. The system addresses several weaknesses identified in the existing academic management process by integrating user management, courses, learning materials, assignments, assessments, attendance, results, communication, notifications, and reporting within one platform.

The implementation and testing presented in Chapter Four demonstrate that the developed system can perform the major functions for which it was designed. The system therefore provides a practical foundation for improving academic information management and supporting teaching and learning activities.

However, the system should be regarded as a developed academic software solution whose wider institutional deployment would require further testing, user training, infrastructure provision, security review, maintenance, and possible integration with additional educational technologies.

Future improvements can build upon the existing API-first architecture to introduce mobile applications, advanced learning analytics, artificial intelligence, cloud deployment, and other technologies that can further enhance the delivery and management of educational services.

REFERENCES

Adzobu, F. K. (2014). Designing a multimedia learning environment to facilitate learning of fractions at the primary school level in Ghana. *International Journal of Educational Research and Development*, 3(1), 1–10.

Al-Fraihat, D., Joy, M., Masa'deh, R., & Sinclair, J. (2020). Evaluating e-learning systems success: An empirical study. *Computers in Human Behavior*, 102, 67–86.
<https://doi.org/10.1016/j.chb.2019.08.004>

Bates, A. W. (2019). *Teaching in a digital age: Guidelines for designing teaching and learning* (2nd ed.). Tony Bates Associates Ltd.

Bass, J. M., & Heeks, R. (2011). Small and medium enterprise systems adoption in developing countries: Learning from a case study in Ghana. *Information Technology for Development*, 17(2), 1–17.

Bates, A. W. (2015). *Teaching in a digital age: Guidelines for designing teaching and learning*. Tony Bates Associates Ltd.

Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R. C., Mellor, S., Schwaber, K., Sutherland, J., & Thomas, D. (2001). *Manifesto for agile software development*. Agile Alliance. <https://agilemanifesto.org/>

Connolly, T., & Begg, C. (2015). *Database systems: A practical approach to design, implementation, and management* (6th ed.). Pearson.

Creswell, J. W., & Creswell, J. D. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). SAGE Publications.

Dabbagh, N., & Kitsantas, A. (2012). Personal learning environments, social media, and self-regulated learning: A natural formula for connecting formal and informal learning. *The Internet and Higher Education*, 15(1), 3–8. <https://doi.org/10.1016/j.iheduc.2011.06.002>

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>

Food and Agriculture Organization of the United Nations. (2022). *Digital learning for agriculture*. FAO.

Garrison, D. R., Anderson, T., & Archer, W. (2000). Critical inquiry in a text-based environment: Computer conferencing in higher education. *The Internet and Higher Education*, 2(2–3), 87–105. [https://doi.org/10.1016/S1096-7516\(00\)00016-6](https://doi.org/10.1016/S1096-7516(00)00016-6)

International Organization for Standardization. (2018). *ISO/IEC 25010:2011 systems and software engineering—Systems and software quality requirements and evaluation (SQuaRE)—System and software quality models*. ISO.

International Telecommunication Union. (2023). *Measuring digital development: Facts and figures 2023*. International Telecommunication Union.

Laudon, K. C., & Laudon, J. P. (2022). *Management information systems: Managing the digital firm* (17th ed.). Pearson.

Moodle. (2024). *Moodle documentation*. Moodle. <https://docs.moodle.org/>

Mozilla Developer Network. (2024). *MDN Web Docs*. Mozilla.
<https://developer.mozilla.org/>

Nielsen, J. (1994). *Usability engineering*. Morgan Kaufmann.

Organisation for Economic Co-operation and Development. (2021). *Digital education outlook 2021: Pushing the frontiers with AI, blockchain and robots*. OECD Publishing.
<https://doi.org/10.1787/589b283f-en>

Pressman, R. S., & Maxim, B. R. (2020). *Software engineering: A practitioner's approach* (9th ed.). McGraw-Hill Education.

Sommerville, I. (2016). *Software engineering* (10th ed.). Pearson.

W3C. (2023). *HTML: The living standard*. World Wide Web Consortium.
<https://html.spec.whatwg.org/>

W3C. (2023). *Cascading Style Sheets (CSS)*. World Wide Web Consortium.
<https://www.w3.org/Style/CSS/>

World Bank. (2020). *Remote learning during COVID-19: Lessons from today's education technology*. World Bank.

World Wide Web Consortium. (2024). *Web standards*. <https://www.w3.org/standards/>

Technical References

Apache Software Foundation. (2024). *Apache HTTP Server documentation*.
<https://httpd.apache.org/docs/>

MySQL. (2024). *MySQL 8.0 reference manual*. Oracle Corporation.
<https://dev.mysql.com/doc/refman/8.0/en/>

PHP Documentation Group. (2024). *PHP manual*. <https://www.php.net/docs.php>

OWASP Foundation. (2024). *OWASP Top 10: The ten most critical web application security risks*. <https://owasp.org/www-project-top-ten/>

OWASP Foundation. (2024). *OWASP API Security Top 10*. <https://owasp.org/API-Security/>

Postman. (2024). *Postman documentation*. <https://learning.postman.com/docs/>

GitHub. (2024). *GitHub documentation*. <https://docs.github.com/>

APPENDICES

The appendices contain supplementary materials that support the design, development, implementation, testing, and use of the **Ghana Senior High School Learning Management System (GSLMS)**. The materials presented in the appendices provide additional technical and practical information that may not be necessary to include in the main chapters but are important for demonstrating the development and functionality of the system.

APPENDIX A: SYSTEM SCREENSHOTS

This appendix presents screenshots of the major interfaces of the Ghana Senior High School Learning Management System.

A.1 Login Interface

Figure A.1: GSLMS Login Page

The login page provides authenticated access to the system. Users provide their registered credentials before accessing the appropriate dashboard.

A.2 Super Administrator Dashboard

Figure A.2: Super Administrator Dashboard

The Super Administrator dashboard provides an overview of system-wide activities and administrative functions.

A.3 Administrator Dashboard

Figure A.3: Administrator Dashboard

The Administrator dashboard provides access to school-level management functions, including users, academic activities, courses, and reports.

A.4 Teacher Dashboard

Figure A.4: Teacher Dashboard

The Teacher dashboard enables teachers to manage courses, learning materials, assignments, quizzes, assessments, grades, attendance, and related activities.

A.5 Student Dashboard

Figure A.5: Student Dashboard

The Student dashboard provides students with access to courses, learning materials, assignments, quizzes, grades, attendance information, announcements, and other relevant academic activities.

A.6 Parent Dashboard

Figure A.6: Parent Dashboard

The Parent dashboard provides parents or guardians with access to relevant information concerning their wards, including academic performance and attendance.

A.7 Course Management Interface

Figure A.7: Course Management Interface

A.8 Learning Materials Interface

Figure A.8: Learning Materials Management Interface

A.9 Assignment Management Interface

Figure A.9: Assignment Management Interface

A.10 Quiz Management Interface

Figure A.10: Quiz Management Interface

A.11 Assessment and Grading Interface

Figure A.11: Assessment and Grading Interface

A.12 Attendance Management Interface

Figure A.12: Attendance Management Interface

A.13 Timetable Interface

Figure A.13: Timetable Management Interface

A.14 Announcement Interface

Figure A.14: Announcement Management Interface

A.15 Messaging Interface

Figure A.15: Messaging Interface

A.16 Notification Interface

Figure A.16: Notification Interface

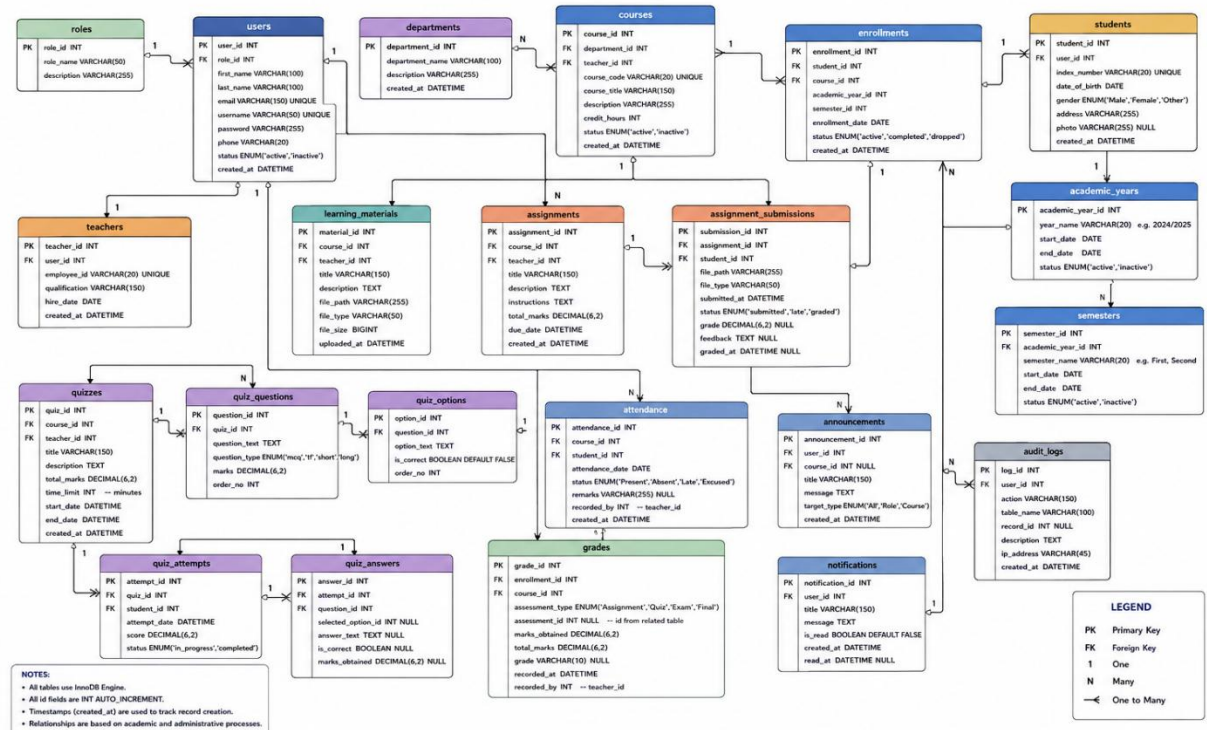
A.17 Reports and Analytics Interface

Figure A.17: Reports and Analytics Interface

This appendix presents the database structure used by the Ghana Senior High School Learning Management System.

B.1 Database Schema

ENTITY RELATIONSHIP DIAGRAM (ERD)



B.2 Major Database Entities

Entity	Purpose
Users	Stores system user information and authentication-related data
Roles	Defines the roles assigned to system users
Institutions	Stores information about participating schools
Students	Stores student-specific information
Teachers	Stores teacher-specific information
Parents	Stores parent or guardian information
Courses	Stores information about academic courses
Course Materials	Stores learning resources associated with courses
Assignments	Stores assignment information
Submissions	Stores student assignment submissions
Quizzes	Stores quiz information
Questions	Stores questions used in quizzes and assessments
Grades	Stores student assessment results
Attendance	Stores student attendance records
Timetables	Stores academic timetable information

Announcements	Stores school and academic announcements
Notifications	Stores system notifications
Messages	Stores communication between authorised users
Audit Logs	Records important system activities

APPENDIX C: API ENDPOINTS

The GSLMS uses a PHP-based Application Programming Interface (API) to facilitate communication between the frontend application and backend services.

C.1 Authentication API

Method	Endpoint	Purpose
POST	/auth/login	Authenticate a user
POST	/auth/logout	End an authenticated session
POST	/auth/refresh	Refresh authentication information
GET	/auth/me	Retrieve authenticated user information

C.2 User Management API

Method	Endpoint	Purpose
GET	/users	Retrieve users
GET	/users/{id}	Retrieve a specific user
POST	/users	Create a user
PUT	/users/{id}	Update a user
DELETE	/users/{id}	Delete a user

C.3 Course Management API

Method	Endpoint	Purpose
GET	/courses	Retrieve courses
GET	/courses/{id}	Retrieve a specific course
POST	/courses	Create a course
PUT	/courses/{id}	Update a course
DELETE	/courses/{id}	Delete a course

C.4 Learning Materials API

Method	Endpoint	Purpose
GET	/materials	Retrieve learning materials
POST	/materials	Upload learning material
PUT	/materials/{id}	Update learning material
DELETE	/materials/{id}	Delete learning material

C.5 Assessment API

Method	Endpoint	Purpose
GET	/assessments	Retrieve assessments
POST	/assessments	Create an assessment
PUT	/assessments/{id}	Update an assessment
DELETE	/assessments/{id}	Delete an assessment

APPENDIX D: SAMPLE API REQUESTS AND RESPONSES

D.1 Login Request

```
{  
  
  "email": "student@example.com",  
  
  "password": "*****"  
}
```

D.2 Successful Login Response

```
{  
  
  "success": true,  
  
  "message": "Login successful",  
  
  "data": {  
  
    "user": {
```

```
    "id": 1,  
  
    "name": "Student Name",  
  
    "role": "student"  
  
  },  
  
  "token": "JWT_TOKEN"  
  
}  
  
}
```

D.3 Course Request

```
{  
  
  "name": "Information and Communication Technology",  
  
  "code": "ICT101",  
  
  "description": "Introduction to ICT"  
  
}
```

D.4 Example API Response

```
{  
  
  "success": true,  
  
  "message": "Course retrieved successfully",  
  
  "data": {
```

```
"id": 1,  
  
"name": "Information and Communication Technology",  
  
"code": "ICT101"  
  
}  
  
}
```

APPENDIX E: SYSTEM TEST CASES

E.1 Authentication Testing

Test ID	Test Description	Expected Result	Status
TC-001	Login with valid credentials	User is authenticated	Pass
TC-002	Login with invalid password	Error message is displayed	Pass
TC-003	Login with empty fields	Validation message is displayed	Pass
TC-004	Logout	User session is terminated	Pass

E.2 User Management Testing

Test ID	Test Description	Expected Result	Status
---------	------------------	-----------------	--------

TC-005	Create user	User account is created	Pass
TC-006	Update user	User information is updated	Pass
TC-007	Delete user	User is removed/deactivated	Pass
TC-008	Retrieve user	Correct user information is displayed	Pass

E.3 Course Testing

Test ID	Test Description	Expected Result	Status
TC-009	Create course	Course is created	Pass
TC-010	Update course	Course information is updated	Pass
TC-011	View course	Course information is displayed	Pass
TC-012	Delete course	Course is removed	Pass

E.4 Assessment Testing

Test ID	Test Description	Expected Result	Status
---------	------------------	-----------------	--------

TC-013	Create assessment	Assessment is created	Pass
TC-014	Submit assessment	Submission is recorded	Pass
TC-015	Record grade	Grade is stored successfully	Pass
TC-016	View grade	Correct grade is displayed	Pass

APPENDIX F: USER MANUAL

F.1 Student User Guide

Step 1: Login

The student opens the GSLMS login page and enters the registered username/email and password.

Step 2: Access Dashboard

After successful authentication, the student is redirected to the Student Dashboard.

Step 3: Access Courses

The student selects the Courses section to view registered courses.

Step 4: Access Learning Materials

The student selects a course and accesses the available learning materials.

Step 5: Complete Assignments

The student opens an available assignment, reviews the instructions, completes the required work, and submits it through the system.

Step 6: Take Quizzes

The student selects an available quiz and completes it within the specified conditions.

Step 7: View Grades

The student can access available academic results through the grades section.

Step 8: View Attendance

The student can review attendance information recorded by authorised teachers.

APPENDIX G: TEACHER USER GUIDE

Teachers can perform the following activities:

1. Log into the system.
2. Access the Teacher Dashboard.
3. Manage assigned courses.
4. Upload learning materials.
5. Create assignments.
6. Review student submissions.
7. Create quizzes.
8. Manage assessment questions.

9. Record or review grades.
10. Record student attendance.
11. View relevant reports.
12. Communicate with students and other authorised users.
13. Publish relevant announcements.

APPENDIX H: ADMINISTRATOR USER GUIDE

Administrators can:

1. Log into the administrative dashboard.
2. Manage students.
3. Manage teachers.
4. Manage parents or guardians.
5. Manage courses.
6. Manage academic years and semesters.
7. Manage institutional information.
8. Monitor academic activities.
9. Manage announcements.
10. View reports and analytics.
11. Monitor system activities.

12. Manage authorised users and permissions.

APPENDIX I: SYSTEM INSTALLATION GUIDE

I.1 Software Requirements

The system requires an appropriate web-server environment capable of running PHP and MySQL.

Recommended components include:

- PHP
- MySQL
- Apache or Nginx
- Web browser
- Composer
- Git

I.2 Backend Installation

1. Clone the project repository.
2. Navigate to the backend directory.
3. Install the required PHP dependencies.
4. Configure the environment variables.
5. Create the required MySQL database.
6. Import the database schema.

7. Configure the web server.
8. Start the backend API.
9. Test the API endpoints.

I.3 Frontend Installation

1. Navigate to the frontend directory.
2. Configure the API base URL.
3. Ensure the frontend can communicate with the PHP API.
4. Open the application through the configured web server.
5. Test user authentication and system functionality.

Security note: Production installations should use secure environment variables and should not expose database passwords, JWT secrets, API credentials, or other sensitive configuration values.

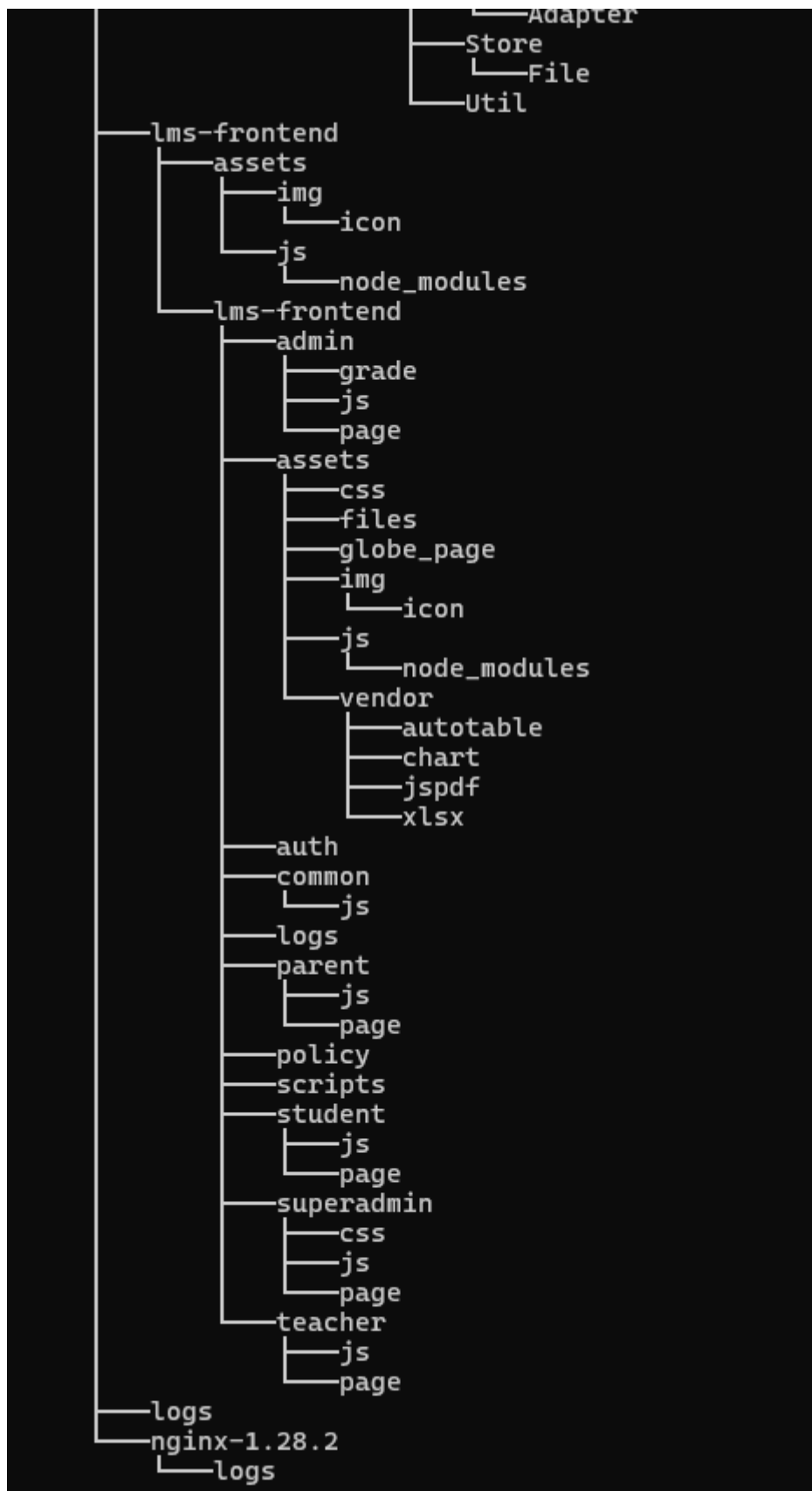
APPENDIX J: PROJECT REPOSITORY STRUCTURE

The project is maintained using Git version control.

The major project components include:

```
D:.\
├── Learning-mangt-sys-by-php
│   ├── .github
│   │   └── skills
│   │       └── design
│   │           ├── data
│   │           │   └── icon
│   │           └── scripts
│   │               └── icon
│   ├── .sql file
│   ├── docs
│   │   └── 1-5
│   └── lms-api
│       ├── config
│       ├── logs
│       ├── migrations
│       ├── public
│       ├── scripts
│       ├── src
│       │   ├── Config
│       │   ├── Controllers
│       │   ├── Middleware
│       │   ├── Repositories
│       │   ├── Routes
│       │   ├── Services
│       │   └── Utils
│       ├── uploads
│       │   ├── assignments
│       │   ├── chat-avatars
│       │   ├── general
│       │   ├── institutions
│       │   │   └── accra-shs
│       │   ├── materials
│       │   ├── profiles
│       │   └── quiz-questions
│       └── vendor
│           ├── composer
│           ├── firebase
│           │   └── php-jwt
│           │       └── src
│           ├── graham-campbell
│           │   └── result-type
│           │       └── src
│           ├── phpmailer
│           │   └── phpmailer
│           │       ├── language
│           │       └── src
│           ├── phpoption
│           │   └── phpooption
│           │       └── src
```

```
└─PhpOption
└─symfony
  └─polyfill-ctype
  └─polyfill-mbstring
    └─Resources
      └─unidata
  └─polyfill-php80
    └─Resources
      └─stubs
└─vlucas
  └─phpdotenv
    └─src
      └─Exception
      └─Loader
      └─Parser
      └─Repository
        └─Adapter
      └─Store
        └─File
      └─Util
└─lms-frontend
  └─assets
    └─img
      └─icon
    └─js
      └─node_modules
  └─lms-frontend
    └─admin
      └─grade
      └─js
      └─page
    └─assets
      └─css
      └─files
      └─globe_page
      └─img
        └─icon
      └─js
        └─node_modules
    └─vendor
      └─autotable
      └─chart
      └─jspdf
      └─xlsx
  └─auth
  └─common
    └─js
  └─logs
  └─parent
    └─js
    └─page
  └─policy
  └─scripts
  └─student
    └─js
    └─page
```



The backend provides API services while the frontend provides the user-facing interfaces.

APPENDIX K: SAMPLE DATABASE OPERATIONS

K.1 Create a Course

INSERT INTO courses

(name, code, description)

VALUES

('Information and Communication Technology',

'ICT101',

'Introduction to ICT');

K.2 Retrieve Courses

SELECT *

FROM courses

ORDER BY name ASC;

K.3 Retrieve Student Grades

SELECT *

FROM grades

WHERE student_id = ?;

K.4 Retrieve Attendance Records

SELECT *

FROM attendance

WHERE student_id = ?;

APPENDIX L: SECURITY CONSIDERATIONS

The system incorporates security mechanisms intended to protect users and academic information.

Key security considerations include:

1. User authentication.
2. Role-based access control.
3. JWT-based authentication.
4. Password protection.
5. Input validation.
6. API authorisation.
7. Session management.
8. Database access controls.
9. Audit logging.
10. Protection against unauthorised access.

Sensitive configuration information should be stored outside publicly accessible source files and should not be committed to the public repository.

APPENDIX M: PROJECT GANTT CHART

Table M.1: Project Implementation Schedule

Activity	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
Requirement gathering	✓				
Literature review	✓	✓			
System analysis		✓			
System design		✓	✓		
Database development			✓		
Backend development			✓	✓	
Frontend development			✓	✓	
System integration				✓	
Testing				✓	✓

Documentation	✓	✓	✓	✓	✓
Final evaluation					✓

APPENDIX N: PROJECT TEAM

The project was developed by:

GROUP ONE (1)

Project Title

DESIGN AND IMPLEMENTATION OF A GHANA SENIOR HIGH SCHOOL LEARNING MANAGEMENT SYSTEM (GSLMS)

The project focuses on the design and implementation of a web-based platform for supporting teaching, learning, assessment, academic administration, communication, and monitoring within Senior High Schools in Ghana.